

Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8
and read off the Standard Model: status assessment and reading guide

Stefan Hamann

Alessandro Rizzo

June 22, 2026 · v5.2 (rev 237)

In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\text{car}} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* (v_{geo} , G_{net} , F_{transfer}), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
5. **tfpt_5_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R.** **tfpt_research_contracts** — the open interfaces (v_{geo} , G_{net} , F_{transfer}) as numbered contracts; **H.** **tfpt_horizon_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B , axion, m_p/m_e)	[C]

The single open *structural theorem* is G_{net} (= SEAM.EQUIV.01: the raw seam *is* the holomorphic $(E_8)_1$ net), a constructive-AQFT theorem whose only open input is the continuum Osterwalder–Schrader reconstruction of the gapped quasi-free collar. QGEO.SYM.01 (the μ_4 deck acting geometrically) is now its *corollary*: a conformal net’s vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The full nonperturbative quantum-gravity measure QG.AMB.01 is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), from which TFPT is rigorously *gap-decoupled* (margin $1.648 > 0$, v76/v330) — an inherited, decoupled problem, *not* a TFPT-specific open item. So the honest residual is *one* solvable theorem (G_{net}) plus the typed v_{geo} and F_{transfer} interfaces. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels (U_{wall} , G_{metric} , F_{frontier}) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.

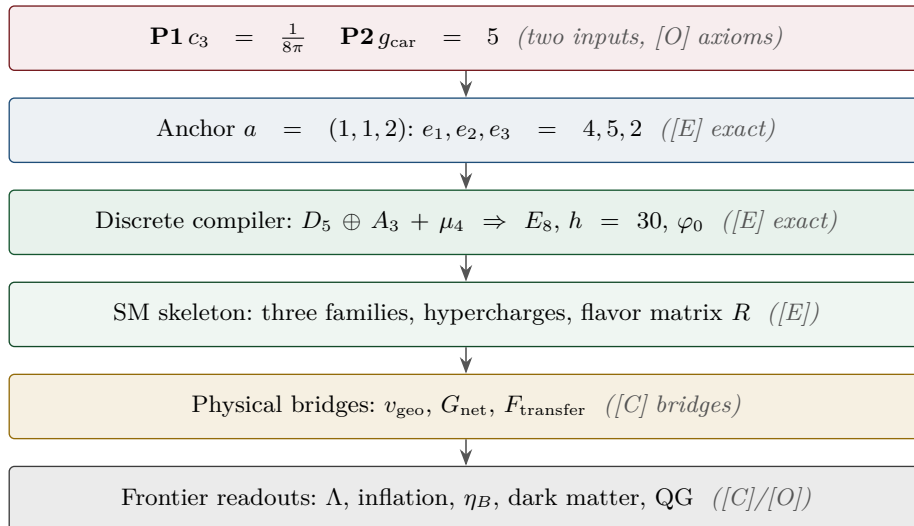
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



Contents

1 What this is about: the jump in construction principle

4

2 The architecture at a glance

6

3	The dependency DAG and the proof ledger	7
4	The compiler: why $30 = 2 \cdot 3 \cdot 5$	11
5	The μ_4 glue: how E_8 is really built	13
6	How derivation now works: the derivation map	13
7	Physical, geometric, topological: why the architecture is plausible	14
8	What role E_8 now plays exactly	15
9	What this means for the theory-of-everything question	15
10	Does this make the theory more probable?	18
11	Predictions for running and upcoming experiments	19
	Conclusion	21
	Appendix: computational verification	21
	Appendix: changelog	73

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

Physical — under hypotheses	RG masses, QG QG.AMB.01
Numerical — fit-free fixed point	$\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01
Lattice — Lie / glue / root system	$E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01
Formal — Lean / exact script	P2 algebra FORM.P2.02
Axiom — declared inputs	P1 c_3 , P2 g_{car} AX.P1.01

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[E]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L) ; (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{1, a\}$; (v) a Plücker coordinate of the anchor plane $\langle 1, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never

spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);
4. the **single status ledger verification**/`status_ledger.csv` — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8 , and E_8 is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number π .

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the big Lie data directly: $p_1=4=|\mu_4|$, $p_2=6=|R^+(A_3)|$, $p_3=10=\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$.
[`verification/v23_anchor_generator.py`]

And a is itself *half-forced* ([E]/[C]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing

to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why $a = (1, 1, 2)$ ” but “why the carrier $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with c_3) is the [O] input layer.

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1, 1, 2)$ plus π . The *single structural bedrock postulate* is the keystone SEAM.EQUIV.01, whose conformal-deck face QGEO.SYM.01 (the carrier μ_4 clock is the seam’s conformal deck, see `tfpt_research_contracts`) forces the μ_4 deck, hence $N_{\text{fam}} = 3$, $\deg E = -4$ and the tripartition $4 = 2+1+1$, of which c_3 and g_{car} are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local $\Rightarrow \omega \circ \rho = \omega$ implication are now Lean-formalised (FORM.QGEO.02/FORM.QGEO.01); the postulate itself stays [O].

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself (`[verification/v197_rr_carrier_clifford_d5.py]`): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3 , D_5 , c_3 and E_8 (`[verification/v189_riemann_roch_carrier.py]`) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0 , α^{-1} , the exponential scales) flows from π ; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad (`[verification/v119_review_validation_2.py]`). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3 + e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\text{rank } E_8 = p_0 + e_2$.

$$a = (1, 1, 2) \quad [\text{E}]$$

elementary $e_k(a)$

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

power sums $p_n = 2 + 2^n$

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

derived motifs

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

The companion documents

The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

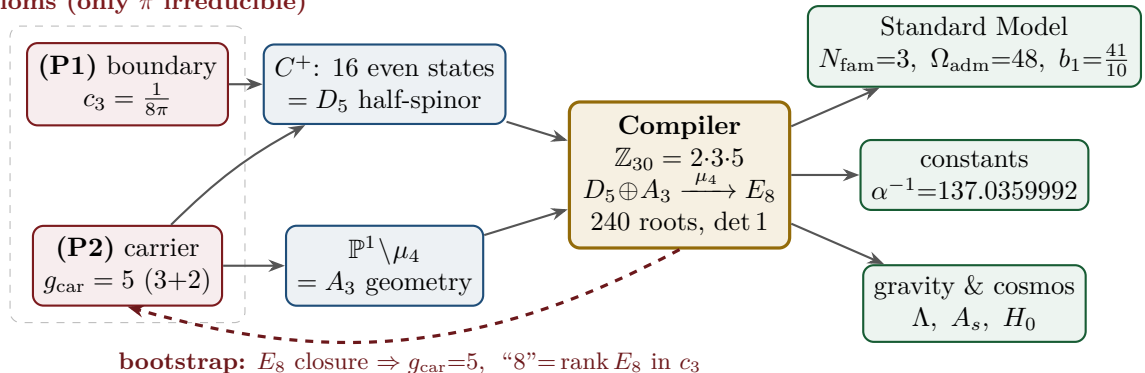
Document	What it contains (former notes now as parts)
tfpt_1_architecture_e8	Core. The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$, the explicit $D_5\oplus A_3+\mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
tfpt_2_standard_model	Standard Model. The full SM in one φ_0^{ret} -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ n_s, r , Starobinsky M , derived θ_{12} , quark c , the H2 splitting type, Λ -typing (Part III).
tfpt_3_e8_audit_bootstrap	E_8 audit & bootstrap. The seven E_8 slices as an audit raster ($E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in c_3 as overdetermined fixed points, only π irreducible (Part III).
tfpt_4_frontier	Frontier. The honest status of η_B , m_p/m_e , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.
tfpt_5_redteam	Red Team. The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
tfpt_research_contracts	Research contracts. The open gates as numbered contracts with lemma chains and closing theorems: (U_{wall}) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
tfpt_horizon_readouts (App. H)	Appendix H — horizon unit system (reframe, not new physics). One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ($1920= W(D_5) $, $ \mu_4 =4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
origin_theory	Origin Theory. Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and π (the dimensionful v_{geo} and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only π irreducible)



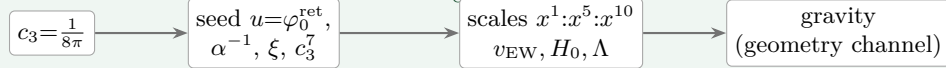
Key statement (the loop closes): previously D_5 , A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

Engine 1 — discrete closure (from $g_{\text{car}} = 5$)

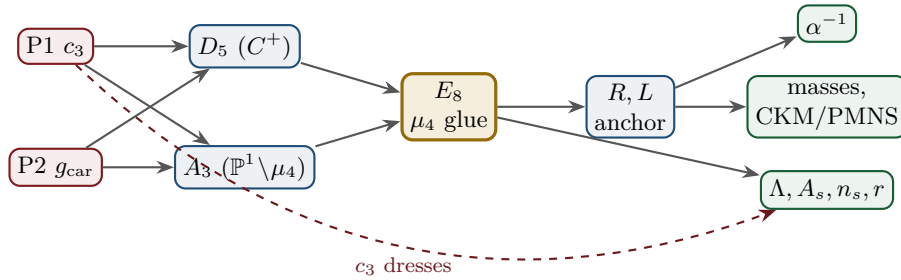


Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$)



3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, $c_3=1/(8\pi)$	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\text{car}}=5$	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+=16, Y$	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\text{fam}}=3, \mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$	D_5, A_3	[E]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[E]	$\det 8, \text{ minors } (2, 3, 5)$	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)=0$	$c_3, M=41$	[E]	137.0359992	no/second root (excl.)
masses	φ_0^{ret} -ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam $\varepsilon=c_3$	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.2 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R -bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, **v75** — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, **v76**) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (**v77**), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (**v78**), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned (**[E]** for the attractor, `[verification/v56_unique_attractor.py]`; the *physical identification* of the transport operator stays **[C]**), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — `[verification/v6_bootstrap.py]`, `[verification/v14_carrier_uniqueness.py]`). (ii) *Newton’s constant is not an independent input:* fixing the physical Λ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ relative to Λ , so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor (**[E]/[C]**, `[verification/v60_lambda_metrology_branch.py]`), not predicted from pure numbers. What stays irreducible is then just π , the discrete E_8 -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \rightarrow v_{\text{geo}}$, **v75**) and gravity ($1/G$, **v68**): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$ are kept only for ledger continuity). In current terms: $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, **v49/v71/v75**), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ — reduced to a single boundary-net theorem, “*the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net*” (holomorphy then follows, **v89/GATE.METRIC.06**); since the QBL chain (**v108–v113**) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (**v110**), the certified channel counts the code identically (**v112**), pair transport is minimally complete (**v111**), and the one quasi-free kernel — rank = c : 5 carrier block, 8 seam hull

— is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [E]; $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers; the four are one typed functor with a four-axiom contract (CONTRACT.F.01, v213) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with F_{pole} at the *seam* rate $(2/3)^6$ and the other three external (v303). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion G_{net} is the index-4 μ_4 simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ (Simple-Current Extension Theorem, v154, exact: index 4, $c=8$, $\mu=1 \Rightarrow$ holomorphic $\Rightarrow E_8$); net existence and full-cone reflection positivity are discharged to [E] (v175); and the **No-Unit Theorem** (v153) makes v_{geo} an irreducible metrology primitive ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). What stays open is the *one seam-side identification*, now packaged as the single named theorem SEAM.EQUIV.01 — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — whose conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face (per v323; the residual is Lean-pinned to named cited steps, v308). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[E]	gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)
Newton constant G_N	[E]/[C]	not <i>independent</i> : branch pins $\bar{M}_{\text{PI}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
flavor absolute scale	[E]/[O]	ratios <i>closed</i> (v49/v71); only the absolute scale remains $= v_{\text{geo}}$ (Gate 1, v75), the same anchor as $1/G$
$R + R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[E]/[C]	gap-decoupled admissible measure ($\Delta_{\text{eff}}=1.648>0$)
ambient QG measure	[O](reduced)	Gate 2 tier B : holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1 \otimes (A_3)_1 \times \langle (1,1) \rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the already-open A_2 net-existence + the keystone SEAM.EQUIV.01 — zero new gates (v160–v165)
v_{geo} metrology unit	[O](primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\text{src}} + \varphi_0/24$ as source $\rightarrow$ pole transfer conjecture
axion DM	[C]/[O]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[C]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[O]	cross-sector QCD/EW ratio
N_*	[C]	reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection $g=5$ (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = c); residue = the G_{net} premise itself — gate closes $\Rightarrow$ carrier [E]
selector establishment $R4'$	[E]/[C]	$(d, n) \Rightarrow R$ columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment
sheet/parity (Q-geometry derived)	[E]/[C]	H^1 characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification, folded into SEAM.EQUIV.01
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born ² Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

The structural open core is the *single* named keystone **SEAM.EQUIV.01** (the G_{net} inclusion: the raw RP seam = $(E_8)_1$ at $\tau=i$); its conformal-deck face **QGEO.SYM.01** is now a *corollary* (a conformal net's vacuum is rotation-invariant by axiom, v323/[`verification/v335_seam_equiv_unify.py`]), not a second item. v_{geo} is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes. G_{net} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\text{eff}}=1.648>0$ isolates the closed admissible sector from the un-built ambient), so the ambient measure **QG.AMB.01** is *not* a TFPT structural item at all: it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited and decoupled — its absence does not affect any TFPT readout, it only blocks certification as a strict physical

TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority on the v_{geo} scale anchor and the G_{net} /SEAM.EQUIV.01 inclusion theorem. [\[verification/v36_spectral_action_g2.py\]](#)

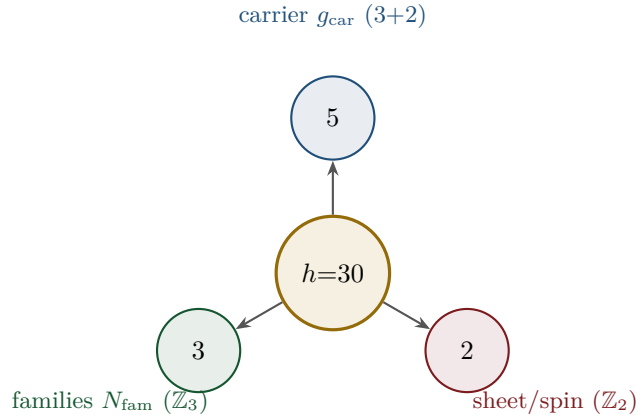
Sharper still: the structural residual is *one* condition. G_{metric} , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “*the seam carries no nontrivial abelian sector*” — holomorphy (μ -index 1), a homology-sphere seam link (Γ perfect $\Leftrightarrow 2I$, [\[verification/v232_e8_kleinian_seam.py\]](#)), exactly one 1-dim irrep ([\[verification/v219_icosahedral_mckay.py\]](#)) — all of which force E_8 ($\#(\text{mark-1}) = |H_1| = 1$ only for E_8 ; holomorphic $c = 8 = g_{\text{car}} + N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic $\Leftrightarrow \det K = 1$, the extension tower $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$ being anyon condensation (the Kitaev E_8 state); the one open analytic step is then “the free RP seam condenses the order- $|\mu_4|$ Lagrangian glue” = the single named keystone SEAM.EQUIV.01 (with QGEO.SYM.01 its downstream conformal-deck face). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ($f_2/f_0 = 1$, v259), the seam, the carrier-16 and E_8 live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone SEAM.EQUIV.01 and adds *no new open item*, while the ambient quantum-gravity measure stays separate by design. [\[verification/v234_seam_holomorphy_selection.py\]](#) [\[verification/v235_seam_chern_simons.py\]](#)

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence*. And even those two tighten: the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8 -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence: E_8 is the exceptional top of the tower of finite $SU(2)$ subgroups, so its branch data is the icosahedron and the three atoms (2, 3, 5) are its rotation-axis orders — the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees equal to the affine- E_8 Kac marks, $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$ ([\[verification/v219_icosahedral_mckay.py\]](#)). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [\[verification/v6_bootstrap.py\]](#) [\[verification/v14_carrier_uniqueness.py\]](#) [\[verification/v23_anchor_generator.py\]](#) [\[verification/v219_icosahedral_mckay.py\]](#)

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

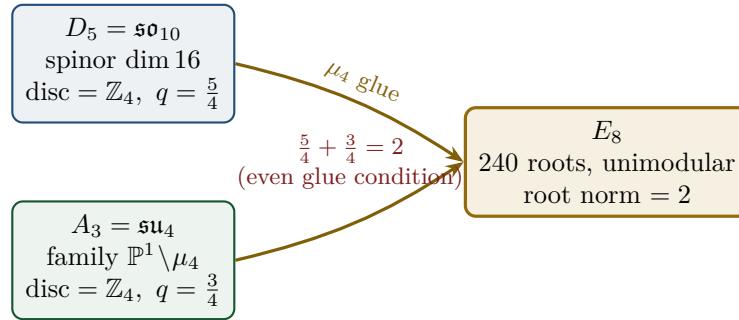


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [E]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30} ; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the A_3 Coxeter element. [E]
- **why these three primes:** E_8 is the icosahedral top of the McKay tower ($2I$, order 120, irrep degrees = affine- E_8 marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the E_8 exponents $(\mathbb{Z}/30)^\times$ carry an order-4 totative character $\langle 7 \rangle$ — the same μ_4 seam clock. [E] [verification/v219_icosahedral_mckay.py] [verification/v223_coxeter_totative_clock.py]
- **a network reading (computational substrate).** On the same McKay (2, 3, 5) graph the affine- E_8 marks are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A + 2I)/4 \cdot m$ (it converges from *any* positive start to (1, 2, 3, 4, 5, 6, 4, 2, 3)); the spherical tower terminates at E_8 and the cascade $E_8 \supset E_7 \supset E_6 \dots$ is nested-subgraph coarse-graining. So E_8 reads as the *universal attractor of a minimal (2, 3, 5) network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing the *full* structure (lattice + recovery + readouts) stays open, and φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction. [O] [verification/v298_e8_network_attractor.py]
- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate $Am \leq 2m$, Collatz–Wielandt = Smith's $\rho \leq 2$; the Perron eigenvector enters only as an audit) *reaches* E_8 as the last finite point ($E_6 \rightarrow E_7 \rightarrow E_8$, then stops at the affine boundary $\hat{E}_8$) — but *only* from a 3-arm (2, 3, $\cdot$) seed (the controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n , so neither family ever reaches E_8). So P_2 is exactly the irreducible seed on which the rule reaches E_8 as the last finite point — a second network reading of the (2, 3, 5) hull, not a derivation of P_2 . [O] [verification/v299_hypergraph_0d_autonomous.py]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ($Am=2m$) and the subleading eigenvalue is exactly $2 \cos(\pi/5)$ = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate $(2/3)^6$ is *not* any eigenvalue of the adjacency and φ_0^{ret} is not a rational edge fraction. So a substrate reproducing the *full* structure must inject exactly two non-graph-spectral data — the cusp weight $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ (the gap $6 \ln(3/2)$) and the analytic seed φ_0^{ret} — a precise delimiting of the open question. [O] [verification/v312_hypergraph_substrate.py]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1_e8_glue.py]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier $3+2$, $\dim S^+ = 16$, $N_{\text{fam}} = 3$, $\Omega_{\text{adm}} = 48$	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \frac{N_{\star}^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4\text{-glue}$	[E]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status_ledger.csv.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

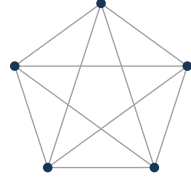
Typing note on Λ (a deliberate danger zone). “ Λ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5 :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit x^1 , Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. “Hubble = slot product, Λ = pair product.” [E]



K_5

- $\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
- $\binom{5}{2} = 10$ edges — Λ density pair x^{10}
- $\binom{5}{3} = 10$ triangles — dual sector
- $\binom{5}{4} = 5$ complements — inverse sector
- $\binom{5}{5} = 1$ determinant — closure

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\text{tr} = 9 = N_{\text{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret} -ladder $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, **v49/v71**), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). (Full derivation: document 2, **tfpt_2_standard_model**.) [verification/v18_quark_yukawa.py] [verification/v20_lepton_c_derivation.py] [verification/v46_grand_mass_volume.py]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1 \setminus \mu_4$ produce D_5 and A_3 , glued to E_8 ; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30} /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain “everything”; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8 .

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R , the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\text{Spec}(Q_+)$ from D_4 , `v69/v70`; since `v136/v139` the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\text{Spec}(Q_+) = (1, 2, 3)$ is the A_3 -exponent grading on the seam curve’s cohomology, `v137`), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, `v49/v71`) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point} , an anchor; Research Contract 1). [\[E\]](#)/[\[O\]](#)
2. **Gravity — exact in the R^2 regime.** Spectral action $\rightarrow R+R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\rightarrow$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n \geq 3} \rightarrow R^2$, valid for curvature $\ll M^2$). [\[E\]](#) in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}=0$ follows from three structural facts (γ_5 -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [\[E\]](#) (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (`v258`), the spectral-action cutoff *is* that KMS weight (`v259`), the seam/carrier/ E_8 live on one Kummer/K3 (`v260`), all certified together (`v261`) — closed modulo the single keystone SEAM.EQUIV.01. *What this does not close*: the full measured pole masses (the absolute scale v_{geo} + standard RG), the absolute QCD scales (m_p/m_e , [\[O\]](#)), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [\[C\]](#)/[\[O\]](#)), and the real 4D interacting QFT / ambient measure ([\[O\]](#)). [\[E\]](#) (transfer model + boundary relative object) / [\[C\]](#) (continuum) / [\[O\]](#) (real 4D)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (3357 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [\[E\]](#)/[\[O\]](#)

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.*)

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam} , hypercharge, b_1	[E]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[E]	— complete
bootstrap $(g, \mu) = (5, 4)$, “8” in c_3	[E]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[E]	— existence+uniqueness proved ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]
fermion masses (φ_0^{ret} -ladder)	[E]/[O]	leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)	[E]/[C]	θ_{12} cond. derived; θ_{23} maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]
flavour matrix / H2	[E]	dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d, n) (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map
seam quantum clock (R1)	[E]/[C]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r , scalaron M	[C]	scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; n_s, r are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)
boundary QFT (Modular Spectral Closure)	[E]/[C]	one relative object TFPT _{QFT} : D_F = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ E_8 on one K3 (v260), certified (v261) — closed modulo SEAM.EQUIV.01
perturbative 4D S-matrix (S_{pert})	[E]/[C]	Epstein–Glaser-constructible (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278) — matter+gauge; gravity R^2/Weyl^2 non-unitary (Stelle ghost)
<i>genuinely open — the actual TOE remainder (what the closure does not cover)</i>		
QFT measure (admissible sector)	[E]	closed: RP from cusp spectrum, OS reconstruction
full measured pole masses	[C]	ladder $\times v_{\text{geo}}$, dressed by standard RG (no new physics); v_{geo} over-determined (v274)
absolute QCD scales (m_p/m_e , Λ_{QCD})	[O]	QCD matching contract (FR.MPME.01); not a compiler number
full PMNS dynamics (mass scale, CP)	[C]/[O]	angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute ν -scale = one seesaw ratio (v272)
(1) SEAM.EQUIV.01 — one named theorem (<i>raw seam</i> = $(E_8)_1$ at $\tau=i$)	[O]	the single arrow unifying the flat- $\tau=i$ geometry and the det $K=1$ holomorphy (v282/v286, import firewall). After the closing arc (v300–v302) its residual is a composition of <i>cited</i> theorems over <i>established</i> TFPT facts: Flat-Away hardened to a discrete obstruction + its pin derived from $(E_8)_1$ (v300); Route A’s invertibility discharged by the free-fermion classification (v301); the last input = the derived Recovery gap $6 \ln(3/2) > 0$ (v302). No undischarged TFPT-internal assumption remains (stays [O]: not machine-proved end-to-end)
(2) QG.AMB.01 — ambient measure (<i>decoupled, non-TFPT</i>)	[O]	<i>not</i> a TFPT structural item: the <i>general</i> Euclidean-QG conformal-factor problem (GHP 1978), gap-decoupled (margin $1.648 > 0$, v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B’s holomorphy bit (det $K=1$, v277) is absorbed into SEAM.EQUIV.01
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[C]	hardened : $c_3 = 1/(\mathbb{Z}_2 \oint_{S^2} K) = 1/(8\pi)$ (Gauss–Bonnet); “8” = rank E_8 ; π stays the one primitive
$\eta_B, m_p/m_e$, Koide, dark matter	[O]	frontier handles only (see document 4)

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status_ledger.csv.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret} -ladder (lepton rationals

exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* TFPT_{QFT} (the *Modular Spectral Closure*, v258–v261: D_F a covariance induction, the cutoff the seam KMS weight, seam/carrier/ E_8 on one Kummer/K3) — closed modulo the single keystone **SEAM.EQUIV.01**. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder $\times v_{\text{geo}}$ + standard RG, [C]), the absolute QCD scales (m_p/m_e , [O]), the complete PMNS dynamics ([C]/[O]), and the nonperturbative *ambient/metric-sector* measure, i.e. *full quantum gravity*. The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278); only the nonperturbative ambient measure stays [O] — its R^2/Weyl^2 gravity sector is renormalizable but non-unitary (the Stelle ghost), the lone perturbatively-irreducible piece, with Tier-B reduced to the single holomorphy bit $\det K=1$ (v277) and the whole bedrock reduced to one geometric postulate, *the raw seam is the flat $\tau=i$ pillowcase* (v276). The single open lemma is the named theorem **SEAM.EQUIV.01** — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* . (The flat $\tau=i$ geometry and the E_8 holomorphy are *two faces of one object*: the $(E_8)_1$ character is $\chi_{E_8}=j^{1/3}$, so $\chi_{E_8}(i)=1728^{1/3}=12$ and $\tau=i$ is at once the order-4 CM point and the $(E_8)_1$ partition-function modulus, v282.) It stays [O] (not machine-proved end-to-end), but its whole residual is now a composition of *standard cited theorems* (Osterwalder–Schrader/quasi-free clustering, the Kitaev free-fermion classification, the Route-A AQFT stack) over *established* TFPT facts, with *no* undischarged TFPT-internal assumption left — Lean-pinned to exactly those named steps plus the derived Recovery gap $\Delta = 6 \ln(3/2) \approx 2.43 > 0$ (v308); the conformal-deck premise **QGE0.SYM.01** is its *downstream* definitional face (v323). This is by design the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; “closing the theory” is therefore the concrete task of proving (or refuting) exactly this one arrow, not an open-ended search. The full sprint-by-sprint reduction that pinned **SEAM.EQUIV.01** is recorded in the changelog, not replayed here. The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive. *4D-QFT fork policy (frozen, v265)*: the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* (E_8 is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

10 Does this make the theory more probable?

Plausibility balance. *Pro*: (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar $\sin^2 \theta_{12}$	angle prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.)	JUNO : first 59.1 d run 0.3092 ± 0.0087 (compatible), precision phase pending	[E]/[C]
reactor $\sin^2 \theta_{13}$	angle $\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$; seed $\times$ carrier-trace $e^{-\gamma}$, $\gamma = \frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2 \theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$ -symmetric limit; octant not selected)	NOvA / T2K / DUNE (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$); scalaron-reheating point 0.0045 ($N_*=51.4$, v86)	now $r < 0.036$ (BICEP/Keck BK18); CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, obs. ~ 2033)	[C]
tilt n_s	$n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ($N_*=51.4$, v86)	Planck (0.9649 ± 0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_*^2}{24\pi^2} c_3^7$; 2.0×10^{-9} at $N_*=55$ (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)	PSI nEDM / SNS (running)	[E]
CP phase δ_{CKM}	$\delta = \frac{\pi}{3} + 3\lambda^2 = 68.65^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs γ_{PDG}	LHCb / Belle II γ combination; decision at $\sigma_\gamma \leq 0.96^\circ$	[E]
cosmic birefringence β	$\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)	ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); Simons Obs. / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	LEGEND / nEXO , DUNE, KATRIN	[C]
flavor invariants	$\det R = h(D_5) = 8$, principal minors $(2, 3, 5)$, $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / DESI / Planck (Hubble tension)	[C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

The two decisive near-term tests

- (1) **JUNO and θ_{12} .** JUNO is *taking data since August 2025* and targets sub-percent $\sin^2 \theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ (misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$). The first 59.1-day JUNO run reports $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~ 0.307 at high significance falsifies the seam mechanism.
- (2) **r .** The R^2 scalaron with M fixed by c_3^7 predicts a small $r = 12/N_\star^2$: the frozen band $r \in [0.0033, 0.0048]$ over $N_\star \in [50, 60]$, with the scalaron-reheating point $r = 0.0045$ ($N_\star=51.4$, **v86**) — already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first observations ~ 2033 –34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which M_{Pl} and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8 ; and the discipline rule “*every load-bearing number appears in at least one E_8 projection*” (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s ;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger REG.FREEZE.01). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, **v49/v69/v71**); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ R^2 branch is lifted from [C] to [E].

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	What it machine-checks
v1_e8_glue.py	glue theorem $E_8 = (D_5 \oplus A_3) + \mu_4$: $\text{disc} = \mathbb{Z}_4$, $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$, $240 = 16 \cdot 5 \cdot 3$, $248 = 240 + 8$
v2_carrier_pascal.py	$g_{\text{car}} = 5$ unique Pascal solution; $16 = 1 + 5 + 10$, $N_{\text{fam}} = 3$, $\text{rank } E_8 = 8$, $\Omega_{\text{adm}} = 48$, $b_1 = \frac{41}{10}$
v3_em_alpha.py	EM closure: unique root of $F_{U(1)}(\alpha) = 0$, $\alpha^{-1} = 137.0359992168$, existence/uniqueness, inverse test, ablation
v4_flavor_matrix.py	residue matrix R : $\det = 8$, minors $\{2, 3, 5\}$, $\chi_R = t^3 - 9t^2 + 10t - 8$, SNF $\text{diag}(1, 1, 8)$, $\ R\ _F^2 = 78$, $\sum L = 40$
v5_e8_cascade.py	cascade $D_n = 60 - 2n$: endpoints, exponent rungs $\rightarrow 240$, IR tail 46080, variance 6552
v6_bootstrap.py	reverse glue $\mu^2 - 5\mu + 4 = 0$; $g_{\text{car}} = 5$ three ways; $8 = \text{rank } E_8 = h(D_5) = \varphi(30)$
v7_gravity_cosmo.py	scalon exponent 7, $M_{\text{scal}}^2 / \tilde{M}_{\text{Pl}}^2 = c_3^7$, A_s, n_s, r , Ω_b , $\sin^2 \theta_{12}$
v8_horizon.py	horizon code: $\frac{1}{2\pi} = 4c_3$, $1920 = W(D_5) $, S_{dS} , Page, $\beta_{\text{rad}} = 0.2424^\circ$
v9_neutrino_texture.py	solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23} = 45^\circ$, $\theta_{13} = 0$
v10_projection_involution.py	Q, Σ algebra: $K = R + Q\Sigma$, $L = R + Q(I + \Sigma)$; χ_Q, χ_K ; det ladder $3 \cdot 4 \cdot 8 \cdot 20 = 1920 = W(D_5) $; $a^\top K a = 41$
v11_unique_KQ.py	K, Q are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
v12_mass_generation_polynomials.py	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
v13_open_gates.py	gate closures: $M = 41 = 10b_1 = \sum L + N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2 = N_{\text{fam}}$
v14_carrier_uniqueness.py	$g_{\text{car}} = 5$ unique ($N_{\text{fam}} \in \mathbb{Z}^+$, $\text{rank } E_8 = 8$); split $(3, 2)$ from $b + s = 5$, $b^2 + s^2 = 13$; $\text{Tr } Y = \text{Tr } Y^3 = 0$
v15_bootstrap_classification.py	$D_5 \oplus A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
v16_solar_dual_anchor.py	$a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$; $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; full PMNS

continued on next page

Script	What it machine-checks (<i>continued</i>)
v17_hexagonal_resolvent.py	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
v18_quark_yukawa.py	quark source ratios $\frac{m_u}{m_d}=0.470085$, $\frac{m_c}{m_s}=13.61$, $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$; full hierarchy
v19_monodromy_moduli.py	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4 -fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
v20_lepton_c_derivation.py	lepton c 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9}\Rightarrow\frac{7}{6}$
v21_solar_product_quark.py	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
v22_open_gates_audit.py	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
v23_anchor_generator.py	anchor $a=(1,1,2)$: $e_k=(4,5,2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n\Rightarrow 240, 248$, binary ladder; inputs $\rightarrow\{a,\pi\}$
v24_quark_ratio_closure.py	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
v25_frontier_conjectures.py	conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\mu\text{eV}$
v26_flavor_frontier_unification.py	the “11” is not uniquely forced ($\Rightarrow$ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
v27_wall_representative.py	explicit balanced wall rep W_{wall} (cols = perm(0,1,2), rows $w=(2,1,1)$, pardeg 0); selector $\det R=8$, $\text{Spec}(Q_+)=\{1,2,3\}$
v28_gravity_fR.py	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_\star=57\rightarrow n_s, r, A_s$; full metric measure open; $248c_3^2<6\log\frac{3}{2}$
v29_research_contract_certs.py	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\ V\ <\Delta$
v30_d4_character_variety.py	$C_U^{(2)}$: full D_4 -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)
v31_R_dictionary.py	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
v32_rh_splitting.py	RH route: $\sum_k U^k A_0 U^{-k}=4\text{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow\text{diag}(A_0)=(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k=I+R$ -bridge stay open (c_u/c_d not forced)
v33_explicit_flat_bundle.py	explicit valid (U_{wall}) flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2 + \ M_\infty-I\ \sim 10^{-9}$ ($\prod M_k=I$) + irreducible (case A)
v34_h2_bridge_attempt.py	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k=I$); honest negative — natural extraction $\neq$ lepton amplitudes, the Γ^{min} dictionary is the remaining input (c_u/c_d not obtained)
v35_quark_qcd_boundary.py	the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
v36_spectral_action_g2.py	G2: spectral action $\rightarrow R+R^2$ ($a_2\sim-R/3$, $a_4 _{R^2}\sim R^2/72$); $M^2/\bar{M}_{\text{Pl}}^2=6(4\pi)^2/f_0$, c_3^7 fixes f_0 ; gap-decoupling $\Delta_{\text{eff}}=1.648>0\Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
v37_plucker_anchor.py	anchor-plane Plücker apparatus: $\ Pl(K)\ _1=11$, $\ Pl_R(K)\ _1=26$, $\sum Pl_R(L)=60$; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$; $K_e-K_u=a$, $L_e-L_u=\text{Exp}(A_3)$; lepton ring $c_e c_\tau=\mathbb{Z}_2 c_\mu$; Koide $u\rightarrow\frac{53}{54}u$ ([C])

continued on next page

Script	What it machine-checks (<i>continued</i>)
v38_uwall_killswitch.py	(U_{wall}) U2 kill-switch hardened: D_4 -cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥ 0.35 , reducible locus measure-zero $\Rightarrow$ case A generic, (U_{wall}) survives; amplitude dictionary still research-level
v39_uwall_selectors.py	(U_{wall}) selector side closed on the explicit point: splitting type = anchor $a=(1,1,2)$, pardeg=0; $\det R=8=n \cdot a$, $\text{Spec}(Q_+)=\{1,2,3\}=3\alpha+1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\min}$ Hitchin) remain
v40_harmonic_metric.py	(U_{wall}) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$): unique invariant form $H>0$, unitarised $\widetilde{M}_k$ cusp-class with $ \widetilde{M}_k \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
v41_leg_assignment.py	(U_{wall}) final leg test: amplitudes $=\delta=\frac{1}{2}$ resolvent (closed); $\Lambda_\mu>1$ rules out holonomy moduli; harmonic holonomy $ \text{diag} =(0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta=\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
v42_exterior_leg.py	(U_{wall}) Exterior Leg Lemma: u, d share $(r, w)=(1,1)$ so any scalar leg gives $c_u/c_d=1$; the u/d split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$, $\ \cdot\ _1=11$ generated $\Rightarrow \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]
v43_exterior_bridge.py	(U_{wall}) $\Lambda^2 F$ bridge: $\Lambda^2(\widetilde{M})=\widetilde{M}$ (exterior leg $=\bar{3}$) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ($\det R=8$, $\text{Spec}(Q_+)$ confirmed), not continuous
v44_carrier_exterior.py	(U_{wall}) Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$; $u^c \in \Lambda^2(5)$, $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2,4)$, sum=6= $ R^+(A_3) $, diff=2= $ \mathbb{Z}_2 $; the exterior leg is the carrier's own Λ^2 grading (no special math)
v45_family_exterior.py	(U_{wall}) the “11” is Pascal: $ \text{Pl}(K) =(1,4,6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$; $\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of μ_4 up to $\deg u^c=2$); $15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$
v46_grand_mass_volume.py	absolute mass-scaling is [E], not a product rule: sector dets $(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \Lambda_\Lambda \Rightarrow \det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$; Q -rows $(4,5,6)$ sum 15= $\dim A_3$; $25+15=40=\sum L$
v47_selection_theorem.py	Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$; 16-spinor $\rightarrow D_5$; $q(D_5)+q(A_3)=2$, glue 4= $ \mu_4 $; only $D_5 \oplus A_3$ meets all four conditions ($D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]
v48_em_ward.py	EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^2 \alpha^2 - \text{transport } 8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$; $8b_1=\frac{4}{5}M=\frac{164}{5}$, $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]
v49_readout_rigidity.py	Readout Rigidity (Thm U2): $c_u/c_d=\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}})=U_{\text{unitary}}+U_{\text{H2}}+U_{\Lambda^2}+U_{\text{point}}$, only U_{point} open
v50_q_geometry.py	Q geometry (Thm Q): $Q_+ = \text{diag}(3,2,1)$ -block $\chi=(t-1)(t-2)(t-3)$, $Q_- \chi=t(t^2-3)$; Q unique under rows(4,5,6)/cols(9,5,1)/spectra; geom. origin [C]
v51_boundary_half_step.py	glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $: $q(D_5)=\frac{5}{4}$, $q(A_3)=\frac{3}{4}$; four ops forced $\rightarrow \{\text{sum}=2$ root norm, diff= $\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$, prod= $\frac{15}{16}$, ratio= $\frac{5}{3}\}$. So $\delta=\frac{1}{2}$ is the carrier-family count over the glue index, not chosen [E]
v52_pencil_endpoints.py	$K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $, $P(0)=4= \mu_4 $, $P(1)=20=\det L$, $P(2)=68=2p_5(a)$; $\det(K-Q)=2$, $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]
v53_compiler_core.py	Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5,3)$ — rank $E_8=8$, $ \mathbb{Z}_2 =2$, Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2 \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]
v54_seam_horizon_keystones.py	The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned $(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1, (2/3)^6, (1/3)^6\}$, $\lambda_2=(2/3)^6$) governs both SM flavor gap and horizon Page recovery [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v55_coxeter_cycle.py	Computed E_8 Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$, eigenphases = exponents = totatives of 30, so $\text{rank } E_8=8=\varphi(30)$ = live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda=32\pi^4$) [E]
v56_unique_attractor.py	Gapped boundary transport (gap $6\log\frac{3}{2}>0$) $\Rightarrow$ <i>unique</i> attracting fixed point, geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes; entropy reset = rank-1 projector [E]
v57_horizon_crosslinks.py	Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4=1/ \mu_4 $ entropy density); Hod QNM $\omega_R/T_H=\ln 3=\ln N_{\text{fam}}$; Hawking power denom $1920= W(D_5) $; one α^{-1} for EM/ S_{dS} / Λ /Hubble [E] (+ [C] identification)
v58_seam_horizon_chain.py	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3=1/(2\cdot 4\pi)=\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}=M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open
v59_area_law_evidence.py	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3}\log L$), by the <i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2 \mu_4 $. The c_3 coefficient stays [O] open
v60_lambda_metrology_branch.py	Λ branch selection: $(8\pi)^2\delta_{\text{top}}=\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\text{top}}=\frac{64\pi^3}{3}=661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215^\circ\pm 0.074^\circ$ vs 0.2424° (0.37σ) [E]
v61_cft_bridge.py	Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$, $c(D_5)=5$, $c(A_3)=3$ = (rank $E_8, g_{\text{car}}, N_{\text{fam}}$); conformal embedding $c_{\text{coset}}=0$; $SU(3)\leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$, $4\rightarrow 3+1$ (one geometry $\mathbb{P}^1\setminus\mu_4$, two sides) [E]
v62_data_scorecard.py	TFPT vs 2024/25 data: α^{-1} , $\sin^2\theta_{12}$, β_{rad} match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r\approx 0.004$ future falsifier [E]
v63_seam_engineering_index.py	Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2\log(3/2)}\approx 0.323$ ($2\ V\ =\frac{31}{4\pi^2}$, $248=\dim E_8$, $31=1+h^\vee$), $\Delta_{\text{eff}}\approx 1.648$; four-class possibility taxonomy [E]/[O]
v64_causal_boundary_nogo.py	Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)
v65_falsification_layer.py	Prediction layer with explicit failure thresholds: 3 core ($v_{\text{GW}}=c$, no 4th gen, seam=8), 4 observational ($\beta, r, n_s, \theta_{12}$), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]
v66_e8_casimir_degrees.py	Organizing simplification: the compiler atoms <i>are</i> the E_8 Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ($2= \mathbb{Z}_2 $, $8=\text{rank}$, $14=\dim G_2$, $20=\det L$, $24= W(A_3) $, $30=h(E_8)$); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]
v67_area_law_coefficient.py	Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$, $k=c_3/2 \Rightarrow S=2\pi c_3A=\frac{1}{4}A$ ($2\pi c_3=\frac{1}{4}$); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]
v68_seeley_dewitt_residual.py	Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. R^2/scalon derived [E]
v69_d4_q_geometry.py	D_4 -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1\setminus\mu_4)=B_1\oplus E$, $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$; Σ -split $=D_4=\mathbb{Z}_4\rtimes\mathbb{Z}_2$ [E]
v70_q_integer_lift.py	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1, i, -i\}$); $\det Q=3=N_{\text{fam}}$, SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$, not closable by D_4 alone [E]
v71_simple_r_bridge.py	<i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\text{point}}=$ absolute norm. = an anchor [E]/[O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v72_q_det_from_cusp.py	$\det Q = N_{\text{fam}}$ derived from the cusp class: $\det Q = \text{coker } Q = \text{cusp-class order} = \text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$); $\text{coker } Q = \mathbb{Z}/N_{\text{fam}} = \text{triality deck group} \Rightarrow \text{integer-lift residual closed}$ [E]
v73_k_c3_half.py	Central coefficient $k=c_3/2$ forced: $(1/2)$ variational $\times 1/(\mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2)=2$), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$; only the absolute 4D Newton scale stays the anchor (v68) [E]
v74_compiler_micro_lemmas.py	Spine quotient ladder $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}) = \text{adjacent quotients of } (2, 3, 4, 5)$; pencil differences $(2, 16, 48) = (\mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L = R + 61e_1^\top$, $a^\top R^{-1}\mathbf{1} = 0$ (Sherman–Morrison rank-one invariance) [E]
v75_upoint_to_vgeo.py	Gate 1 complete: $U_{\text{point}} \rightarrow v_{\text{geo}}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = φ_0^{ret} -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale v_{geo} = the same anchor as $1/G$ (v68); two $[A] \rightarrow \text{one}$ [E]/[O]
v76_gmetric_reduction.py	Gate 2: Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, $\ V\ \leq 248c_3^2 = \frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]
v77_e8_conformal_net.py	G6 route: the seam boundary measure = the $(E_8)_1$ lattice net (rigorous, holomorphic $c=8$); level-1 $c(E_8) = \frac{248}{31} = 8$, $c(D_5)=5$, $c(A_3)=3$, embedding $5+3=8$ (coset $c=0$); imports G6 into RCFT/conformal-net rigor [E]/[C]
v78_vgeo_floor.py	v_{geo} floor: no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times \text{one } \bar{M}_{\text{Pl}}$; $S_{\text{dS}}\rho_\Lambda = 32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [E]/[O]
v79_review_identities.py	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n = (3^n - (-2)^n)/5 = 1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1} = 1/\text{atom}$, $a^\top M^{-1}a = 1$ for R, K, L); MacWilliams $(1, 10, 5)$; $6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)$. Rejects m_p/m_e , η_B near-formulas (firewall) [E]
v80_operator_pencil_geometry.py	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ} = (N_{\text{fam}}x + \mathbb{Z}_2)(N_{\text{fam}}x + g_{\text{car}}) = (3x+2)(3x+5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3, 10, 16, 20)$; type checker $\det B_{Q,K,R,L} = (9, 10, 16, 40)$; audit: differential spectrum, $F_4 \times G_2$ shadow $248 = 52 + 14 + 26 \cdot 7$ [E]/[C]
v81_singular_pencil_matrices.py	Double cover $y^2 = \det B_{K+xQ} = (3x+2)(3x+5)$: Koide $x = -\frac{2}{3}$ and carrier $x = -\frac{5}{3}$ are the two branch points (deck $2 = \mathbb{Z}_2 $; disc $= 81 = N_{\text{fam}}^4$; separation $= 1$ transport period). Clearing matrices $C_{2/3} = 3K - 2Q$ ($D_5 \oplus A_3$ glue, $B = (5, 7)^\top (9, 11)$, $\sum = 240$), $C_{5/3} = 3K - 5Q$ (neutral, $\chi = (\lambda+2)(\lambda^2 + \lambda + 6)$); scalaron $7 = \text{mixed anchor disc}/N_{\text{fam}}$ [E]
v82_koide_attractor_splitting.py	Koide attractor forced: the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6 = \lambda_2$ of the established transfer operator $\Rightarrow F'(2) < 1$ attractor, $F'(5) > 1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$: the clean split is non-generic (anchor-first) [E]/[C]
v83_e8net_holomorphic_uniqueness.py	Holomorphy pins $(E_8)_1$: #primaries = det Cartan ($D_8:4$ vs $E_8:1$), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $); carrier Pascal truncation $K = (g-1)/2 = \text{half-spinor split}$ [E]
v84_frozen_registry.py	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> θ_{12} prediction of record; r , n_s as $N_\star$ bands; conditional entries carry their hypothesis by name [E]
v85_master_cover.py	Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $= N_{\text{fam}}^4 \det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3} = -\text{rank } E_8/N_{\text{fam}} = \text{carrier} - \text{one period}$); μ_4 is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v86_nstar_reheating.py	$N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128$ GeV, $T_{\text{reh}}=9.6\times 10^9$ GeV, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$ (inside the frozen band); honest tensions: $n_s+0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]
v87_bulk_uniqueness_reduction.py	Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow$ Rep = Vect $\Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both E_8 pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]
v88_cp_phase_audit.py	Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs γ_{PDG} ; the 0.07° -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma\leq 0.96^\circ$ [E]
v89_carrier_index_lemma.py	Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1\times(A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); μ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]
v90_conical_defect_chain.py	Fursaev–Solodukhin factor 4π <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$, $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]
v91_spine_tetrahedron.py	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $; $240= \mu_4 E(K_4) E(K_5) $ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]
v92_glue_uniqueness.py	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$, nothing else [E]
v93_koide_relaxation_toy.py	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho=-\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]
v94_sheet_diamond.py	Sheet diamond: all four flavor operators on <i>one</i> surface $M(s,t)=R+Q \text{ diag}(s,t,t)$ (pencil = the cut $(1+x, x-1)$; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all s ; cofactor seam normal $(5, -9, 6)$; Plücker canonicalisation: 11 and 26 both live in K ; spine-quotient firewall <i>rejected</i> [E]
v95_centered_diamond.py	Centered cross: $Q=U+V$, $R/L=C\mp U$, $K/F=C\mp V$ (five objects $\rightarrow$ three); Spec $V=\{0,1,2\}=N_{\text{fam}}\cdot\text{cusp}$; $\det C=14$, $\sum C=31=2^{g_{\text{car}}}-1$; $\text{Pl}_R(C)=7(2,3,1) \parallel \text{Pl}_R(L)=10(2,3,1)$; G_2 reading audit-typed [E]
v96_branch_kernel_selection.py	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$: up = $-$ down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
v97_sheet_conjugation_bridge.py	Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer T_A , $\det=-1$; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$; $\langle T_A, \Sigma \rangle = D_4$; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
v98_discriminant_dictionary.py	Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A\Sigma$ acts integrally on the cusp basis ($Ge_1=-e_1$, $Ge_3=e_2$, $Ge_2=-e_3 = B_1\oplus E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A GT_A^{-1}=G^{-1}$ ($k\mapsto -k$); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$, $q_{A_3}(2)=\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing Q -geometry gate [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v99_koide_flow_time.py	Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}})\det B(q)$, time-1 map = F [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: $n=2$ excluded (-2.9σ), $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427\text{ MeV}$ ($+0.14\sigma$), decision at $\sigma(m_\tau)\sim 0.01\text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)
v100_numerology_null_mc.py	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2\times 2\cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, <i>conditional on the declared grammar</i> [E]
v101_horizon_anchor.py	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1, 1, -2) =$ traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}] =$ the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)
v102_seam_orientation.py	One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]
v103_trisection_normal_form.py	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — <i>one</i> cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N=-\frac{8}{9}=-\text{rank } E_8/N_{\text{fam}}^2$; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, missing = one <i>clock</i> [E]/[C]
v104_nariai_clock.py	The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2=-2\Lambda=- \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$; entropy-deviation rate $2H= \mathbb{Z}_2 H$; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
v105_residual_inventory.py	The residual inventory: <i>one constant</i> $(2/3= \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + π ; a sixth gap would fail the script [E]/[C]
v106_review_validation.py	External-review validation: seed normal form $\varphi_0^{\text{ret}}=\frac{ \mu_4 }{N_{\text{fam}}}c_3+\Omega_{\text{adm}}c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2=120=5!$ computed from the charge table; factorial spine $5!= R^+(E_8) =\sum \text{exponents}$, $240=2\cdot 120$, $1920=2^4\cdot 5!$; degree-2 inventory (Pascal $K=2$ unique at $g=5$, glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$, $\binom{5}{2}=10$); named [C] hypothesis <i>Quadratic Boundary Locality</i> ($\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240=16\times 15$ non-unique [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v107_quantum_clock_target.py	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/M_{\text{Pl}}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
v108_pascal_ladder.py	Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- g midpoint identity + even- g straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]
v109_sheet_pairing.py	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$; odd g flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ($256=1+45+210$, zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms ($2\Lambda^1 + 2\Lambda^3 + \Lambda^5$); tower tops at Λ^{2K} , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]
v110_calderon_sheet.py	Calderón-sheet selection theorem: a Calderón involution ε on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2= \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3, 5, 7$): sheet-oddness forces $K=(g-1)/2$, <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ ε sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees ≤ 2) [E]/[C]
v111_quadratic_transport.py	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$, the Λ^2 term of $1+45+210$) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length 2 $\Rightarrow$ <i>transport degree exactly 2</i> (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]
v112_selfcounting_channel.py	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels = 2^{g-1} <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]
v113_quasifree_kernel.py	One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 = \mu_4 $); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$, seam level $8=\text{rank } E_8$ ($c = \text{rank of the kernel}$); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]
v114_torsion_delta.py	Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}}) \mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ($\mu_4 = \text{order of the twisted cusp generator}$); on the involutive branch ($T^2 = \mathbf{1}$) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]
v115_anchor_residue.py	The anchor pins the residue: μ_4 -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $; off-weights uniquely $(8, 0, 5)/144$ ($= (\text{rank } E_8, 0, g_{\text{car}})/(\mu_4 N_{\text{fam}})^2$, total $13 = \Delta_Q$); <i>exact</i> residue A_0^* with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ($\ M_\infty - \mathbf{1}\ \sim 10^{-10}$); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v116_resonance_uniqueness.py	Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted μ_4 averages collapse to $B_1=4a_{12}E_{12}+4\bar{a}_{13}E_{31}$; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\bar{a}_{13} \Rightarrow M_\infty=\mathbf{1} \Leftrightarrow a_{13}=0$; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact $A_0^* - U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\ \sim 8.5 a_{13} $ [E]
v117_monodromy_weyl_a3.py	The monodromy is $W(A_3)$: the unitarised holonomy of A_0^* is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$), $\text{diag } \widetilde{M}_0=(0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta=\frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$, characters $(3, -1, 0, 1) = S_4=W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match 3.5×10^{-10} [E]
v118_hexagon_family_dictionary.py	First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6=\mathbb{Z}_2 \times \mathbb{Z}_3$); $\det(\mathbf{1}-t\widetilde{M}_0)=1-t^3 \Rightarrow \det_{\mp}=(\frac{7}{8}, \frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e= \mu_4 /(2\det_-)$, $c_\mu=N_{\text{fam}}/(2\det_+)$, $c_\tau= \mu_4 \det_-/N_{\text{fam}}$, product = $ \mu_4 /\det_+=\frac{32}{9}$); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48=\Omega_{\text{adm}}$; the address table stays open [E]/[C]
v119_review_validation_2.py	Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0=(\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5)=(e_3, e_2))$; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a)=121=11^2=\ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}}=2p_1p_2$, $10b_1=1+p_1p_3$, $\Delta_Y=e_2^2=p_0^2+2\text{rank}$; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]
v120_address_table.py	Address table (frozen v18/v20 integers): lepton words $(8, 5, 3)=(\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$; addresses = divmod by the hexagon ($p_2=6$): $e \rightarrow (2, 1)$, $\mu \rightarrow (5, 0)$, $\tau \rightarrow (3, 0)$, $\sum r=p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3 , down = p_1+p_3 , all quarks = $24= W(A_3) $, all nine = $40=p_1p_3=a^T Ra$; top at the vacuum site $(0, 0)$; the assignment derivation stays open [E]/[C]
v121_address_pinning.py	Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a ; margins = atoms (rows (e_1, rank, p_3) , columns $(e_1, \Delta_Q, g_{\text{car}})$); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is R (also unique with SNF $(1, 1, 8)$; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C]
v122_margin_theorem.py	Margin theorem: the <i>established</i> selectors — the D_4 annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$, $\det K=4$ (the v71 quartet) — pin R <i>uniquely</i> among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]
v123_inventory_update.py	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 Q realisation — + v_{geo}, π ; a sixth class fails the script [E]
v124_resummed_clock.py	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$, the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]
v125_glue_qsystem.py	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ($q(k(1, 1))=k^2 \in \mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step, } 4=2 \times 2$; R2 sharpens from “find” to “identify” [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v126_clock_wall_bridge.py	Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly spec A_0^* (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed <i>classically</i> ($p_2/N_{\text{fam}}=2= \mathbb{Z}_2 $ = the v104 entropy rate $2H$); $\det(\mathbf{1}-A_0^*)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]
v127_ring_resummation.py	The clock is a log determinant: rate = $-p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$; v107's "O(1) resummation" is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]
v128_graded_hull.py	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [E]/[C]
v129_entropy_power_law.py	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); rate = $p_2 \ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}}$ = MS weight = entropy-deficit share $\Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]
v130_born_square.py	Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability = $ A ^2=(S/S_{dS})^6 \Rightarrow h=n_{\text{zero}}/2=3=N_{\text{fam}}$, $p_2=2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of S^2 ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C]
v131_measure_is_area.py	The measure is the area: $\ Y_{1m}\ ^2=r^2=A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude S^3 , Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3 , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ [E]/[C] (<i>cancellation since derived within the SdS family, v144</i>)
v132_detratio_anomaly.py	Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) = $a_1-3=\frac{7}{3}-3=-\frac{2}{3}=- \mathbb{Z}_2 /N_{\text{fam}}$ exactly ($a_1=\frac{1}{N_{\text{fam}}}+ \mathbb{Z}_2 $; numerical continuation $\sim 10^{-7}$) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL)=c^{\zeta(0)} \det'(L)$; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) [E]/[C]
v133_zeta_budget.py	The ζ budget, both routes exact (Euclidean Nariai = $S^2 \times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}=-e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2=\frac{161}{45}$, $\zeta(0)=-\frac{109}{45}$ (<i>no</i> atom; continuation $\sim 10^{-8}$); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C]
v134_dual_anchor.py	Dual anchor lemma: $d:=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$, $d \cdot \mathbf{1}=0$, $d \cdot a=1$, $(1, 1, -2)=-2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1, 1, -1)$); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); (d, n) = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge [E]/[C]
v135_det_surface.py	Determinant surface: $\det M(s, t)=3s(t+1)(t+2)+t^2+5t+8$ — two s -independent iso-volume walls $t=-2 \Rightarrow 2= \mathbb{Z}_2 $, $t=-1 \Rightarrow 4= \mu_4 $ ($\det K=4$ is a <i>layer</i>); winding line $6s+8 \Rightarrow (8, 14, 20)$, $\det F=32=2^{g_{\text{car}}}$; $\det B(s, 0)=2 \det M(s, 0)$ exactly, B -wall only at $t=-2$; row-sum axes $C=(7, 11, 13)$, $U=(3, 3, 3)$, $V=(1, 2, 3)=\text{Spec}(Q_+)$; micro-identities $e_1^2+e_2^2=41=10b_1$, $p_1^2+p_2^2=52$ = the carrier coset root count (v128) [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v136_dual_normal_selector.py	Dual-normal selector: (d, n) pins R columnwise — $d \cdot c_j = a_j$, $n \cdot c_j = 8\delta_{1j}$, kernel = primitive $(6, 8, 7)$, census: each column unique (1 of 216) $\Rightarrow$ $R4'$ shrinks from 6^9 to three column problems; the A_0^* zero mode carries $\sigma = (2, -9, 5) = (\mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ($\ v\ ^2 = \frac{16}{9}$, $16 = \dim S^+$; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element [E]/[C]
v137_qplus_cohomology.py	The Q_+ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ($\omega_k = z^{k-1} dz / (z^4 - 1)$, character i^k , residues $\zeta^k/4$); degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents ($h=4= \mu_4 $); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = Q_+ : one spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E]/[C]
v138_vw_firewall.py	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); edge match : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E]/[C]
v139_selector_triangle.py	Selector triangle: $d = \frac{3}{2}a - 2 \cdot 1$ exactly (pure anchor data, no R) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(1, a, \sigma)$ has $\det = 11 = \ \text{Pl}(K)\ _1$, and n is the <i>unique</i> covector with atom pairings $(2, 8, 121) = (\mathbb{Z}_2 , \text{rank } E_8, 11^2)$; on the constrained line the pairing is $11(11+t)$, a square only at the true n ; review lift exact (audit): $n = \text{reverse}(\sigma) + \mu_4 e_3$; $R4'$ residue = three pairings [E]/[C] (since v142: two — the third pairing is integrality-forced)
v140_canonical_map.py	The canonical map exists and is rigid: the plain deck U (characters $(0, 1, 3)$) cannot match H^1 ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential i^{Q+} $((1, 2, 3))$ both carry the H^1 character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q+} pulls the Euler degree back to Q_+ , $-U$ to $Q_+ \circ \rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice [E]/[C] (the choice since derived, v141)
v141_deck_selection.py	Deck Selection Theorem (R5): the established integer deck $G = T_A \Sigma$ (v97/v98) pairs the $Q_+ = 1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; i^{Q+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+ \circ \rho$ [E]/[C]
v142_frame_integrality.py	Frame Integrality ($R4'$): pairing triples of <i>integer</i> covectors against $(1, a, \sigma)$ form an index-11 sublattice ($x - 3y + z \equiv 0 \pmod{11}$, $11 = \ \text{Pl}(K)\ _1$); with $(x, y) = (\mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \pmod{11}$, primitivity forces even t ; Cramer reductions for the two line pairings; $R4'$: three pairings $\rightarrow$ two [E]/[O] (since v145: the values are atom identities — residue = one lift map)
v143_graded_frobenius.py	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k = C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$; glue average = carrier projector (index $ \mathbb{Z}_4 = 4$); each sector one $W(D_5) \times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement [E]/[C]
v144_detratio_family.py	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2 -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0) = -\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27} = \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply [E]/[C] (the bend since shown $\det'$ -clean, v147)
v145_pairing_atoms.py	$R4'$ values are <i>atom identities</i> : with the lift $n = w_0(\sigma) + \mu_4 e_3$ (w_0 = the A_3 exponent duality), $n \cdot 1 = \mu_4 - \mathbb{Z}_2 = \mathbb{Z}_2 $, $n \cdot a = \mu_4 a_3 = 8$ via the new closure $ \mu_4 + g_{\text{car}} = N_{\text{fam}}^2$ ($4 + 5 = 9$, $\sigma \perp w_0(a)$), $n \cdot \sigma = \ \sigma\ ^2 - N_{\text{fam}}^2 + \mu_4 g_{\text{car}} = 121$; residue = derive the <i>one</i> lift map [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v146_moebius_d4.py	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — ι^* swaps $\chi_1 \leftrightarrow \chi_3$, fixes χ_2 with $+1$, $\det = -1$ ($= T_A$ in the cusp basis); $\delta\iota = \Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification [E]/[C]
v147_clock_gaussian.py	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born ² Γ -ratio $(1-\alpha)^{p_2}$, rate $= -p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ <i>det'-clean</i> ; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference [E]/[C]
v148_fock_census.py	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ($2^8=256$) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection ($128 = \text{one } SO(16) \text{ chirality}$), $248 = 120_{\text{NS}} + 128_{\text{R}}$; R2 is irreducibly a twisted-sector net statement [E]/[C]
v149_cusp_normal.py	Both dual normals are <i>anchor data</i> (R4'): in the cusp frame n pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line ($d = \frac{3}{2}a - 2 \cdot 1$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the w_0 -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$, sum $14 = \dim G_2$; residue = <i>one</i> assignment [E]/[C]
v150_replica_eh_model.py	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$); the gap makes the ζ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$; linearisation $= (\ln m / 12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] (<i>Calderón transfer since answered — the DtN kernel is conically clean</i> , v151)
v151_bfk_split.py	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ($C_N = C_D$, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}} = C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]
v152_norm_is_anchor.py	R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; $k=c_3/2 \Leftrightarrow m/\mu = e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, π }; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]
v153_no_unit_theorem.py	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$, a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π } [E]/[O]
v154_simple_current_theorem.py	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): $A = (D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L = \langle (1, 1) \rangle$ has index $ L =4$, $c=5+3=8$, $\mu(B) = \mu(A)/ L ^2 = 16/16=1 \Rightarrow$ holomorphic $\Rightarrow B = (E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]
v155_quasifree_boundary.py	Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression PTP (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v156_seam_net_construction.py	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$; the E_8 currents are $248=120_{\text{NS}}+128_{\text{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]
v157_rigid_fixed_point.py	Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\hbar=0 \Rightarrow$ rigid, isolated, no room for an interaction); (iii) the same $ \mathbb{Z}_2 $ that halves 8π in c_3 IS the Ramond projection giving $\mu=1$; (iv) holomorphic $c=8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C]
v158_fixed_point_stable.py	The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$, bilinear current $h=1$, quartic $h=2$; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]
v159_pyrate_gauge_crosscheck.py	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1=\frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1=g_{\text{car}}2^{g_{\text{car}}-2}+1=41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (arXiv:2007.12700) which writes $\beta_{g_1}=(\frac{41}{10})g_1^3$ verbatim; the $M=41$ split $10b_1=40_{(\text{ferm})}+1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in <i>verification/pyrate</i> . The transfer functor’s gauge inputs are not free knobs [E]/[C]
v160_seam_gaussianity_from_pf.py	Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> κ_{2n} is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap}>0$ — the value $(2/3)^6$ only sets the rate. The load [O]: v56’s gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}=1820$, $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement ‘the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum’ [E]/[O]
v161_one_particle_reduction.py	The cone reduces to one particle (discharging v160’s scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$, the second quantisation of a one-particle contraction t ; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\text{spec}(\Gamma(t) _{\text{deg } m}) =$ products of m one-particle eigenvalues [E]; (2) the cone gap equals the one-particle gap — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) [E]; (4) quasi-freeness <i>forced</i> not assumed — the $120=\dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. Residual [O]: v160’s three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v162_seam_transport_identification.py	The final identification, forced as far as it goes — the v161 one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$, $p_2=6= R_+(A_3) $, all carrier data [E] ; (2) μ_4 -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \mu_4 \text{diag}(X)$, commutant of U = the diagonal algebra — so the gapped μ_4 -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ [E] ; (3) the cusp grading <i>is</i> the A_3 grading on the 16 Majoranas (v137) [E] ; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) [E] . Closure [C]/[O] : with v160+v161 , (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, HOR.CLOCK) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) [E]/[O]
v163_rg_stability_flavor.py	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are <i>RG-stable</i> — PMNS running is y_τ^2 -suppressed ($\sim 1.8 \times 10^{-5} \ll \text{precision}$), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b \sim 41$) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries y_{max}^2 (PyR@TE SM RGEs) [E]/[C]
v164_qcd_scale.py	the QCD confinement scale is parameter-free from the carrier $b_3 = -7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b) \sim 0.22$, $\alpha_s(m_c) \sim 0.38$ and confinement ($\alpha_s \sim 1$) at ~ 0.5 GeV — so the QCD-scale <i>numerator</i> of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ($m_p = O(1)\Lambda$ is lattice) [E]/[O]
v165_premise_a_closure.py	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics [C] — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B) = 16/16 = 1$, $248 = 120 + 128$, character $E_4/\eta^8 = j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $= k $ (v156/v157); (2) R1 (seam clock) reduces to GATE.QGEO [C] — μ_4 -equivariance forces diagonal-in-cusp, the flat μ_4 -equivariant generator is <i>unique</i> $= A_0^*$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary’ = the μ_4 -punctured sphere $\mathbb{P}^1 \setminus \mu_4$ = GATE.QGEO (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$, a <i>theorem</i> by the No-Unit Theorem (v153). Final accounting : (A) = A2 assembly [C] + R1= GATE.QGEO + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O]
v166_higgs_free_seam.py	the Higgs quartic from the <i>free seam</i> (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \sim 0.002$, $\beta_\lambda(\bar{M}_{\text{Pl}}) \sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 to ~ 0.002). The double condition predicts $m_H \sim 129\text{--}134$ GeV; at the scalaron scale it gives ~ 107 GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v167_inventory_post_closure.py	The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123's FIVE classes to: Tier 0 — the irreducible core $\{\pi, v_{\text{geo}}\}$, a <i>theorem</i> (No-Unit, v153), nothing to close; Tier 1 — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1=N_{\text{fam}}=3$); Tier 2 — folded ($\rightarrow v_{\text{geo}}, \rightarrow G_{\text{net}}$); Tier 3 — F_{transfer} , one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
v168_qgeo_rigidity.py	The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z \rightarrow iz, z \rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P} \setminus \mu_4)$ has rank $4-1=3=N_{\text{fam}}$; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ($k=1, 2, 3$) are μ_4 -eigenforms with $z \rightarrow iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3 , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P} \setminus \mu_4$ stays the realisation premise) [E]/[C]
v169_etaB_boltzmann_interface.py	The η_B leptogenesis transfer <i>operationalised</i> (Tier-3 F_{transfer} , not a new class): type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$. Davidson–Ibarra $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3 / v^2 \sim 8.5 \times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1) \sim 0.03\text{--}0.19$. Kill test: over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B spans $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$ which <i>brackets</i> the observed 6.1×10^{-10} (canonical point gives 6.0×10^{-10} , not tuned) $\Rightarrow$ the route is viable; M_1 /washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]
v170_diamond_slice_compression.py	E8 Slice Compression Lemma: the seven E_8 audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ($p_n=2+2^n$) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$. The K_4 edge products (12, 18, 30, 24, 40, 60) are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2 \Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]
v171_os_moment_cluster.py	Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi=0.41105$, Källen–Lehmann masses $6 \log \frac{3}{2}, 6 \log 3$; gap safety $\Delta_{\text{eff}}=6 \log \frac{3}{2} - \frac{31}{4\pi^2} > 0$ with $31=1+h^\vee(E_8)$, $c((E_8)_1)=248/31=8$ [E]/[C]
v172_trace_anomaly_seed.py	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 ($\chi=2$) the $(E_8)_1$ ($c=8$) Weyl anomaly is $c\chi/6=\frac{8}{3}$, halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$. Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]
v173_pfaffian_cp_car.py	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$, γ_5 -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ($\arg \in \{0, \pi\}$), and reflection positivity ($Z>0$) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v174_seam_fock_readings.py	Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16=2^{11}$), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$, with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig g -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]
v175_net_existence_full_cone.py	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) full-cone RP for all m — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}=65536$); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$, verified on all 2^{16} products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces <i>for all m</i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) A2 net existence assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$, E_8 Cartan even unimodular ($\det 1$); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O]
v176_seam_collar_realisation.py	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, $b_1=N_{\text{fam}}=3$), Calderón DtN symbol $ k $ (v156), H^1 character = generation grading $(1, 2, 3)=A_3$ exponents= $\text{Spec}(Q_+)$ (v137/v168), μ_4 -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar’s boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$), left honestly [O][E][O]
v177_seam_marking_kernel.py	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , the reflection $z \mapsto 1/z$ gives $\sigma \rho \sigma = \rho^{-1}$ (D_4 , order 8) — a genus-0 four-marked faithful- D_4 boundary <i>is</i> $(\mathbb{P}, \mu_4)$; [E] COHOM: $\rho^* \omega_k = i^k \omega_k$ so the H^1 grading is $(1, 2, 3)=A_3$ exponents= $\text{Spec}(Q_+)$; [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); [E] NET (v154/v175): index-4 μ_4 extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$. The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked D_4 boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the μ_4 -equivariant free gapped contraction as an <i>operator</i>); the whole structural residual is now exactly these two doors [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v178_marks_kernel_attempt.py	An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock ρ ($\sim z \mapsto iz$, fixed points $\{0, \infty\}$) and reflection τ forces the marks to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$, $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O]
v179_conformal_realisation.py	The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 is P1 — $c_3=1/(\mathbb{Z}_2 2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$, the same $\chi=2$ that gives the ‘8’ (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$, order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$, a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to ‘the carrier clock acts conformally on the genus-0 double’. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]
v180_clock_is_mobius.py	Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: ‘the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric’. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v181_clock_is_conformal_symmetry.py	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4= \mu_4$, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i>, not a missing theorem: ‘the seam’s conformal structure <i>is</i> the μ_4-deck structure of the carrier clock’ (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam’s conformal geometry, not reducible further without relabeling. The chain $\text{REALIZE(v175)} \rightarrow \text{theorem(v176)} \rightarrow \text{MARKS+KERNEL(v177)} \rightarrow \text{finite cores(v178)} \rightarrow \text{CONF.01(v179)} \rightarrow \text{ISO.01(v180)} \rightarrow \text{SYM.01(v181)}$ has reached bedrock; we stop here honestly [E][O]
v182_reviewer_residual_map.py	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \rightarrow 4D$) [E]/[C]: the Plücker readout $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$. B (UV gravity) [E]: ‘gravity = geometric boundary readout’ <i>is</i> QGEO.SYM.01. C (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a=(1, 1, 2)$ in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck, $t \sim 2.84$ only locates the point, and the missing RG generator <i>is</i> F_{transfer}, so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$. NET: three reduce onto $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$, one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C]
v183_koide_f_corner_transfer.py	The Koide source $\rightarrow$ pole factor $\frac{53}{54}$ has an <i>operator</i> origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2 \cdot 27) = \frac{53}{54}$. (1) $\mathbf{1}^T Ra = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}}$); (2) $F=R+Q$ is the <i>missing</i> Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52+1$; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly ($54=2 \cdot 27$); (4) negative control: only F gives 53 ($R/Q/K/L \rightarrow 32/21/35/56$). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source $\rightarrow$ pole reading the F corner stays an F_{transfer} conjecture [E][C]
v184_etaB_anchored_boltzmann.py	Honest test of the external-review η_B proposal ($M_1=M_R\phi_0^4$, $\tilde{m}_1=m_3/A_\Lambda$) — firewall verdict: <i>sharper scenario</i>, not a closure. The washout anchors plausibly ($\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5$ meV, $A_\Lambda=10= E(K_5)$, in the strong-washout window) [C]; but M_1 does <i>not</i> — M_R is the seesaw scale $\sim v^2/m_3 \approx 6 \times 10^{14}$ GeV and the nearest clean TFPT powers of $\tilde{M}_{\text{P1}}$ both miss it (c_3-power off $\sim 75\%$, ϕ_0-power off $\sim 40\%$), so $M_1=M_R\phi_0^4$ merely relocates the free input $M_1 \rightarrow M_R$ (the reviewer’s 6.0×10^{-10} uses $M_R=1.3 \times 10^{15}$; the seesaw value gives 3.4×10^{-10}, the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays [C], a sharper scenario [C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v185_axion_relic_solver.py	Axion relic, honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ($m_a \approx 23.8 \mu\text{eV}$) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this f_a (prefactor $\sim 0.026/\theta^2$, $\theta \sim O(1)$ gives $\sim 21\%$ of Ω_{DM} , need $\theta_{\text{eff}}^2 \sim 4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128 = (16, 4) + (16, 4)$) [C][O]
v187_ftransfer_laws.py	The F_{transfer} discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3 = -7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]
v188_frontier_wording_guard.py	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” (M_R is the seesaw scale, not a compiler power), “relic scale f_a, m_a open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q = 0.664 \dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
v189_riemann_roch_carrier.py	The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8$, v1), $N_{\text{fam}} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$, but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
v190_nariai_entropy_bound.py	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2 + 1)/\Phi_3(x)$ ($x = r_b/r_c$; $\Phi_3 = x^2 + x + 1$ the $N_{\text{fam}} = 3$ cyclotomic, already in <code>tfpt_horizon_readouts</code>) satisfies $S_{\text{tot}} \geq \frac{2}{3} S_{\text{dS}}$ with equality exactly at the Nariai merge $x = 1$, via the perfect square $3(x^2 + 1) - 2\Phi_3(x) = (x - 1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x > 0$ is $2/3 = Z_2 /N_{\text{fam}}$) [E]
v191_universal_branch_line.py	The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ($m = 0 \rightarrow 7/2$). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v192_qgeo_energy_clock.py	The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ($\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda = k $. But ‘ ρ preserves Λ ’ is plausibly equivalent to ‘ ρ is the conformal deck’, so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays [O]
v193_qgeo_energy_commutator.py	QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, order $4= \mu_4 $, v117), not imported from the seam; (2) [O] the non-circularity hinge — Λ_Σ must be the <i>raw</i> RP-seam DtN, not the μ_4 normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the (1, 2, 3) grading on H^1 (QGEO.COHO.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O]
v194_raw_seam_dtn_rp.py	QGEO.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i> , not a closure. (1) [E] finite core: $\rho: z \mapsto iz$ on $H^1=\langle w_1, w_2, w_3 \rangle$, $w_k=z^{k-1}dz/(z^4-1)$, acts as $\rho^* w_k=i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma]=0$ on H^1 (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] residual relocates: full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGEO.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]
v195_marks_lefschetz_character.py	QGEO.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are <i>forced</i> to be a free μ_4 orbit. [E]: $\rho: z \mapsto iz$ (order 4) fixes only $\{0, \infty\}$; $H^1=\chi_1+\chi_2+\chi_3$ (the A_3 exponents (1, 2, 3), v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; rank $H_1=n-1=3 \Rightarrow n=4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks $=\mu_4$ (cross-ratio 2); $\text{Tr}(\rho H^1)=i+i^2+i^3=-1$, $L(\rho)=1+1=2=\#\text{fixed}$. [O] residual relocates from “the seam is $\mathbb{P} \setminus \mu_4$ (geometric)” to “ H^1 carries the A_3 -exponent rep” — less circular [E][O]
v196_seam_energy_functional.py	QGEO.VARI.01 — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: $E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2$, with $E_{\text{fail}}=0 \Leftrightarrow$ the QGEO.SYM.01 conditions. [E] on the finite H^1 block ($\rho=\text{diag}(i, -1, -i)$, Λ diagonal, $\Theta=\text{conj}$) all three vanish $\Rightarrow E_{\text{fail}}=0$; [E] a μ_4 -breaking Λ or non-order-4 ρ gives $E_{\text{fail}}>0$ (μ_4 a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]
v197_rr_carrier_clifford_d5.py	ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space $C_{\text{car}}:=H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10)=D_5$ half-spinor by its even Clifford algebra. [E] $\dim \Lambda^{\text{even}}(\mathbb{C}^5)=\binom{5}{0}+\binom{5}{2}+\binom{5}{4}=1+10+5=16=2^{g_{\text{car}}-1}=\dim S^+$; the $\mathfrak{so}(10)=D_5$ half-spinor $2^4=16=S^+=\Lambda^{\text{even}}(C_{\text{car}})$; closes the chain $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ from one four-point divisor (cycle side rank $H_1=3=N_{\text{fam}}(A_3)$; function side $h^0=5=g_{\text{car}}$; Clifford spinor $16=D_5$ half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined $g_{\text{car}}=5$ [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v198_modular_commutator_reduction.py	QGE0.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual ($[\rho, \Lambda_\Sigma]=0$ on full L^2 , framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $ k =\sqrt{-d^2}$ is $\text{diag}(n)$ and the clock $\rho:z\mapsto iz$ is $\text{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho, k]=0$ exactly on <i>all</i> of L^2 ; [E] Tomita–Takesaki: $[\rho, \Lambda_\Sigma]=0$ follows from $\omega\circ\rho=\omega$ (the modular Δ is intrinsic to (M, Ω) ; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite H^1 block is already μ_4 -invariant (v177). [O] residual: the full raw seam state is μ_4 -invariant ($\omega\circ\rho=\omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational QGE0.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O]
v199_seam_state_invariance.py	QGE0.STATE.01 (work package A) — attacking $\omega\circ\rho=\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular (ρ = Coxeter of $W(A_3)=S_4$, $\rho^4=1$, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega\circ\rho=\omega \Leftrightarrow [\rho, H_1]=0$ ($C_\Sigma=\frac{1}{2}(1+\text{sgn } H_1)$); [E] character criterion: $[\rho, H_1]=0 \Leftrightarrow H_1$ block-diagonal in the four μ_4 -character classes $\{n=r \bmod 4\}$; principal symbol $ k =\text{diag}(n)$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure [E][O]
v200_seam_variational_scan.py	QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}}=0$ at the μ_4 config on the fixed block; here the clock angle is free and E_{fail} is minimised over it. [C] scanning $E_{\text{fail}}(\theta)=\ \rho(\theta)^4-1\ ^2+\text{faithfulness}$ over $\theta\in(0, \pi)$ gives the unique minimum at $\theta=\pi/2$ ($E_{\text{fail}}=0$: order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the μ_4 normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) [C][O]
v201_seam_subprincipal_marks.py	QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda= k +M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n M_f n'\rangle=f_{n-n'}$. [E] M_f is μ_4 -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ($ k $ already commutes, v198). [E] a μ_4 -mark sum $f(\theta)=\sum_{j=0}^3 g(\theta-2\pi j/4)$ has $f_m=4g_m[m\equiv 0 \bmod 4]$ for <i>any</i> profile g — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks= μ_4 orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega\circ\rho=\omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v202_rare_kaon.py	The rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)= V_{cb}$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu})=9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu})=3.33 \times 10^{-11}$. [E] CKM point + exact apex $(\bar{\rho}, \bar{\eta})=(0.1374, 0.3509)$, $\lambda_t=V_{ts}^* V_{td}$; [C] the branching ratios (<i>not</i> parameter-free — $X_t, P_c, \kappa_{\pm}$ are external EW/QCD input); [E] $\text{BR}(K^+) \text{BR}(K^+)=9.45 \times 10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6_{-1.8}^{+1.9}) \times 10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an F_{transfer} bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6 \pm 0.4) \times 10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$; [X] dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X]
v203_eht_achromatic.py	The EHT achromatic polarization-rotation intercept β_{BH} around a horizon-scale black hole — the surviving core of the old UFE/BH notes (<i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\text{BH}}(r)=Q_e^{\text{eff}} Q_m^{\text{eff}}/(256\pi^4 r^2)=16c_3^4(Q_e Q_m/r^2)$, $16c_3^4=1/(256\pi^4)$ exactly. [E] α-kernel cross-link: $16c_3^4=\delta_{\text{top}}/3$, $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ — the <i>same</i> top-form coefficient that fixes the α-kernel precision-zone correction (one row from $\beta_{\text{rad}}=\varphi_0/4\pi$). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E \cdot B$ reversal; the amplitude $Q_e Q_m$ is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X]
v204_muon_seam_g2.py	The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon-$g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma \delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2$ ($\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4$, $\sqrt{2}/\sqrt{23}$; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9) \approx 2.879 \times 10^{-9}$. [E] the value is an exact compiler number ($\delta_2=4! \text{Tr}_{S^+}(X^2)c_3^8$, $\text{Tr}=120=5!$); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi=4c_3$) is a physical bridge; [E] 0.81σ vs dispersive $\Delta a_\mu=(2.49 \pm 0.48) \times 10^{-9}$; [C] HONEST: lattice/CMD-3 HVP shrinks it ($\sim 1.5 \times 10^{-9}$), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged Δa_μ outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ excludes the mechanism [E][C][X]
v205_xi_threequarter.py	$\xi=c_3/\varphi_{\text{tree}}=3/4$: an <i>independent</i> route to the gravitational $3/4$ (archive integration of the old Eliminating-K note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$), and at tree level $\varphi_{\text{tree}}=1/(6\pi)$ this is exactly $3/4$. [E] tree identity $\xi=c_3/\varphi_{\text{tree}}=3/4$; [E] Einstein-limit reduction $8\pi c_3^2=c_3 \Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)$ (v152 $k=c_3/2$) — the torsion-compression ξ and the gapped EH replica are two independent appearances of the same gravitational $3/4$; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ (-0.226%); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v206_h0_lambda_branch.py	The cosmological-constant Letter branch: the seam number $\delta_\Sigma = m_1 e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1 = \Omega_{\text{adm}} = 48$, the Planck-convention split, and the quantitative $H_0 = 66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative H_0 -vs- Λ row). [E] $\delta_\Sigma^{\text{lead}} = 48 e^{-2\pi \cdot 137.036/3} \approx 1.085 \times 10^{-123}$; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4 = (8\pi)^2 = 631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda = M_{\text{Pl}}^4 \delta_\Sigma \approx 2.4 \times 10^{-47} \text{ GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10} \delta_\Sigma \approx 122.96$, consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}} = 66.51$, $H_0^{\text{disc}} = 67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; H_0 uses imported $\Omega_b, \omega_{\text{DM}}$, a [C] readout [E][C]
v207_asymptotic_safety.py	Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close G_{metric} . [E] the axion coupling is fixed and small $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.025330$ exactly ($g_{a\gamma\gamma} = -4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g = 2 + \eta_R$ is $+2 > 0$ as $g \rightarrow 0$ and < 0 at large g (torsion K^2 damping) $\Rightarrow$ a root $g_* > 0$; [E] $g_* = 2/a_1 > 0$ finite, Reuter-type $O(1)$, no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]
v208_bh_thermodynamics.py	Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular T_H : $\Delta^{it} = e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta = 2\pi$ and $T_H = \kappa/(2\pi)$; the 2π is the seam unit $1/(4c_3)$ ($1/2\pi = 4c_3$, v58), reproducing $T_H = c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R) = R + R^2/(6M_s^2)$ ($M_s = c_3^{7/2} \bar{M}_{\text{Pl}}$, v28), $S_W = (f_R A)/(4G) = (A/4G)(1 + R_h/3M_s^2)$ — leading $A/4G$ is the c_3 area law ($1/4 = 1/ \mu_4 $), the correction an exact scalaron consequence; [E] SdS $T_H = (1/4\pi r_h)(1 - \Lambda r_h^2)$, $1/4\pi = 2c_3$; [E] area law $S_{BH} = M^2/(2c_3) = 4\pi M^2 = A/4G$ [E][C]
v209_bh_regular_core.py	The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2 = (2/3)^6 = 64/729$ with gap $\Delta = 6 \log(3/2) > 0$ (v56) — the core is a fixed point, $\rho \rightarrow \infty$ replaced by $\varphi \rightarrow \varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3 - 3t + 2 = (t-1)^2(t+2)$, roots $(1, 1, -2)$ = the traceless projection of $a = (1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2 = (2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH} = A/4G$, $1/4 = 1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v210_mark_local_dtn.py	A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega \circ \rho = \omega$, not just $[\rho, \Lambda] = 0$); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four μ_4 marks $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = D_\theta + M_f$. [E] every off-(mod 4) Fourier coefficient of the real f vanishes to $< 10^{-16}$ (exact 4-fold cancellation), although g has all modes. [C] $\ [\rho, \Lambda]\ < 10^{-15}$. [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ satisfies $\rho C \rho^\dagger = C$ to $< 10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. [O] certifies ‘mark-local DtN $\Rightarrow \omega \circ \rho = \omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O]
v211_axion_spine_angle.py	Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$, no fit); [E] the v185 finite-T solver ($\theta=1 \rightarrow 0.03$) reaches Ω_{DM} at $\theta \approx 106^\circ$, so 108° is on target (harmonic $0.03\theta^2 = 0.107$); [C] <i>robust</i> : 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i = \pi(1 - \varphi_{\text{seam}}) \approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes the branch [E][C][O][X]
v212_leptogenesis_decouple.py	Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda = 10 = E(K_5) $ — $M_1 = M_{\text{scal}} \varphi_0^2 / A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3 / A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1 = M_R \varphi_0^4$ (which only relocated the free input to the un-pinned seesaw M_R). [E] shared decouple $A_\Lambda = 10$; [E] $M_1 = 8.65 \times 10^9$ GeV inside the thermal window $[3 \times 10^9, 3 \times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5 \times 10^{-7}$, $\eta_B \approx 1.2 \times 10^{-9}$ brackets observed 6.1×10^{-10} ; [C] HONEST: M_1 uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]
v213_ftransfer_functor.py	The F_transfer <i>functor contract</i> (CONTRACT.F.01): the four frontier transfers are ONE typed functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]; consolidates v183 (F_{pole}), v212 ($F_{\text{Boltzmann}}$), v211 (F_{relic}), v164/FR.MPME (F_{QCD}), the structural complement of the v187 typing guard. [E] axiom 1 μ_4 -deck equivariance ($\lambda_2 = (2/3)^6 = 64/729$ the deck transfer eigenvalue); [E] axiom 2 Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\ \text{Pl}(K)\ _1 = 11$); [E] axiom 3 positivity/stochasticity ($\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f \in (0, 1)$); [E] axiom 4 external modules explicit ($b_3 = -7$ a carrier output, $ b_3 = \Omega_{\text{adm}} - 10b_1 = 7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v214_seam_pillowcase.py	The <i>pillowcase</i> reduction of QGEO.SYM.01 (QGEO.PILLOW.01): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the μ_4 marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. [E] $\chi_{\text{orb}}(S^2(2, 2, 2, 2)) = 2 - 4(1 - \frac{1}{2}) = 0 \Rightarrow$ flat (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits π sum to $4\pi = 2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}} = 0$, flat away from the marks). [E] NEW LINK: cross-ratio(μ_4)=2 (v168) $\Rightarrow j(\lambda) = 256(\lambda^2 - \lambda + 1)^3 / (\lambda^2(\lambda - 1)^2)$ with $j(2) = 1728$ (all six harmonic cross-ratios $\{2, -1, \frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. [E] $j = 1728 \Leftrightarrow \tau = i$ (square torus) $\Leftrightarrow$ CM by $\mathbb{Z}[i] \Leftrightarrow \text{Aut} = \mathbb{Z}/4$ ($j=0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). [E] explicit CM $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$ has order $4 = z \mapsto iz$. [E] negative control $\lambda = 3 \Rightarrow j = 21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut} = \mathbb{Z}/2$, no clock. [E] UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. [O] does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s [E]
v215_seam_deck_killtest.py	The bedrock QGEO.SYM.01 ($\omega \circ \rho = \omega$) recast as a <i>falsifiable</i> kill-test (QGEO.KILL.01), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$; the principal symbol k commutes exactly, so the residual is the sub-principal M_f with $[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant; measure form = the raw Gaussian bulk form A is μ_4-deck-invariant. Four levers (carrier-side [E], seam-side [O]): K1 exactly 4 marks (order-4 clock free orbit $\mu_4 + b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = \mu_4$); K2 square config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); K3 mod-4 sub-principal support (μ_4-mark sum $\Rightarrow f_m = 0$ for $m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f] = 0$); K4 transfer gap $(2/3)^6 = 64/729$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$). Freeze-bound (freeze_file.csv seam_deck_symmetry). [O] the decisive kill is <i>deferred</i> (QGEO.REALIZE.01): the raw $\mathbb{P} \setminus \mu_4$ collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure
v216_marks_gauss_bonnet.py	The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = 2χ: $\mathbb{Z}_2$ branch points (deficit π) on a flat sphere ($\chi=2$) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \mu_4 = e_1(a) = N_{\text{fam}} + 1$ (the same $\chi=2$ as the 8 in $c_3 = 1/(8\pi)$). [E] the Euclidean sphere 2-orbifolds ($\chi_{\text{orb}}=0$) are exactly $(2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2)$. [E] all-order-2 (the $\mathbb{Z}_2$ sheet branch) $\Rightarrow$ uniquely $(2, 2, 2, 2)$; [E] $N_{\text{fam}} = 3$ (rank $H^1 = \# \text{marks} - 1$) $\Rightarrow$ uniquely $(2, 2, 2, 2)$, picking the 4-mark square over the 3-mark hexagonal $(3, 3, 3)$; [E] a free order-4 deck $\Rightarrow$ uniquely $(2, 2, 2, 2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio $2 \Rightarrow j = 1728$, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01

continued on next page

Script	What it machine-checks (<i>continued</i>)
v217_marks_emergence_scan.py	The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks n and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\ \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$, only $n-1=3=N_{\text{fam}}$; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$); [C] gap: μ_4 -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6 = 64/729$; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square ~ 0.4); [C] negative controls $n=3, 5, 6$. [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only
v218_diamond_axis_geometry.py	Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: C center, U winding, V sheet), F the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$; $\det B(C+xU)=2 \det(C+xU)$, anchor-block sheet curvature $6= R^+(A_3) $. [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$, steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2 \text{Disc}(q)$ for Q, K, C, F ; squares $(1, 3, 4, 6)$, kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$, $105=N_{\text{fam}}g_{\text{car}} \cdot 7$. [C] heuristic: F the transfer-completion corner, a search principle, not a hard filter; the G_2/F_4 pair-sum labels stay audit-only (cf. v94/v95) [E][C]
v219_icosahedral_mckay.py	McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$. E_8 is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$, containing the $2, 3, 5$ axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$, sum of squares $120= R^+(E_8) $); the natural-rep McKay adjacency <i>is</i> the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]
v220_cp_hexagonal_modulus.py	CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$, $\text{Aut}=\mathbb{Z}/4$ (the μ_4 clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$, $\rho=e^{i\pi/3}$, $\text{Aut}=\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg \rho=\pi/3$. [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]
v221_seam_qecc.py	The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$; the gapped transport T on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$; on the trace-zero subspace $\ T\delta\ \leq (2/3)^6 \ \delta\ $ hence $\ T^n \delta\ \leq (2/3)^{6n} \ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]
v222_cm_norm_duality.py	CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, $j=1728$): $N(g_{\text{car}}+i \mu_4)=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, $j=0$): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=\text{scalaron}$. The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21 \neq 41$). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v223_coxeter_totative_clock.py	The μ_4 clock is the order-4 character of the E_8 Coxeter cycle. [E] $(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ = the E_8 exponents = live Coxeter phases; $\varphi(30)=8=\text{rank } E_8$, $30=2\cdot 3\cdot 5$. [E] $(\mathbb{Z}/30)^\times \cong \mathbb{Z}/4 \times \mathbb{Z}/2$ with order-4 generator 7 $(7^2=19, 7^4=1 \bmod 30)$: a literal μ_4 clock $\langle 7 \rangle = \{1, 7, 13, 19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes. Only E_8 has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$; the seam-clock identification is [C][E][C]
v224_diamond_fttransfer_path.py	F_{transfer} as one path on the Sheet Diamond, not four external interfaces. [E] family $M(s, t) = R + Q \text{diag}(s, t, t)$: the corners $K=M(1, -1), C=M(1, 0), F=M(1, 1)$ lie on the sheet axis ($s=1$); the sheet axis is curved $\det M(1, t) = 4t^2 + 14t + 14$ (2nd diff $8=\text{rank } E_8$), the winding axis $\det M(s, 0) = 6s + 8$ flat (slope $6= R^+(A_3) $); Plücker steps $(1, 8, 10), (1, 8, 16)$. [C] F_{transfer} is the $K \rightarrow C \rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius projection (v37/v183) [E][C]
v225_dual_normal_frame.py	The dual normal frame (d, n) : one boundary compass for flavor and horizon, with CP as orientation. [E] $d = a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}(1, 1, -2)$ (the Nariai/horizon normal; $d \cdot \mathbf{1} = 0, d \cdot a = 1$); $n = (5, -9, 6)$ with $n^\top R = (\det R, 0, 0) = (8, 0, 0)$, $n^\top L = (20, 0, 0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$. [C] CP as orientation: $\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$ for $\rho = e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]
v227_degree_exponent_channel_split.py	$248=120+128$ as a magnitude/phase channel typing (replaces the over-strong S^- -dark-matter reading). [E] E_8 exponents sum $120= R^+(E_8) $ (<i>magnitude</i> channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank} \cdot \dim S^+$ (<i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$; $\sum \deg - \sum \exp = \text{rank} = 8$. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]
v228_rr_index_gate.py	P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ; bedrock language for P2, not a proof from nothing). [E] $\deg D = 4 = \mu_4 $ on $\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D)) = 5 = g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1 \setminus 4) = 3 = N_{\text{fam}}$; $h^0 + \text{rank } H_1 = 8 = \text{rank } E_8$, $h^0 - \text{rank } H_1 = 2 = \mathbb{Z}_2 $; $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]
v229_lepton_frobenius_algebra.py	The charged leptons as an étale Frobenius algebra over the μ_6 resolvent. [E] exact coefficients $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = \mathbb{Z}_2 c_\mu$, product $\frac{32}{9} = 2g_{\text{car}} / N_{\text{fam}}^2$. [E] $A = \mathbb{Q}[t]/(m)$, $m = \prod(t-c)$: the trace-form Gram is nondegenerate ($\det = \text{disc}(m) \neq 0$, A étale) — a commutative Frobenius algebra; the C_6 shift has characteristic polynomial $t^6 - 1$ (spectrum $\mu_6 = \mu_3$ family $\times \mu_2$ sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C]
v230_center_budget_norms.py	The Sheet-Diamond center budget $(7, 11, 13)$ is the three <i>local norms</i> of the theory. [E] $C = R + Q \text{diag}(1, 0, 0)$, row sums $(7, 11, 13)$: $7 = N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13 = N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = Δ_Q), $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 1+4+6$ (QBL boundary Fock count on the 4 μ_4 corners = $\dim S^+ - g_{\text{car}} = 16 - 5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v231_cp_mu6_phases.py	Both CP phases are μ_6 powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] ρ is a primitive 6th root ($\rho^6=1$, $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] δ_{CKM} leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i>); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the C_6/μ_6 monodromy has charpoly t^6-1 (spectrum μ_6 carries ρ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$, $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG μ_3 has no -1 (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]
v232_e8_kleinian_seam.py	The seam as the E_8 <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine E_8 ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> E_8 Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$'s = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \# \text{ nontrivial } 2I \text{ irreps}$ — a fourth reading of the 8 in $c_3=1/(8\pi)$. [E] the negated intersection form is the E_8 Cartan (even, $\det=1$, positive definite; each $\mathbb{P}^1$ a (-2) -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$: the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $. [C] seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes E_8); bedrock [O][E][C][O]
v233_cp_triality_phase.py	The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$: the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$, the sheet $\mu_2=\{1, -1\}$; $\rho=\omega^2 \cdot (-1)$. [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$, so the quark (ρ^1) and lepton (ρ^4) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase ω^2 (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] ω is the μ_3 factor of the order-30 $(2, 3, 5)$ Milnor/Coxeter monodromy that builds E_8 (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]
v234_seam_holomorphy_selection.py	The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing E_8 . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic (μ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) = \Gamma^{\text{ab}} = H_1(S^3/\Gamma) = \#(1\text{-dim irreps})$: $A_n \rightarrow n+1$, $D_n \rightarrow 4$, $E_6 \rightarrow 3$, $E_7 \rightarrow 2$, $E_8 \rightarrow 1$ — only E_8 gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = E_8 . [E] the three faces (v83/v154, v232, v219) are the SAME E_8 -selector, so Target A, G_{net} , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v235_seam_chern_simons.py	The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer K -matrix with $\# \text{anyons} = \det K $, edge $c = \text{signature}(K)$, holomorphic $\Leftrightarrow \det K =1$. The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ($\det 16$, 16 anyons) $\rightarrow SO(16)_1 = D_8$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$, holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$. [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev E_8 quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$). [E] NEG: D_8 is free+bosonic but has 4 anyons; an odd K is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]
v236_brieskorn_capstone.py	The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $(\mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$. [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, m the E_8 exponents (charpoly Φ_{30} , $\deg \varphi(30)=8=\mu$) — so it <i>is</i> the order-30 E_8 Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the μ_3 triality (CP, v233) as h^{10} ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the μ_4 clock ($\times 7$, v223). [E] Milnor lattice = E_8 , link = Poincaré sphere. [C] one singularity generates atoms + 8 + E_8 + order-30 + both clocks + seam; a unification, not a closure [E][C]
v237_seam_sre_closure.py	The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the K -matrix ground-state degeneracy on a genus- g surface is $ \det K ^g$ (carrier $\rightarrow 16$, $D_8 \rightarrow 4$, $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$. [E] among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev E_8), edge holomorphic; D_8 /carrier are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk (D_8) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev E_8 phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O]
v238_modular_lindblad_dynamics.py	DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the μ_4 clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v239_kms_thermal_time.py	DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. [E] KMS at $\beta = 1$: $\omega(A\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \text{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.
v240_gns_os_reconstruction.py	DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data (M, ω) , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6 \log(3/2), 6 \log 3\} \geq 0$, gap = $6 \log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.
v241_dhr_sectors.py	DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\text{carrier}} = \dim S^+$), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$. [E] tower = v235/v237: Lagrangian $\langle (1, 1) \rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$; order-2 $\langle (2, 2) \rangle \rightarrow 4 = D_8$; carrier $16 \rightarrow D_8 \rightarrow E_8$. [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact.
v242_spin_statistics_modular.py	DHR.MODULAR.01 — spin-statistics and the modular (S, T) data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4 \times \mathbb{Z}_4$, $q = (5x^2 + 3y^2)/8$, $\theta = e^{2\pi i q}$). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) = $\{(0, 2), (2, 0)\}$, 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.
v243_haag_ruelle_braiding.py	DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. [E] mass gap $\Delta = 6 \log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M = 1$, diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: M is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$, $M(a, -b) = M(a, b)^*$. [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding ($\det 1/\text{SRE}$). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).

continued on next page

Script	What it machine-checks (<i>continued</i>)
v244_spectral_action_contract.py	CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net $\rightarrow$ 4d physics). A single Dirac operator whose $\text{Tr } f(D/\Lambda)$ yields gravity AND matter. [E] carrier spinor = one generation: $SO(10)$ half-spinor = $\dim S^+ = 2^{g_{\text{car}}-1} = 16$ ($15 + \nu_R$), $N_{\text{fam}} = 3$. [E] SM gauge group inside the carrier: $SU(3) \times SU(2) \times U(1)$ ($\dim 12$) $\subset SU(5)$ (24) $\subset SO(10) = D_5$ (45); colour $SU(3) \subset A_3 = SU(4)$ (Pati-Salam); $\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8$. [E] one Dirac operator $\rightarrow$ gravity ($a_2 \sim R$, $a_4 \sim R^2$, $\sqrt{36}$) + matter ($a_4 \sim F^2$ Yang-Mills + $D\phi ^2$ Higgs, Gilkey). [X] hooks: $g_3 = g_2 = \sqrt{5/3} g_1$ + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam $\rightarrow$ Dirac D; finite geometry = carrier; spectral action reproduces SM couplings) $\rightarrow$ the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.
v245_spectral_matter_content.py	QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the $SO(10)$ 16. [E] 16 = one generation: $SO(10)$ 16 = $SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})$; SM dims $6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+$, $N_{\text{fam}} = 3$ (15 Weyl fermions + ν_R). [E] electric charges $Q = T_3 + Y$ are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: $\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2 U(1) = 0$, $[SU(3)]^2 U(1) = 0$ (exact). [E] unification: $\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8$, i.e. $g_3 = g_2 = \sqrt{5/3} g_1$. [E] Higgs = the $(1, 2)_{1/2}$ inner fluctuation (one doublet). [O] residual: the seam $\rightarrow$ Dirac realisation + measured-coupling reproduction stay the open contract.
v246_unification_data_crosscheck.py	QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1 = g_2 = g_3$, $\sin^2 \theta_W = 3/8$ at Λ survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, M_Z): $\alpha_i^{-1} = (59.01, 29.59, 8.47)$, SM 1-loop $b = (41/10, -19/6, -7)$ (v159). [E] the prediction: $\alpha_1 = \alpha_2 = \alpha_3$, $\sin^2 \theta_W = 3/8$ at Λ. [X] tension: pairwise crossings spread ~ 4 decades ($\sim 10^{13} - 10^{17}$ GeV), residual spread at 2×10^{16} $\Delta \alpha^{-1} \sim 8.8$ — $g_1 = g_2 = g_3$ not realised by the plain SM. [X] $\sin^2 \theta_W$ 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread ~ 3.2 at 1.7×10^{14} GeV). [O] no rescue: SM gauge content + no new states (the E_8 spine is arithmetic, not a threshold tower) $\Rightarrow$ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.
v247_e8_branching_no126.py	PS.E8BRANCH.01 — the E_8 audit hull forbids the 126 and supplies exactly $\{1, 10, 16, 45\}$ for $SO(10)$. Carrier $D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8$ (via $SO(16)$); 248 = $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248$. [E] ranks $5 + 3 = 8$; [E] dims = 248 ($SO(16)$: 120 adj + 128 spinor); [E] $SO(10)$ content $\{1, 10, 16, 45\}$, the 126 ABSENT $\Rightarrow 126_H$ algebraically forbidden, $10 + 16 + 45$ E_8-supplied not fitted; [E] caveat only one 45/one 15 $\rightarrow$ proton-marginal; [O] the field-content principle + NCG complement (v250).
v248_ps_algebra.py	PS.ALGEBRA.01 — the Pati-Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ and its exact SM matching. [E] unimodular group $SU(2)_L \times SU(2)_R \times SU(4)_c$ (rank 5, $\dim 21$); [E] 16 = $(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8$; [E] $Y = T_{3R} + (B - L)/2$ gives the exact SM hypercharges and $Q = T_{3L} + Y$ the exact charges; [E] matching $\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$. [O] A_F uniquely from seam+sheet+μ_4; [O] caveat $SU(4)_c \subset SO(10) = D_5$ is not the carrier $A_3 = SU(4)$ (family).
v249_ps_unification.py	PS.RGTEST.01 — carrier-native Pati-Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread ~ 8.8 @ 2×10^{16}); [C] $M_{PS}/M_s = 1.0 - 1.7$ (1-loop; 2-loop $\sim 1.0 - 1.2$) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match ($b_{2R} > 0$; also E_8-forbidden v247); [X] NEG proton decay selects content (minimal $M_{GUT} \sim 2 \times 10^{15}$ excluded; $+(15, 1, 1)$ safer). [O] the gauging fork, exact κ, the one-45 marginality.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v250_dirac_triple_contract.py	CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target (A, H, D, J, γ) almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$, D_F from $(R, L, \varphi_0, \text{Majorana})$. [E] $H_F = 3 \times 16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + ν_R Majorana; $M_{PS} = \kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.
v251_first_order_sigma.py	PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ σ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + σ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: σ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$. [E] order-zero; [E] real-even ($J^2 = +1$, $\gamma^2 = 1$, $D\gamma = -\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R \nu_R^c$ blocks (σ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).
v252_full_finite_triple.py	PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ 16 -plets (dim 96). [E] dim 96; D_F Hermitian+odd; reality $J^2 = +1$, $JD = DJ$; grading $\gamma^2 = 1$, $[\gamma, a] = 0$, $J\gamma = -\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2 / M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion.
v253_kappa_scalaron_ps.py	PS.KAPPA.01 — pinning κ in $M_{PS} = \kappa M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] κ pinned by RG (scan PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval ~ 1.3 (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$, a ratio of heat-kernel traces over the 96-dim H_F (v252) — Λ cancels, κ scale-free and $O(1)$. [X] the 126 -content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments f_2/f_0 fixed.
v254_inner_fluctuations.py	PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not 126) + the no-junk obligation + the σ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no 126 : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$); [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam — σ is the CCvS beyond-first-order scalar; [E] σ is an SM singlet $(1, 1, 0)$ with $B-L = 2$; [C] $\sigma = \text{scalaron consistency}$.
v255_spectral_action_expansion.py	PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4 \Lambda^4 a_0 + 2f_2 \Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl ² + R^2 +Higgs-quartic, $N=96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with σ ; [E] scalaron = R^2 spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v256_orientability_ poincare.py	PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$, full orientability is a degree-4 cycle on $M \times F$; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.
v257_quasiorient_ modpd.py	PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); [E] quasi-orientability holds (left/right A -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- L/R -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.
v258_dirac_ covariance_ induction.py	PS.DIRAC.03 — D_F is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance C (contraction, $\text{spec} \subset (0, 1)$) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ($\beta=1$, v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ($D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$); [E] unitarily equivalent to v252 (inducing C_F from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$); [C] $C_F = P_F C_\Sigma P_F$, so D_F is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of C_Σ .
v259_modular_ cutoff_kappa.py	PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$, $f_2=\int f=1$, $f_4=\int u f=1$, so $f_2/f_0=f_4/f_2=1$; [E] seam-fixed $\beta=1$ (no new parameter beyond c_3); [E] non-vacuous neg control: a Gaussian cutoff e^{-u^2} gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$, no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ($O(1)$); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v260_k3_kummer_unification.py	ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the E_8 lattice as facets of one smooth object, the geometric unification of the v1 ‘glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$’ step (the K3/Kummer proposal). The seam square torus $\tau=i$ ($j=1728$, v214) under the elliptic involution $z \mapsto -z$ squares to the Kummer K3. [E] the E_8 Cartan is even, $\det=1$, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3}=U^3 \oplus E_8(-1)^2$ has rank $22=b_2$, $\det=-1$ (unimodular), is even, signature $(3, 19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam = $T^2/(z \mapsto -z)$: $4= (\mathbb{Z}/2)^2$ fixed points, pillowcase $S^2(2, 2, 2, 2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $A[2] =2^4=16 \rightarrow 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$ (v197), $16=4 \times 4$; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of E_8 in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global H^2 summand); [C] unification (lattice [E], carrier$\leftrightarrow$node identification [C]).
v261_modular_spectral_closure.py	QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object TFPT_{QFT} = $(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B:A]=4$ (v89) = $\mu_4 =4$ (v92) = seam marks $2\chi=4$ (v216) = pillowcase/Kummer fixed points $(\mathbb{Z}/2)^2 =4$ (v260); [E] one carrier-16: $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16$, $H_F=2N_{\text{fam}} \cdot 16=96$; [E] one gap $\Delta = -\log((2/3)^6) = 6\log(3/2) > 0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1-C)C^{-1})=H$, v258; KMS cutoff $f_2/f_0=1$, v259; K3 lattice signature $(3, 19)/\det -1/\text{even}$, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.
v262_fqcd_mp_me.py	FR.MPME.02 — the F_{QCD} transfer pipeline for m_p/m_e, implemented as a real solver (the one F_{transfer} pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3 = -7 = -(11-2n_f/3)$ at $n_f=6$ (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z)=0.1179$ run with thresholds gives $\alpha_s(2\text{ GeV}) \sim 0.30$ (real ODE solve); [C] $\Lambda_{\text{MS}}^{(3)} \sim 0.33\text{ GeV} \times$ the lattice bridge $C_p = m_p/\Lambda$ (external $O(1)$) over the closed m_e gives $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget; [X] firewall: m_p/m_e is not an admissible compiler integer ($SU(9)$ absent from $D_5 \oplus A_3$). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
v263_mnu_from_df.py	FLAV.PMNS.02 — the light neutrino matrix M_ν as the type-I seesaw of D_F, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from D_F’s blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of D_F); [E] a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the v9 texture is in the seesaw image); [C] diagonalising gives $\theta_{23}=45^\circ$ and $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675$ under the seam misalignment; [O] $\theta_{13}=0$ in the leading texture (physical $\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}$ a separate readout), the CP magnitude $\delta_{\text{PMNS}}=240^\circ$ is the $\mu_6/\text{triality}$ phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v264_raw_seam_ marklocal.py	QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit π each, sum $4\pi=2\pi\chi(S^2)$), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the μ_4 orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] M_f then commutes with the clock $\rho=\text{diag}(i^n)$; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the μ_4 orbit) breaks it. A reduction/sharpening, not a closure.
v265_qft4d_fork_ freeze.py	QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread ~ 8.8 at 2×10^{16} , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch (E_8 content $\{1, 10, 16, 45\}$ no 126 v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake τ_p window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.
v266_ps_threshold_ proton.py	PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$: the carrier-native PS UV branch survives proton decay <i>only</i> with the E_8 -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$, $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$; [X] minimal KILLED ($\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K 2.4×10^{34} , even high-band); [C] $+(15, 1, 1)$ survives now ($\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> (τ_p at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.
v267_qgeo_rigidity_ minimal_axiom.py	QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier μ_4 clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of a); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$, only $\mathbb{Z}_2$; hexagonal $j=0$ has $\mathbb{Z}_6$, order $6 \neq 4$); [C] arithmetic rigidity ($\{1, i, -1, -i\}$ over $\mathbb{Q}$, $j=1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom.
v268_theta13_ carrier_trace.py	FLAV.TH13.01 — the reactor angle θ_{13} <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG ~ 0.0222 ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ θ_{13} from M_ν ’ route); [O] CP magnitude + absolute ν scale remain open.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v269_spert_paqft_skeleton.py	QFT4D.SPERT.01 — the perturbative 4D S-matrix (S_{pert}) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix (S_{top} 2d DHR braiding v243 / S_{pert} 4d EG of the spectral action / S_{phys} LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action a_4 operator has mass dimension ≤ 4 in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ($\Delta=6 \log(3/2)>0$) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.
v270_pmns_jarlskog_assembly.py	FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog J_{PMNS} (the CP <i>strength</i>), assembled from the four TFPT channels ($\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$, $\sin^2 \theta_{23}=\frac{1}{2}$, $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$, $\delta=4\pi/3$) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative sign reproduced; [O] δ is the μ_6 /triality phase (assembled input, <i>not</i> from M_ν/D_F) and the absolute ν -mass scale (Σm_ν , M_R) stays open. Assembles the PMNS matrix + CP strength; does not derive δ from D_F nor fix the scale.
v271_eg_oneloop_quartic.py	QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 S_{pert} skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=L_{\text{int}}$: needs no extension, the first extension is T_2 ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$, the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d,d)=1$, renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action a_4 geometric quartic; [C] the ϕ^4 one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number.
v272_nu_mass_scale.py	FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a Σm_ν number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like v_{geo} (v153), while the ratios (m_2/m_3 , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology (< 0.072 – 0.12 eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number.
v273_eg_gauge_running.py	QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $\text{sd}=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop b_i from the explicit carrier/SM content: $b_3=-7$, $b_2=-19/6$, $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact b_i mirrored in Wolfram.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v274_scale_ overdetermination.py	SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\text{Pl}}^d$ ($\rho_\Lambda/M_{\text{Pl}}^4$, $M_{\text{scal}}/M_{\text{Pl}}=c_3^{7/2}$, the spectrum); [C] two independent routes to M_{Pl} agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\text{Pl}}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit (G), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.
v275_qgamb_ roadmap.py	QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2)-2\cdot 248\,c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c=248/31=8$; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.
v276_qgeo_ flat_closes_ commutator.py	QGEO.SYM.03 — the flat $\tau=i$ pillowcase premise <i>closes</i> the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to <i>one</i> geometric postulate (it does <i>not</i> prove the premise). For a quasi-free seam state $\text{QGEO.SYM.01} = (\omega \circ \rho = \omega) \Leftrightarrow [\rho, C]=0 \Leftrightarrow [\rho, H]=0$. [E] isometry $\Rightarrow$ commutator: if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \rightarrow iz)$ is a flat- $\tau=i$ isometry then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ <i>exactly</i> (all orders), verified on the discretised square torus ($\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full H (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded; v214/v267); [O] QGEO.SYM.01 collapses to the single statement ‘the raw seam is the flat $\tau=i$ pillowcase quasi-free state’ — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form.
v277_seam_calderon_ e8_match.py	QGAMB.TIERB.01 — the seam-Calderón $\rightarrow (E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to <i>one</i> bit (it does <i>not</i> close it). [E] $c=8$ (Sugawara, $248/31=g_{\text{car}}+N_{\text{fam}}$), but $c=8$ alone does not pin the net; [E] the discriminator is holomorphy — $\det \text{Cartan}(E_8)=1$ (1 primary, holomorphic) vs the $c=8$ counterexample $\det \text{Cartan}(D_8=SO(16))=4$ (4 primaries, non-holomorphic), so the single discriminator is $ \det K =1$; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248=\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic $\Leftrightarrow \det \text{Cartan} 1 \Leftrightarrow$ unique even unimodular rank-8 (v83) $\Leftrightarrow$ homology-sphere link $\Leftrightarrow \det K =1$ (v235); [C] seam-side match via v276 (flat $\tau=i$) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, <i>not</i> closed.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v278_lszy_bridge_unity.py	QFT4D.SPERT.04 — the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (S_{phys} = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: S_{phys} = amputate+on-shell-residue of the connected time-ordered products, and the EG T-products of S_{pert} are those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states so LSZ is valid; [E] <i>one-loop unitarity</i> (optical theorem): the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$ <i>exactly</i> = the two-body phase space, so $2\text{Im } M = \sum \int d\Pi M ^2$ (cutting rule) holds $\Rightarrow$ matter+gauge unitary; [E] threshold: $\text{Im } B=0$ below $s=4m^2$, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary $\rightarrow$ QG.AMB.01; [C] finite field-strength Z from the EG scheme.
v279_qgeo_obligation_lemma.py	QGE0.OBLIG.01 — the QGE0.SYM.01 bedrock written as <i>one</i> precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does <i>not</i> close it). Lemma: given the premise ‘the raw seam carries the flat $\tau=i$ pillowcase metric g_{flat} (Trojanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$’, then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$ mark-locality $\rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has 8 closed nodes and <i>exactly one</i> open leaf (‘the raw seam state is the flat state’); [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders, Lean FORM.QGE0.03), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Trojanov + OS); [E] the all-orders step is a Lean theorem (SeamDeckClosure.flat_all_orders_clock, standard axioms, no sorry); [O] one constructive-QFT statement closes QGE0.SYM.01.
v280_pillowcase_steklov.py	QGE0.STEKLOV.01 — a direct numerical experiment on the flat $\tau=i$ pillowcase orbifold (the seam geometry): the self-investigable test of the QGE0.SYM.01 obligation (does <i>not</i> prove the premise). [E] pillowcase = the Z_2-even sector of the flat torus, 4 cone points (2-torsion, deficit π each, sum $4\pi=2\pi \cdot 2$); [E] the order-4 deck $\rho(z \rightarrow iz)$ descends ($[\rho, Z_2]=[\rho, P_{\text{even}}]=0$) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact: $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.
v281_holomorphy_modular_data.py	QGAMB.MODULAR.01 — the modular-data / anyon-condensation layer of the QG.AMB.01 Tier-B holomorphy bit. [E] #anyons = $\det \text{Gram} : E_8 \rightarrow 1$ (holomorphic), $D_8=SO(16) \rightarrow 4$, carrier $D_5 \oplus A_3 \rightarrow 16$; [E] the anyon-condensation tower $D_5 \oplus A_3 (16) \rightarrow D_8 (4) \rightarrow E_8 (1)$, each step condensing a Lagrangian Z_2 boson, the Kitaev E_8 endpoint (v92/v235); [E] Gauss–Milgram recovers $c=8$; [E] E_8 is the unique holomorphic $c=8$ ($\det 1$) — $SO(16)$ has the same c but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to $\det K=1$? Reduced to one bit.
v282_e8_tau_i_unification.py	QGAMB.UNIFY.01 — the E_8-at-$\tau=i$ unification: the candidate <i>simplification</i> reducing the two open obligations to <i>one</i> object (the raw seam is the $(E_8)_1$ holomorphic CFT at $\tau=i$). $\chi_{E_8}(\tau)=\Theta_{E_8}/\eta^8=j^{1/3}$. [E] $\tau=i$ is the order-4 CM point ($j=1728$, the QGE0.SYM.01 flat geometry); [E] $\chi_{E_8}=j^{1/3}$ ($\chi^3=qj$ as q-series), so $\chi_{E_8}(i)=1728^{1/3}=12$ (the $(E_8)_1$ character at the SAME $\tau=i$, the QG.AMB.01 holomorphy); [E] one object, two faces; [C] obligation count $2 \rightarrow 1$: prove ‘the raw seam is $(E_8)_1$ at $\tau=i$’ and <i>both</i> follow; [O] the unified premise stays open but is now one object.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v283_live_killtest_scorecard.py	PRED.LIVE.01 — a live kill-test scorecard of the recent round’s computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2 \theta_{12}=0.30675$ ($+0.31\sigma$), $\sin^2 \theta_{13}=0.02311$ ($+1.45\sigma$), θ_{23} maximal; [C] leptonic CP $J_{\text{PMNS}}=-0.0297$, $\delta=240^\circ$ vs $\sim 232(30)$; [X] Σm_ν NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p=1.55\times 10^{35}$ yr for $+(15, 1, 1)$ (Hyper-K decisive); [E] closed readouts α^{-1} , Ω_b , β ; all rows consistent $< 2\sigma$, two sharp near-term kill tests. A consolidation, not new physics.
v284_route_i_rp_uniqueness.py	QGEO.ROUTEI.01 — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does <i>not</i> prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip DtN(k)/ $k \rightarrow 1$, principal symbol $ \xi $ with no curvature term), L4 covariance $C = (1 + e^H)^{-1}$ from the DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat ($\tau=i$), so L_{open} reduces to ‘the raw seam is the constant-curvature geometric state’; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step).
v285_route_ii_seam_condensation.py	QGAMB.ROUTEII.01 — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma <i>coincides</i> with Route (i)’s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow \det K =1 \Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic $c=8 \Leftrightarrow (E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower $16 \rightarrow 4 \rightarrow 1 =$ the unique μ_4 -Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share <i>one</i> open lemma — both say ‘the raw seam is the $(E_8)_1$ -at- $\tau=i$ object’ (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. $M_{\text{open}} = \text{QGEO.SYM.01}$.
v286_seam_equivalence_contract.py	SEAM.EQUIV.01 — the Seam Equivalence contract: concentrates the whole open structure on <i>one</i> named arrow (does <i>not</i> prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau=i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] import firewall : no Route-i module imports any Route-ii module or vice versa, so the routes converge only through SEAM.EQUIV.01 (kills the ‘ E_8 into the geometry and back’ circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary’.
v287_free_rp_bulk_to_holomorphic_boundary.py	SEAM.EQUIV.A01 — Route A (AQFT) theorem-dependency checker for the open Gral arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary net’: types the 5-lemma chain and isolates the <i>one</i> missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality $\Rightarrow$ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk $\Rightarrow$ single-sector boundary (μ -index 1) — the one missing standard theorem (the AQFT form of $\det K=1$); [E] L4 $\mu=1+c=8 \Rightarrow$ holomorphic (v234); [E] L5 holomorphic $c=8 \Rightarrow (E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces SEAM.EQUIV.01 to one cited theorem.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v288_full_l2_subprincipal_z4.py	SEAM.EQUIV.B01 — Route B (DtN): the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the ‘probably provable’ step and lifting v201/v284 from the finite block to L^2 . [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod{4}$; [E] full- L^2 block-diagonality: $\Lambda = n + M_f$ commutes with $\rho = \text{diag}(i^n)$, $\ [\rho, \Lambda]\ \sim 2 \times 10^{-15}$ on $-50..50$; [E] negative control: a generic curvature gives $\ [\rho, \Lambda]\ = 4.44 > 0$; [E] Lean consistency (geom_sum_fourth_root + markLocal_blockDiagonal); [O] the sharper residual — why is the raw seam sub-principal term mark-local?
v289_marklocal_raw.py	MARKLOCAL.RAW.01 — decompose the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $= D_\theta + M_f + \text{cone sources}$ (b-calculus/Melrose, imported); [O] L2 Flat-Away — the one open analytic lemma: $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$ (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one μ_4 orbit (v195/v216); [E] L4 Fourier-support: μ_4 -orbit curvature $\Rightarrow$ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: $\Rightarrow [\rho, \Lambda] = 0$ on full L^2 (v288). 4/5 discharged; Route B reduces to exactly L2.
v290_z4_smooth_curvature_adversary.py	SEAM.ADVERSARY.01 — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is <i>not</i> mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$ -symmetric off-mark curvature $f = \varepsilon \cos(4\theta)$: [E] it is $\mathbb{Z}_4$ -symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$) but <i>not</i> mark-local ($f(\pi/4) = -\varepsilon \neq 0$); [E] it <i>passes</i> the v288 commutator $\ [\rho, \Lambda]\ = 0$ (same as mark-local) — so $[\rho, \Lambda] = 0$ is necessary but <i>not</i> sufficient; [E] but it shifts the DtN spectrum off the flat ladder ($\max \Delta\lambda = 0.155$) and the heat trace ($\Delta = 0.019$), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input.
v291_flataway_contract.py	FLATAWAY.RP.01 — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four μ_4 marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] <i>spectral-rigidity</i> : scanning $f = \varepsilon \cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon=0$ — flat is the unique spectral match; [E] <i>heat-kernel</i> : the heat-trace deviation is 0 only at $\varepsilon=0$; [C] <i>RP-energy</i> : $\text{RP} + \text{gap}$ is a Dirichlet-energy minimisation in the Troyanov conformal class $\Rightarrow$ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B.
v292_flataway_heat_reduction.py	FLATAWAY.HEAT.01 — the Heat-Kernel route made precise: reduces Flat-Away to <i>one</i> named spectral invariant. [E] the heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta \text{Tr} = 0 \Leftrightarrow f = 0$; leading invariant $\ f\ _{L^2}$ (Parseval). [O] reduction: Flat-Away $\Leftarrow$ ‘the seam fixes the a_2 heat coefficient to its flat value’ $\Rightarrow \int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.
v293_flataway_spectral_hessian.py	FLATAWAY.SPEC.01 — the Spectral-Rigidity route upgraded from one direction (v291) to the <i>full</i> smooth- $\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $ n $ is split by mode $4k$ at $ n =2k$ by $\pm\varepsilon/2$; [E] $S = \sum (\Delta\lambda)^2 \sim 0.50\varepsilon^2$; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$, so the flat seam is a <i>strict</i> isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non- $\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam’s Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation $\Rightarrow$ Flat-Away.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v294_flataway_rp_energy.py	FLATAWAY.ENERGY.01 — the RP-energy / Trojanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: S^2 with 4 cone angles π , $\sum(2\pi - \theta) = 4\pi = 2\pi\chi$; [C] Trojanov/Euclidean-cone uniqueness (Trojanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f] = \int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: ‘RP+gap realises the energy-minimiser’, converging with v292/v293 on one selection principle; closing any one closes Route B.
v295_flataway_a2_exact.py	FLATAWAY.A2.01 — the <i>exact</i> analytic a_2 coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \text{Tr } e^{-t\Lambda(\varepsilon)} _0 = -t\hat{f}(0) \sum e^{-t n } = 0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x) = e^{-tx}$ strictly convex $\Rightarrow \Lambda \mapsto \text{Tr } e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon=0$ is the global minimum and $\Delta \text{Tr} \geq 0$ for all ε , $= 0$ iff $f = 0$; [E] exact a_2 kernel via the divided difference of e^{-tx} (Duhamel), matched to numerics to rel.err $< 10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1) \geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ ‘the seam fixes a_2 ’. Lean-mirrored (FORM.FLATAWAY.01).
v296_flataway_a2_closed.py	FLATAWAY.A2.CLOSED.01 — the exact a_2 coefficient in <i>closed form</i> . [E] the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$), giving $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$; [E] a finite $(4k-1)$ -term middle (incl. the degenerate diagonal $\Phi(2k, 2k) = (t^2/2)e^{-2kt}$); [E] closed form $W_k(t) = \frac{t(1-e^{-4kt})}{8k(1-e^{-t})} + \frac{1}{2} \sum_{m=1}^{4k-1} \Phi(m, 4k-m)$ $(W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8)$, reproducing v295 to machine precision; [E] manifestly positive; [E] small- t : $W_k = t/2 + O(t^3)$ (mode-independent leading $1/2$, the t^2 term cancels).
v297_route_a_literature_stack.py	SEAM.EQUIV.A.LIT.01 — Route A’s ‘one standard import’ as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c=8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into ‘invertible bulk $\Rightarrow (E_8)_1$ ’, all legs established; [O] the one open hypothesis is the input of LIT-A, ‘the raw quasi-free seam bulk is invertible (SRE)’ = Route B’s Flat-Away. [E] non-circular with Route B (the v286 firewall holds).
v298_e8_network_attractor.py	<i>E8 as a network attractor</i> — a computational-substrate (Wolfram-program) reading of the $(2, 3, 5)$ /McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the <i>unique</i> attractor of the minimal lazy graph update $m \leftarrow (A + 2I)/4 \cdot m$ on the affine- $E_8/(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at E_8 ($2T \rightarrow E_6$, $2O \rightarrow E_7$, $2I \rightarrow E_8$, $h = 30$); [E] the cascade is nested-subgraph coarse-graining ($240 \rightarrow 126 \rightarrow 72 \rightarrow 40 \rightarrow 20$). [C] Wolfram reading: E_8 as the universal attractor of a minimal $(2, 3, 5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph <i>rewrite</i> whose attractor is the full TFPT structure stays open; $\varphi_0 = (4/3)c_3 = 1/(6\pi)$ is the analytic seed, not an edge fraction.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v299_hypergraph_0d_autonomous.py	The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth <i>reaches</i> E_8 as the last finite point ($E_6 \rightarrow E_7 \rightarrow E_8$, then stops at the affine boundary $\hat{E}_8$) — <i>not</i> ‘P2 derived’. [E] icosahedron = 3-uniform hypergraph ($ \text{Aut} = 120 = I_h $; the McKay group is the $SU(2)$ binary lift $2I$, a different order-120 group); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; [E] Smith: $\rho \leq 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \leq 2 \Leftrightarrow$ a positive witness m with $Am \leq 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \leq 2m$) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ under the $(2, 3, \cdot)$ seed; [E] A_n ($\rho = 2 \cos \frac{\pi}{n+1}$) and D_n ($\rho = 2 \cos \frac{\pi}{2n-2}$) controls are < 2 for all n , never reaching E_8 ; [O] the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).
v300_flataway_rigid.py	FLATAWAY.RIGID.01 — the attack on Flat-Away. [E] the flat fingerprint: $\text{spec}(D_\theta)$ is the integer ladder, every level $n \geq 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \rightarrow 1+1$) and changes the spectral multiset, so preserving the flat ladder <i>with</i> its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 ‘deviation grows’ scan; [E] with the v295 convexity, flat is isolated analytically <i>and</i> discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer q -coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries <i>zero</i> new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact.
v301_route_a_invertible.py	SEAM.EQUIV.A.INV.01 — discharge Route A’s last open hypothesis (‘the raw quasi-free seam bulk is invertible/SRE’) via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of SEAM.EQUIV.01. [E] carrier = 16 Majoranas, $c = 8$ ($g_{\text{car}} + N_{\text{fam}} = 5 + 3 = 8 = 16/2$, v148); [E] $SO(16)_1 \supset (E_8)_1$: $\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248 = $\dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$ -classified by the chiral Majorana number and is <i>always</i> invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $ \det K = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A’s ‘invertible bulk’ now follows from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input ‘is the quasi-free bulk gapped?’; [O] with v300 the whole open content of SEAM.EQUIV.01 reduces to the single spectral statement ‘the quasi-free seam bulk is gapped’.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v302_seam_gap.py	SEAM.EQUIV.GAP.01 — the last lever: after v300/v301 the whole open content of SEAM.EQUIV.01 was ‘the quasi-free seam bulk is gapped’; that gap is <i>not</i> a free input but the established Recovery gap of the seam transfer operator. [E] $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1 - (2/3)^6 = 665/729$; $6 = 2N_{\text{fam}}$, $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); [E] the positivity margin $\Delta - 31/(4\pi^2) = 1.6476 > 0$ (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap $6 \ln(3/2) > 0$, and SEAM.EQUIV.01’s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end).
v303_ftransfer_dynamics.py	FR.DYNAMICS.01 — the discrete→dynamic lens on F_transfer: the main-branch mechanism (a gapped, positivity-preserving relaxation to a <i>unique</i> attractor; the local-averaging update → the E_8 marks, gap $6 \ln(3/2)$) is the shared <i>shape</i> of all four instances. [E] F_pole (Koide) <i>is</i> the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue (v82/v54); $F' < 1 \Rightarrow$ a unique attractor at the seam rate; [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0, 1)$, H-theorem; compiler fixes $A_\Lambda = 10$, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed $\theta_i = 3\pi/5$, rate cosmological); [E] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom $\Rightarrow$ the Gaussian UV attractor, Λ_{QCD} generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so F_transfer is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).
v304_idg_nonlocal_ghost.py	QGAMB.IDG.01 — the KMS spectral-action cutoff suggests a <i>nonlocal</i> (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a <i>truncation artefact</i> — a [C] narrowing of the red-team rt_F perturbative-gravity attack, <i>not</i> a closure of QG.AMB.01 (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator $1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]$ has a healthy massless graviton (residue $+1/M^2$ at $p^2=0$) and a massive spin-2 mode with <i>negative</i> residue $-1/M^2$ at $p^2 = -M^2$ (the Stelle ghost, reproducing v278); [E] for an IDG operator $p^2 a(p^2)$ with $a(z) = e^{z/M^2}$ entire and nowhere zero, the dressed propagator $1/(p^2 a)$ keeps its <i>only</i> pole at $p^2=0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation $a(z) = 1 + z/M^2$ (the $R+R^2$ order) has a real zero at $z = -M^2$ = the ghost pole, so ‘entire vs polynomial’ is the discriminator; [C] the cutoff is the $\beta=1$ seam KMS weight $f(u) = e^{-u}$ (v259), not a choice — truncating $\text{Tr} e^{-D^2/\Lambda^2}$ at a_4 is what makes the local $R+R^2$ and the ghost; [C] conditional narrowing: <i>if</i> the resummed form factor is entire (Biswas–Mazumdar–Siegel class) <i>then</i> the perturbative graviton is ghost-free and rt_F’s attack is a truncation artefact; [O] open: (i) the trace regulator f is not the kinetic form factor $a(\square)$ — entire analyticity of the resummation is unproven; (ii) a ghost-free <i>perturbative</i> graviton is not the nonperturbative ambient measure — QGAMB.IDG.01 stays [C]/[O], not a QG.AMB.01 closure.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v305_witness_independence.py	Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ($e=(4,5,2)$, $p_n=2+2^n$; v23/v228), π the only transcendental primitive (ARCH.CORE.01) — 13 atoms / 2 free primitives = $6.5\times$ compression made explicit (the recurring $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ and $(2/3)^6$ are <i>one</i> ratio); E_8/g_{car} /anchor are overdetermined (≥ 3 structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); α^{-1} 's '137' is foreign to the anchor family = an independent witness [E]
v306_seed_crossval.py	Leave-one-out cross-validation of the shared seed φ_0 : back-solving φ_0 independently from the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β , Ω_b , λ_C) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, $\sim 2\sigma$); a data $\leftrightarrow$ formula shuffle explodes the χ^2 by $> 100\times$ [E]
v307_data_watchdog.py	Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live n_σ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core $\sin^2\theta_{13}$ (2.0σ); the prediction-of-record $\sin^2\theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) [E]
v308_seam_equiv_chain.py	TOE attack 1 — the SEAM.EQUIV.01 keystone as a composition certificate: the v297 stack as one well-typed chain $\text{gap}>0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$ with exact discriminators $ \det \text{Cartan}(E_8) =1$ vs $ \det \text{Cartan}(D_8) =4$, carrier tower $4\cdot 4=16 \rightarrow 4 \rightarrow 1$, $c=g_{\text{car}}+N_{\text{fam}}=8$; the input gap is the derived recovery gap $6\ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure [E][O]
v309_modular_bedrock.py	TOE attack 2 — the bedrock $\omega\circ\rho=\omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) $\Rightarrow \mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega\circ\rho=\omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of QGEO.SYM.01 [E][O]
v310_carrier_sm_anomaly.py	TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5 \rightarrow SU(5) \subset SO(10)$ as $10+5+1 = \text{one generation}$; gauge-anomaly-free ($\sum Y=0$, $\sum Y^3=0$, $[SU(2)]^2U(1)=[SU(3)]^2U(1)=0$, exact) and reproduces $(b_1, b_2, b_3)=(\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1 , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ β sub-question; the full EG S -matrix + Yukawa stay open [E][C]
v311_gap_decoupled_measure.py	TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$, correlation length $1/(6\ln(3/2))$, finite susceptibility $729/665$; the v76 margin $6\ln(3/2) - 31/(4\pi^2)\approx 1.648>0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=\text{EH}/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); QG.AMB.01 stays [O] (gap-decoupled, inert) [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v312_hypergraph_substrate.py	TOE attack 5 — the hypergraph substrate sharpened: the affine- E_8 Kac marks are the Perron eigenvector ($A m=2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)=$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is <i>not</i> any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ (gap $6\ln(3/2)$) and the analytic seed φ_0 . The open question is precisely delimited, not closed [E][O]
v313_golden_atoms_spectral.py	The golden ratio is the $g_{\text{car}}=5$ signature: the $(2,3,5)$ atoms <i>are</i> the affine- E_8 network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, the golden minimal polynomial x^2-x-1 divides it ($\phi=2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k \in \{2,3,5\}=\{ \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}}\}$ (giving $0, 1, \phi, 1/\phi$), so the golden ratio is the $g_{\text{car}}=5$ (5-fold) signature — a new spectral witness for g_{car} (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2+1/3+1/5=31/30$ ties the v63/v76 margin numerator $31=2^{g_{\text{car}}}-1=248/8=1+h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: ϕ is structural, not a readout. [C] direction: ϕ is the E_8 -quasicrystal (Elser-Sloane) ratio, a candidate substrate [E][C]
v314_rate_translation.py	Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine- E_8 adjacency spectrum splits into the rational part $\{0, \pm 1, \pm 2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi, \pm 1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$; the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2\cos(\pi/2)=0= \mathbb{Z}_2 $, $2\cos(\pi/3)=1=N_{\text{fam}}$ build the recovery rate $(2/3)^6=(\mathbb{Z}_2 /N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has <i>no</i> rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ($30=g_{\text{car}}(2N_{\text{fam}})=5\cdot 6$); the ϕ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C]
v315_coxeter_coupling.py	Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5,6)=1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: E_8 exponents = totatives(30) = totatives(5) $\times$ totatives(6), $\phi(30)=4\cdot 2=8=$ rank E_8 ; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}]=8=$ rank E_8 , $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) (\ni \sqrt{5})$ and the family $\mathbb{Q}(\zeta_3)$; [E] GALOIS = $\mu_4 \times \mathbb{Z}_2$: $(\mathbb{Z}/30)^\times = (\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2$ (order 8, max order 4), $\mu_4=(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2=(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}=\phi+1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] $30=5\cdot 6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C]
v316_galois_readout.py	Does the Galois $\mu_4 \times \mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho=\zeta_6=\zeta_{30}^5$ is the order-6= $2N_{\text{fam}}$ FAMILY factor c^5 (the carrier is $\zeta_5=\zeta_{30}^6$); [C] $\delta_{\text{CKM,lead}}=\pi/3=\arg \zeta_6$, $\delta_{\text{PMNS}}=4\pi/3=\arg \zeta_6^4$ (power $4= \mu_4 $), $\zeta_6^4=-\zeta_6$ the sheet flip ($\mu_6=\mu_3 \times \mu_2$); [E] the Galois $\mathbb{Z}_2=\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q})=\{\zeta_6 \mapsto \zeta_6^5=\bar{\zeta}_6\}$ IS CP conjugation ($\delta \mapsto -\delta$); [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (static); [E] the magnitudes ($\sin^2 \theta_{12}=1/3-\varphi_0/2$, transcendental φ_0) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v317_galois_family.py	Are the 3 generations a μ_3 /Galois orbit? Yes (μ_3), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ ($\text{Spec}(Q_+) = \{1, 2, 3\} = A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive μ_3 orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$). [E] the Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2$, = CP conjugation v316) fixes $w=0$ (rational) and swaps $w=1/3 \leftrightarrow 2/3 \Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C]
v318_arithmetic_capstone.py	The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}] = \varphi(30) = 8 = \text{rank } E_8$, $30 = \mathbb{Z}_2 N_{\text{fam}} g_{\text{car}} = h(E_8)$, Galois $(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$; [E] rigidity: forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] magnitude reduction: $\varphi_0 = (4/3)c_3 + 48c_3^4$ is a pure function of π (no free parameters), so the magnitudes are $\{a, \pi\}$ -determined, not free; [E] free inventory: 0 dimensionless free parameters, π the unique transcendental, v_{geo} the one dimensional unit. [O] residual: the raw-seam realisation stays open (QGEO.SYM.01/SEAM.EQUIV.01) [E][O]
v319_translation_clock.py	The translation clock: the discrete $\leftrightarrow$ dynamic bridge is the order-30 Coxeter element, a clock with two coprime hands 5×6 , read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, $\text{gcd}(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ($\mathbb{Z}/6 = 2N_{\text{fam}}$, ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6$; order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$). [E] LAW VS LIVE: the resummed clock rate(n) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from $n=1$; the E_8 live phases (totatives of 30) start at 1, so 0, 1, 2, 3, 4, 5 is law-inclusive, 1, 2, 3, 4, 5 live-only. [E] STATIC HAND ($\mathbb{Z}/5 = g_{\text{car}}$, ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$, field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$, $ \mu_4 =4$ planes, v55); the “it is one clock” reading is [C].
v320_galois_cp_relation.py	A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$. [E] both phases are powers of one hexagonal unit $\rho = \zeta_6$ (v316): $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ (240°), $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} - \delta_{\text{CKM}}^{\text{lead}} = \pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: δ_{PMNS} robustly away from 240° ($> 3\sigma$; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X]
v321_killtest_board.py	The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a freeze_file.csv kill row: #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r > 0.01$), #3 DESI/Euclid $w = -1$, #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β . A forward decision ledger complementing v307; no new physics [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v322_cp_lock_sharpened.py	The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$; $\delta_{\text{CKM}}^{\text{lead}} = 60^\circ$ (node 1), $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (node 4). [E] the node selector is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = \mu_4 \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3) = 4\pi/3 = \delta_{\text{CKM}}^{\text{lead}} + \pi$ ($\rho^4 = -\rho$). [C] sub-leading budget: $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2$ ($\sim 68.7^\circ$); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7^\circ \Rightarrow$ band $240 \pm \sim 9^\circ$. [C] current tension: vs NuFIT 6.0 NO ($\delta_{CP} = 212_{-41}^{+26^\circ}$) the leading 240° sits at $+1.08\sigma$, band edge $\sim 231^\circ$ at $+0.74\sigma$. [X] kill: the nearest wrong node is 60° away $\gg$ budget; DUNE/Hyper-K ($\sim 5\text{--}15^\circ$) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic δ_{CP} hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X]
v323_bw_geometric_modular.py	The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow <i>geometric</i> , so the μ_4 deck invariance is not an independent axiom. [E] the clock $\rho = \text{diag}(i^n) = \exp(i(\pi/2)L)$ is a geometric rotation ($L = \text{diag}(n)$, $\rho^4 = I$, $\mu_4 \subset U(1)_{\text{rot}}$). [E] for a rotation-covariant $C = f(L)$ the modular Hamiltonian $K = \log((1 - C)/C) = g(L)$, so $[K, L] = 0$ (the modular flow commutes with all rotations) and $[\rho, K] = [\rho, C] = 0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is <i>not</i> geometric ($[K, L] \neq 0$); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ QGEO.SYM.01 ($\omega \circ \rho = \omega$) is downstream of SEAM.EQUIV.01 + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]
v324_hypergraph_fiber.py	The minimal hypergraph substrate that works: a $(2, 3, 5)$ -network <i>fibered</i> by a 3-fold cusp channel reproduces both the E_8 skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}} = (A + 2I)/4$ fixes the Kac marks (eigenvalue $1 = E_8$ skeleton), subleading $(\varphi + 2)/4$ (golden); $(2/3)^6$ is <i>not</i> in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}} = \text{diag}((1 - w)^{2N_{\text{fam}}})$, $w \in \{0, 1/3, 2/3\}$ ($N_{\text{fam}} = 3$, v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$. [E] the fibered substrate $T = T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ($w = 0$), skeleton $\times$ democratic cusp) <i>and</i> $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ($T_{\text{net}} \otimes I_3$) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$. [E] minimality: the smallest fiber is the 3-node cusp ($= N_{\text{fam}}$) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of $2/3$ from a pure rewrite [E][C]
v325_pillowcase_keystone.py	The seam keystone as ONE citable theorem: the raw seam state <i>is</i> the flat $\tau = i$ pillowcase state, with exactly one open lemma. IF (raw RP-collar seam state = flat $\tau = i$ pillowcase / rotation-invariant state) THEN 4 square marks + μ_4 deck + $\omega \circ \rho = \omega$ + holomorphic $c = 8 \Rightarrow (E_8)_1$, with the carrier recovery gap as input. [E] pieces all exact: $n = 2\chi(S^2) = 4$ (v216); cross-ratio $2 \Rightarrow j = 1728$ (v214/v267); flat orbifold $S^2(2, 2, 2, 2)$, $\chi_{\text{orb}} = 0$ (Trojanov); DtN sub-principal at $m \equiv 0 \pmod 4$ (v201/v280); $\det \text{Cartan}(E_8) = 1$ vs $D_8 = 4$ (v83/v143/v308); gap $6 \ln \frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau = i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (QGEO.SYM.01, SEAM.EQUIV.01; v323), machine-pinned in Lean (FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01); an assembly/reduction certificate (like v176/v261), not a closure [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v326_ftransfer_suite.py	The F_{transfer} solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] F_{pole} (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}(\eta_B)$: washout contraction to the balance 6.1×10^{-11} , monotone (H-theorem), external thermal rate. [C] F_{relic} (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$), cosmological rate. [E] $F_{\text{QCD}}(m_p/m_e)$: 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, Λ_{QCD} generated. [C] F_{RareKaon} : $\text{BR}(K^+)=9.45 \times 10^{-11}$ on NA62 2016-2024 (v202). [O] only F_{pole} has the seam rate $(2/3)^6$; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O]
v327_hypergraph_rewrite.py	Deriving the cusp weight $2/3$ from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ($w=0$) and $ \mathbb{Z}_2 =2$ surviving — supplies the cusp fiber. [E] M has spectrum $\{1, 2/3, 0\}$; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}= \mathbb{Z}_2 /N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order- $6=2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): $2/3$ is NOT a root of the affine- E_8 charpoly ($p(2/3)=-3520/19683 \neq 0$), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2, 3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]
v328_theta13_pressure.py	The θ_{13} pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO (0.02195 ± 0.00058) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] θ_{13} is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets θ_{12} — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is $\sim 5\%$ below the axiom, PMNS θ_{13} RG running is $< 1\%$ (NO), and $5/6$ is exact — the prediction stays frozen (REG.FREEZE.01). [X] pre-registered kill: a stable $\sin^2 \theta_{13}$ away from 0.023108 at $> 3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]
v329_os_gap_reduction.py	Closing the OS step: the bulk gap <i>is</i> the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy = $6 \ln \frac{3}{2}$, independent of system size — exactly “OS bulk gap = transfer gap” (the v308 open input), discharged at the many-body level. [E] $H \Rightarrow \omega \circ \rho = \omega$: under H the covariance is rotation-covariant $C=f(L)$, $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$: $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise H (Lean-pinned FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01). [O] residual: the necessity of H on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v330_qgamb_admissible_measure.py	Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer T (v221/v311; spectrum $\{1, (2/3)^6, (1/3)^6\}$) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \rightarrow \lambda_2 = (2/3)^6$ ($\xi=1/\Delta$, $\Delta=6 \ln \frac{3}{2}$). [E] tightness $\Rightarrow$ projective limit: $\chi = \sum_n C(n) = 729/665$ finite, so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu = \lim \mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{OS} = -\log T \geq 0$, unique vacuum, gap $6 \ln \frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi \rightarrow \infty$, tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v275) the physical readouts do not need it, but QG.AMB.01 stays open. A genuine PARTIAL construction, NOT a closure [E][O]
v331_necessity_of_H.py	The <i>necessity</i> of H reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ($\tau=i$), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329’s sufficiency). [E] square $\Rightarrow [\rho, \Lambda]=0$: marks at $\{0, \pi/2, \pi, 3\pi/2\}$ make the curvature $\mathbb{Z}_4$ -symmetric (Fourier support $0 \bmod 4$), M_f couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $ k +M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho, \Lambda] \neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio($1, i, -1, -i$) $=2 \Rightarrow j=1728$, the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$. [E] NEG: a $\mathbb{Z}_2$ mark set commutes with ρ^2 but not ρ . [O] residual: the dynamical <i>selection</i> of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]
v332_qgamb_metric_sector.py	The metric sector of QG.AMB.01 characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330’s metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\text{conf}}(D) = -(D-1)(D-2)/4$ is negative for $D \geq 3$ ($c_{\text{conf}}(4) = -3/2$), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so e^{-S} diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76). [C] IDG conditional route: the entire form factor e^{z/M^2} (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial $(R+R^2)$ truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full QG.AMB.01 needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O]
v333_cm_selection.py	The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\text{SL}(2, \mathbb{Z})$ has $S^2 = -I$, $S^4 = I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\text{PSL}(2, \mathbb{Z})$ (the other, ρ , is order 3). [E] deck order $\rightarrow$ CM point: order-4 $\rightarrow$ cross-ratio 2 $\rightarrow j=1728 \rightarrow \tau=i$ (square); order-3 $\rightarrow j=0 \rightarrow \tau=\rho$ (hexagon); $ \mu_4 =4$ selects $\tau=i$. [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2 =$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects ρ , not the square. [O] residual: only “the deck acts <i>geometrically</i> (= QGEO.SYM.01) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v334_conformal_resolution.py	The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a CONDITIONALLY convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). [E] the obstruction: $c_{\text{conf}}(4) = -3/2 < 0$, the unrotated conformal Gaussian diverges. [E] GHP rotation: $\phi \rightarrow i\phi$ flips the sign, $\int e^{- c \phi^2} = \sqrt{\pi/ c }$ FINITE (GHP 1978). [E] IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. [E] NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). [C] conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). [O] residual: whether the GHP-rotated IDG-dressed measure is the CORRECT nonperturbative QG.AMB measure (the contour’s nonperturbative validity); gap-decoupled (v76), blocks no test, NOT a closure [E][C][O]
v335_seam_equiv_unify.py	The keystone unification: there is ONE open theorem (SEAM.EQUIV.01), QGE0.SYM.01 is its COROLLARY (a conformal-net axiom), and QG.AMB.01 is a decoupled general QG problem, NOT a TFPT gate. [E] QGE0.SYM.01 = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a NET AXIOM); rotations $U(1) < \text{Moebius} \Rightarrow$ rotation-invariant vacuum; the μ_4 clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$. So SEAM.EQUIV.01 $\Rightarrow$ QGE0.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom); the two bedrock items collapse to ONE . [E] the one keystone theorem: SEAM.EQUIV.01 = a construction theorem ($c=8$, $\det \text{Cartan}(E_8)=1$, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma. [E] QG.AMB.01 decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. [E] collapsed status: structural items $2 \rightarrow 1$; live residual $= v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$. [O] the one residual = SEAM.EQUIV.01 ’s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does NOT close SEAM.EQUIV.01 [E][O]
v336_continuum_limit.py	The ONE open lemma of SEAM.EQUIV.01 , stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now CITABLE . [C] continuum leg: Morinelli–Morsella–Stottmeister–Tanimoto (<i>CFT from Lattice Fermions</i> , doi:10.1007/s00220-022-04521-8) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct c); WZW range $\text{rank} \leq c \leq D$. $(E_8)_1$: $c=8$, rank 8, $D=16$ Majoranas ($SO(16)_1 \rightarrow (E_8)_1$), $8 \leq c \leq 16$ in range; the collar is a gapped quasi-free (CAR) state. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. [E] target pinned: $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, the $D=16$ Majorana $SO(16)_1$ IS the carrier-16 and the μ_4 extension IS the seam glue. [O] open residual: $c=8$ alone does NOT pin the net ($SO(16)_1$ also $c=8$, $\det 4$), so the one open lemma = the holomorphy discriminator $\det K=1$. A literature-mapping/reduction; does NOT close SEAM.EQUIV.01 [C][E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v337_decoupling_theorem.py	THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the QG.AMB.01 reclassification (v335) honest. [E] gap margin: $\Delta - 2 \dim(E_8) c_3^2 = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$. [E] factorization: every readout (mass ratios, mixings, α^{-1}, $R+R^2$) is a function of the admissible gapped transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ + the discrete kernel alone. [E] the theorem: margin $> 0 \Rightarrow$ exponential clustering ($\lambda_2=(2/3)^6$), finite susceptibility $\chi=729/665$, so the ambient contribution is gap-suppressed $\Rightarrow$ no readout depends on the ambient measure $\Rightarrow$ QG.AMB.01 decoupled. [E] neg control: gap closed $\Rightarrow \chi$ diverges $\Rightarrow$ fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]
v338_theta13_budget.py	THETA13.BUDGET.01: the θ_{13} sub-leading budget vs the current global fit – the $+2\sigma$ pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: $\sin^2 \theta_{13} = (\varphi_0^{\text{ret}}) e^{-5/6} = 0.023108$ is $+2.0\sigma$ vs the no-SK central (0.02195) and $+1.7\sigma$ vs the with-SK NO best fit (0.02215) – not a clean 2σ. [E] within 3σ: $0.023108 <$ the with-SK 3σ upper 0.02388 (not falsified). [C] budget: moving to the central needs -4.1%; RG $< 1\%$ and the exponent $5/6$ is exact $\Rightarrow$ no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns $1.7\text{--}2.0\sigma$ into $\sim 6\sigma$ – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]
v339_firstprinciples_boundary.py	The honest first-principles boundary, mapped precisely – answers (a) can $M_{\text{Planck}}/v_{\text{geo}}$ be computed? and (b) can F_{transfer} be made first-principles? Both NO, with the exact reason per residual. [E] $v_{\text{geo}}/M_{\text{Planck}}$ FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] m_p/m_e structure in place (v262): the carrier $b_3=-7$ drives the QCD run + the closed electron source; two inputs stay external: $\alpha_s(M_Z)$ and the lattice $C_p=m_p/\Lambda^{(3)} \sim 2.83$. [X] $\alpha_s(M_Z)$ external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread ~ 8.8 at 2×10^{16} GeV, v246). [C] C_p parameter-free but lattice-only (v35). [C] η_B/axion = cosmological initial conditions; Koide $Q=2/3$ structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

TFPT verification suite: the journey of ~300 machine-checked scripts

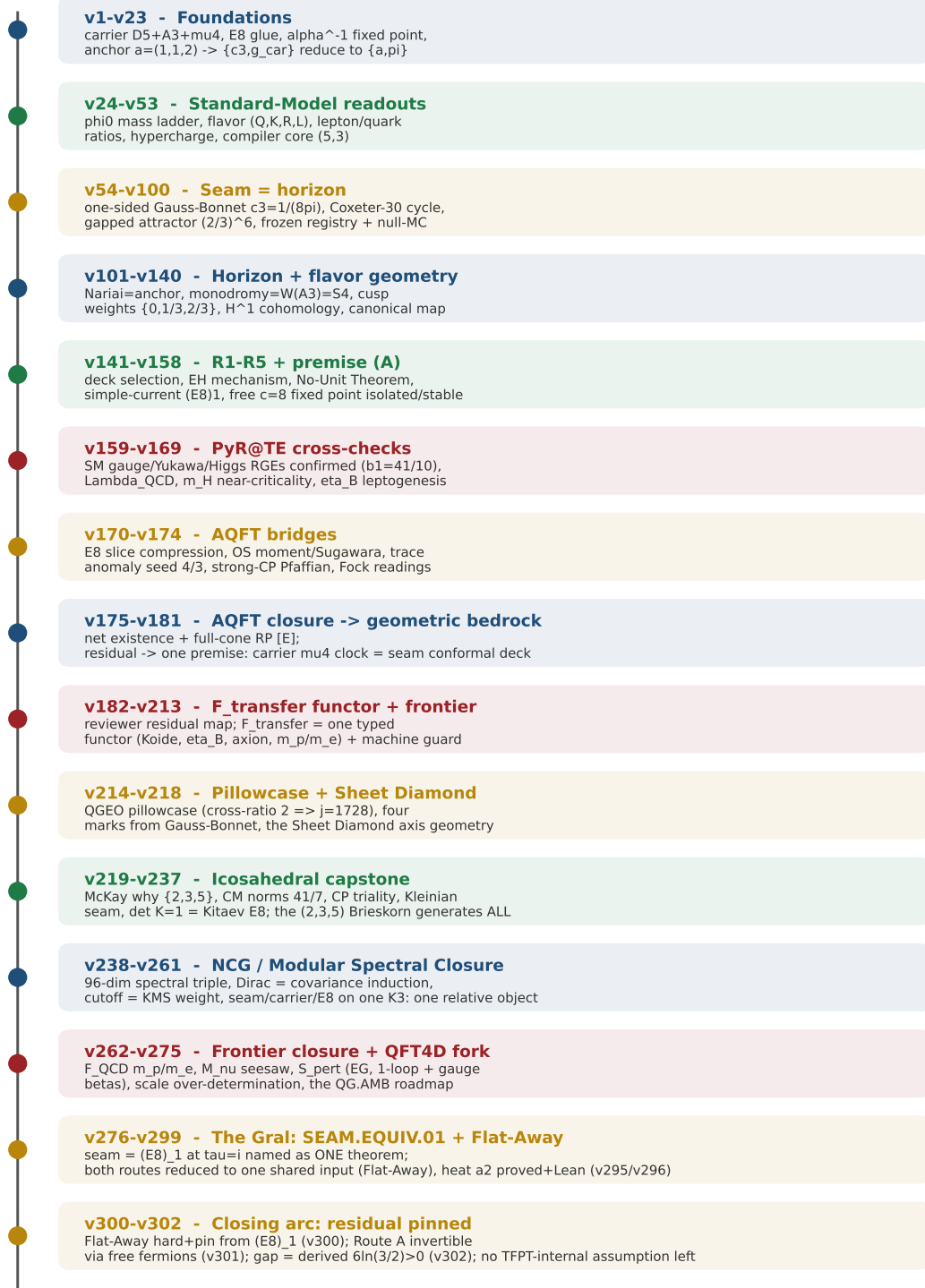


Figure 1: The development timeline of the verification suite — the eleven phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1} , the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{\text{car}}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flux geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the F_{transfer} functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, to the *icosahedral capstone* — where the single $(2, 3, 5)$ Brieskorn singularity is shown to generate the whole skeleton and the closing step becomes the physical question “is the seam short-range-entangled ($\det K = 1$)?”. The master script $\rightarrow$ check map follows.

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers vN refer to `verification/vN_*.py`.

2026-06-22 (the first-principles boundary, mapped: can M_{Planck} be computed? can F_{transfer} be first-principles? both NO, with the exact reason: v339)

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01)**. An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* — $v_{\text{geo}}/M_{\text{Planck}}$ is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E] *F_{transfer} cannot be made first-principles* either: for m_p/m_e the structure is in place (v262, carrier $b_3 = -7$ + closed electron) but two inputs stay external. [X] $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ~ 8.8 at 2×10^{16} GeV, v246). The lattice $C_p \sim 2.83$ is parameter-free but lattice-only (v35); η_B /axion ride on cosmological initial conditions; Koide $Q=2/3$ is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: v336, v337, v338, FORM.CARTAN.DET.01)

- **v336 (SEAM.EQUIV.CONTINUUM.01)** — **the one open lemma, literature-anchored**. SEAM.EQUIV.01's only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Morinelli–Morsella–Stottmeister–Tanimoto (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8; CMP 387 (2021) 299; PRL 127 (2021) 230601) give, by operator-algebraic (wavelet) renormalization, the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct c); WZW range $\text{rank} \leq c \leq D$, and $(E_8)_1$ ($c=8$, rank 8, $D=16$ Majoranas, $SO(16)_1 \rightarrow (E_8)_1$) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$. [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic $(E_8)_1$ net” ($\det K=1$), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01)** — **the Decoupling Theorem**. [E] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1 - (2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure $\Rightarrow$ QG.AMB.01 decoupled; negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01)** — **the θ_{13} budget, fenced**. The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [E] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01)**. `CartanDeterminants.lean` extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same $c=8$) but $4 = \det \text{Cartan}(D_8)$ is genuinely not a unit while

$\det \text{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. **[E] QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01** (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4=1$) fixes it $\Rightarrow \omega \circ \rho = \omega$. So $\text{SEAM.EQUIV.01} \Rightarrow \text{QGEO.SYM.01}$ with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. **[E] the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ($c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). **[E] QG.AMB.01 decoupled, not a gate:** margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) $\Rightarrow$ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. **[E] collapsed status:** open structural items $2 \rightarrow 1$; live residual = $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$. **[O]** the one residual = SEAM.EQUIV.01 ’s continuum OS reconstruction.
- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance` ($=\text{QGEO.SYM.01}$) follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, `FAQ`, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)

- **v333_cm_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual: $\tau=i$ (the square) is the *unique* order-4 modulus — $S=\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $S^4=I$ and fixes i , and an order-4 deck selects $j=1728$ (vs the order-3 hexagon $j=0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau=i$ is forced; the residual sharpens from “why the square” to “the deck acts geometrically” ($=\text{QGEO.SYM.01}$). NOT a closure.
- **v334_conformal_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \rightarrow i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R+R^2$ ghost as control). Together they give a

conditional mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.

- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|=1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel **decide** / the integer-inverse argument. The remaining content (Witt’s uniqueness: E_8 is the only even unimodular lattice in $\dim < 16$) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|=1$ and the μ_4 commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)

- **v331_necessity_of_H.py** (QGEO.NECESS.01, [E]/[O]). The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam $\Lambda=|k|+M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau=i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho, \Lambda] \neq 0$. The square has cross-ratio 2 ($j=1728$), the unique 4-config with an order-4 automorphism (v214/v267), so $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$ and “necessity of H ” sharpens to: the raw dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.
- **v332_qgamb_metric_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D-1)(D-2)/4$ ($=-\frac{3}{2}$ at $D=4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial $R+R^2$ ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator $|\det \text{Cartan}(E_8)|=1$ (E_8 has one anyon, vs D_8 ’s four) is now proved principally and kernel-only: **cartanE8 * cartanE8inv = 1** by **decide** on the 8×8 PRODUCT gives **IsUnit (det E8)** via $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$ — no 8! determinant, no **native_decide** (Mathlib itself leaves $E_8_det = 1$ a **proof_wanted**).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4=1$, $i^2=-1$, and (on concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel **decide**. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)

- **v329_os_gap_reduction.py** (QGEO.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $dT(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6 \ln \frac{3}{2}$, *independent of system size*. Together with $H \Rightarrow \omega \circ \rho = \omega$ and $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.

- **v330_qgamb_admissible_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275's roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n-1) \rightarrow (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\text{OS}} \geq 0$ with gap $6 \ln \frac{3}{2}$. The full QG_{amb} (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det \text{Cartan}(A_3)|=4=|\mu_4|$ (3×3 , `det_fin_three`) and $|\det \text{Cartan}(D_5)|=4$ (5×5 , `decide`), giving the carrier $|\det \text{Gram}|=4 \cdot 4=16$ (the v281 anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib's own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum` $\rightarrow$ `GeometricModularFlow` $\rightarrow$ `Mu4ModularSymmetry` $\rightarrow$ `StateInvariance` ($=\text{QGEO.SYM.01}$) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325_pillowcase_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau=i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho = \omega$, holomorphic $c=8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\text{orb}}=0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326_fttransfer_suite.py** (FR.SUITE.01, [E]/[C]). The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B , H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e , RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327_hypergraph_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight $2/3$ is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}=|\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6=64/729$ is the recovery gap; with a PROOF that $2/3$ is not a root of the affine- E_8 charpoly ($p(2/3)=-3520/19683 \neq 0$, sharpening v312). v324's injected datum is reduced to the anchor arity $\{2, 3\}$.
- **v328_theta13_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified: $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}=0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent $5/6$ exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.

- **Wolfram mirror** 285 → 288. v325 ($\chi_{\text{orb}}=0$) and v327 (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6=64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to SEAM.EQUIV.01.** Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates (U_{wall}) and G_{metric} ” framing was replaced by the current $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/212→v324/322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)

- **Bug fix — δ_{CKM} display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD = 1.198232` rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to 68.65° . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202_rare_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an F_{transfer} bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new F_{transfer} bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.

- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ($\pi/3$) component of the quark phase”: the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$, so $\delta_{\text{PMNS}} = 240^\circ$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305_witness_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (v314/v315).
- **v203_eht_achromatic.py — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \pmod{4}$ (the μ_4 fingerprint of the four marks, v201); a μ_4 -orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.
- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed φ_0 , the error-weighted mean matching the axiom to 0.04%, with $\sin^2 \theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, $275 \rightarrow 285$, *all passing*): v313 (the affine- E_8 characteristic polynomial with the golden factor $x^2 - x - 1$; the $(2, 3, 5)$ atoms as $2 \cos(\pi/k)$; the 31/30 selection), v314 (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), v315 ($30=5 \cdot 6$, $\varphi(30)=8$, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), v316 ($\rho=\zeta_6=\zeta_{30}^5$, $\rho^4=-\rho$, $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$), v317 ($N_{\text{fam}}=3$ as the μ_3 orbit, the $1+2$ Galois split), v318 ($\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$ a pure function of π) and v320 (the CP lock $\delta_{\text{PMNS}}=240^\circ$). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322_cp_lock_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$, and *which* one is selected is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$ (node 4); [C] the sub-leading correction is bounded by the quark analogue $3\lambda^2 \approx 8.7^\circ$, so the prediction is the band $240^\circ \pm \sim 9^\circ$; [C] against NuFIT 6.0 (NO, $\delta_{\text{CP}} = 212_{-41}^{+26^\circ}$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; [X] the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim 5\text{--}15^\circ$) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323_bw_geometric_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the μ_4 clock is a geometric rotation $\rho = \text{diag}(i^n) = \exp(i\frac{\pi}{2}L)$, $\mu_4 \subset U(1)_{\text{rot}}$; [E] for a rotation-covariant covariance $C=f(L)$ the modular Hamiltonian $K = \log((1-C)/C) = g(L)$, so $[K, L] = 0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K, L] \neq 0$); [C] given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega \circ \rho = \omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.

- **v324_hypergraph_fiber.py** (HYP.FIBER.01, [E]/[C]). The constructive v312 follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. [E] the carrier network $T_{\text{net}} = (A+2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\text{cusp}} = \text{diag}((1-w)^{2N_{\text{fam}}})$, $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (spectrum $\{1, (2/3)^6, (1/3)^6\}$); the fibred $T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes (w=0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes (w=\frac{1}{3})$) – the $N_{\text{fam}}=3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \times 6$ of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of $2/3$ from a pure rewrite. Cited in `origin_theory`; ledger HYP.FIBER.01.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321_killtest_board.py** (KILL.BOARD.01, [E]). The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r > 0.01$), #3 DESI/Euclid w (kill $w \neq -1$), #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$ assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Muger / Kawahigashi–Longo–Muger holomorphy; Conway–Sloane E_8 uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a `decide`-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the $|\det \text{Cartan}|$ that separates the holomorphic E_8 from the same- c rival $D_8 = SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$: v320)

- **v320_galois_cp_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ (the $\mathbb{Z}_2$ sheet flip); [C] this upgrades the previously assigned $\delta_{\text{PMNS}} = 240^\circ$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a δ_{PMNS} robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6 , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319_translation_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, v315). [E] the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\text{fam}}$ (ticks $0, \dots, 5$): the recovery exponent is exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, v316). [E] position 0 is the *conserved law* — the resummed clock rate(n) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from $n=1$, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \dots, 29\}$), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is $\mathbb{Z}/5 = g_{\text{car}}$ (ticks $1, \dots, 5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2 \cos(\pi/5)$), field-obstructed (no rational dynamic image, v314) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (v55). [C] verdict: the discrete $\rightarrow$ dynamic translation *is* the order-30 clock = (static 5-ring) $\times$ (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\text{fam}}$), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin_theory](#); ledger row TRANSLATE.CLOCK.01.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$ + Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ — zero dimensionless free parameters: v318)

- **v318_arithmetic_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 = \text{rank } E_8$, $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$), forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] the magnitude seed $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (ARCH.CORE.01), v_{geo} the unique dimensionful input (No-Unit Theorem) — complete input $\{a, \pi, v_{\text{geo}}\}$. [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGE0.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin_theory](#); ledger row ARITH.CAPSTONE.01.

2026-06-21 (are the 3 generations a μ_3 /Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **v317_galois_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\text{Spec}(Q_+) = \{1, 2, 3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w + \frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$); [E] the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the v316 CP conjugation) refines the orbit to 1+2 — fixing $w=0$ (rational) and swapping $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited

in `origin_theory`; ledger row `GALOIS.FAMILY.01`.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2 = \text{CP conjugation}$: v316)

- `v316_galois_readout.py` (`GALOIS.READOUT.01`, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-6 = $2N_{\text{fam}}$ factor c^5 of v315 (the carrier being $\zeta_5 = \zeta_{30}^6$); [C] the leading CP phases are its cyclotomic arguments $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$ and $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$ (power 4 = $|\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits v231/v233); [E] the family Galois $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) is CP conjugation $\delta \mapsto -\delta$; [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in `origin_theory`; ledger row `GALOIS.READOUT.01`.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: v315)

- `v315_coxeter_coupling.py` (`COX.COUPLE.01`, [E]/[C]). Answers the v314 question — does $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family (6 = $2 \cdot 3$, rational) sectors, or just factorize? [E] the cyclic group *factorizes*: $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the v314 field obstruction); [E] the E_8 exponents = totatives(30) split as totatives(5) $\times$ totatives(6), $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$; [E] but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree 8 = rank E_8 ; [E] the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining v223); [E] $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row `COX.COUPLE.01`.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: v314)

- `v314_rate_translation.py` (`RATE.TRANSLATE.01`, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- E_8 network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate $(2/3)^6$) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$ plus the golden part $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles $2 \cos(\pi/2) = 0 = |\mathbb{Z}_2|$ and $2 \cos(\pi/3) = 1 = N_{\text{fam}}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$, same field, same atoms), but [E] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction ($2/3$ is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2 - \phi) \approx 6.37$ not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, and the ϕ -quasicrystal $E_8 = H_4 + \phi H_4$ ($240 = 2 \cdot 120$) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the (2, 3, 5) atoms are the affine- E_8 network spectral angles, $\phi = 2 \cos(\pi/5)$ the $g_{\text{car}}=5$ signature: v313)

- `v313_golden_atoms_spectral.py` (`GOLD.ATOMS.01`, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- E_8 adjacency characteristic polynomial factors as

$x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, so the golden minimal polynomial x^2-x-1 divides it and $\phi = 2\cos(\pi/5) = (1+\sqrt{5})/2$ is an *exact* eigenvalue; [E] the non-trivial eigenvalues are $2\cos(\pi/k)$ for exactly $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ ($2\cos(\pi/2)=0$, $2\cos(\pi/3)=1$, $2\cos(\pi/5)=\phi$, $2\cos(2\pi/5)=1/\phi$), so the golden ratio is *specifically* the $g_{\text{car}}=5$ (5-fold) signature and $g_{\text{car}}=5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the $(2, 3, 5)$ spherical excess $1/2+1/3+1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the v63/v76 gap-margin numerator $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [C] direction: ϕ is the E_8 -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in `origin_theory`; ledger row GOLD.ATOMS.01.

2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **v308_seam_equiv_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic $c=8$ (E_8)₁ net” assembled into ONE well-typed logical chain $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8$ (Müger/KLM) $\rightarrow (E_8)_1$ (Conway–Sloane), with the exact discriminators $|\det \text{Cartan}(E_8)|=1$ vs $|\det \text{Cartan}(D_8=SO(16))|=4$, the carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ and $c=g_{\text{car}}+N_{\text{fam}}=8$ all machine-checked; the input gap is the derived recovery gap $6 \ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309_modular_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310_carrier_sm_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$ is exactly one anomaly-free Standard-Model generation ($SU(5) \subset SO(10)$: $10+5+1$), with $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1 , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ β sub-question; the interacting Epstein–Glaser S -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311_gap_decoupled_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6 \ln(3/2))$, finite susceptibility $729/665$), and the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312_hypergraph_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- E_8 Kac marks are the Perron eigenvector ($Am=2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5) =$ the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$

and the analytic seed φ_0 . The open question is precisely delimited, not closed. Cited in introduction; ledger row HYP.INJECT.01.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305_witness_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ($|\mu_4|=4$, $g_{\text{car}}=5$, $|\mathbb{Z}_2|=2$, $N_{\text{fam}}=3$, $\dim S^+=16$, $\text{rank } E_8=8$, 240, 248, $|2I|=120$, $h(E_8)=30$, $|W(D_5)|=1920$, $\Omega_{\text{adm}}=48$, the gap exponent 6) is a documented image of the single anchor $a=(1,1,2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, $6.5\times$ compression made explicit), so the recurring $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by v100, never multiplied); the negative control shows α^{-1} ’s “137” is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, v3). Cited in the `tfpt_5_redteam` body; ledger row WITNESS.ECON.01. Python-only.
- **v306_seed_crossval.py** (SEED.CROSSVAL.01, [E]). A leave-one-out cross-validation turning the “many observables, one seed φ_0 ” worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$, β , Ω_b , λ_C ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2 \theta_{13}$, $\sim 2\sigma$); a data $\leftrightarrow$ formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row SEED.CROSSVAL.01. Python-only.
- **v307_data_watchdog.py** (DATA.WATCHDOG.01, [E]). The operational decision pipeline over the frozen registry (v84): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2 \theta_{13}$ (2.0σ , the honest current pressure point); $\sin^2 \theta_{12} = 0.306747$ is compatible (0.02σ , the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and $w=-1$ is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row DATA.WATCHDOG.01. Python-only.
- **Target-A holomorphy kill test (freeze_file.csv)**. A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same- c rival $SO(16)_1$) would break holomorphy and kill “seam net = $(E_8)_1$ ” — since holomorphic $c=8 \Leftrightarrow E_8$ is machine-proven (v83/v143) and $\det K=1 \Leftrightarrow$ invertible/SRE (v237/v301/v302).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of c ” framing of SEAM.EQUIV.01: v304)

- **v304_idg_nonlocal_ghost.py** (QGAMB.IDG.01, [C]/[O]). A conditional, parameter-free narrowing of the red-team `rt_F/v278` Stelle-ghost attack on the perturbative gravity sector. [E] the local $R+R^2/\text{Weyl}^2$ four-derivative propagator $1/(p^2(p^2+M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2 a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close QG.AMB.01: the trace regulator is not yet shown to resum to an entire $a(\square)$,

and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row QGAMB.IDG.01.

- **Double-pushout figure (tfpt_1_architecture_e8).** A new shared TikZ macro (TFPTdoublepushout in tex-artefacts/tfpt_figures.tex) draws the commuting square showing that the order- $|\mu_4|=4$ glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (v1) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (introduction, tfpt_research_contracts).** SEAM.EQUIV.01 is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. v267). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on F_transfer: one shape, one seam-rate, v303)

- **v303_ftransfer_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $\rightarrow$ the E_8 marks, gap $6 \ln(3/2)$) — is also the shape of the four **F_transfer** instances, or an unrelated bolt-on. It is the *same* shape: [E] **F_pole** (Koide) *is* the seam dynamics (Möbius multiplier $F'=(2/3)^6=\lambda_2$, the seam-clock eigenvalue $\Rightarrow$ a unique attractor at the seam rate, v82/v54); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0, 1)$, H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point; [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; **F_pole** has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite v303 5/5, AUDIT OK.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$, v302)

- **v302_seam_gap.py** closes the chain on the single statement v301 had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1 - (2/3)^6 = 665/729$; $(2/3)^6$ the μ_4 -deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, SEAM.EQUIV.01's entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6 \ln(3/2)$) — *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite v302 5/5, AUDIT OK.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301_route_a_invertible.py** attacks the single remaining *unconditional* gap of SEAM.EQUIV.01. Since v300 collapsed Route B into Route A's rationality, that gap

is Route A’s open hypothesis (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev’s periodic table, class D). [E] the carrier is 16 Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$ (v148); [E] $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant $|\det K| = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce. [O] LIT-A’s ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of SEAM.EQUIV.01 reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. [E] suite v301 6/6, AUDIT OK.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, v300)

- **v300_flataway_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum (“the spectral mismatch grows with ε ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint — $\text{spec}(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy ($2 \rightarrow 1+1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* [C] the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q -coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches E_8 as the last finite point, v299)

- **v299_hypergraph_0d_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ($|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8 ; 30 edges $= h(E_8)$); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; [E] Smith ($\rho \leq 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \leq 2 \Leftrightarrow$ a local witness $Am \leq 2m$); [E]* the autonomous rule (diffusion-generated witness + *local* gate $Am \leq 2m$, Perron only as audit) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$; [E] the negative controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8 . Honest type: [O] the $(2, 3, \cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* [E] build green, AUDIT OK.

2026-06-18 (E_8 as a network attractor — the Wolfram-program reading, v298)

- **v298_e8_network_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$

are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4m$ on the $(2,3,5)$ McKay graph (converges from any positive start; $Am = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ E_8 is the universal attractor of a minimal $(2,3,5)$ network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **v296_flataway_a2_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$ -term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small- t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.
- **v297_route_a_literature_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) $\rightarrow$ LIT-B (Müger; Kawahigashi–Longo–Müger) $\rightarrow$ LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295_flataway_a2_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\text{Tr } e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon=0$ is the *global* minimum — so $\Delta \text{Tr} \geq 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal $\Delta=0 \Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [E] build green, AUDIT OK.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292_flataway_heat_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact “the seam fixes the a_2 heat coefficient”.
- **v293_flataway_spectral_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294_flataway_rp_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles π , Gauss–Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique

minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290_z4_smooth_curvature_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon \cos(4\theta)$ [E] passes the v288 commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [E] shifts the DtN spectrum ($\max |\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]=0$ — which is exactly why Flat-Away cannot be read off the commutator.
- **v291_flataway_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon=0$) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289_marklocal_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks $\Rightarrow$ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]=0$ (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- L^2 QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk $\Rightarrow$ single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Mger/Kawahigashi–Longo–Mger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286_seam_equivalence_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ E_8 into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287_free_rp_bulk_to_holomorphic_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible $\Rightarrow \mu=1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary* (the AQFT form of $\det K=1$). 4/5 discharged.

- **v288_full_l2_subprincipal_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod{4}$, so $\Lambda = |n| + M_f$ commutes with $\rho = \text{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$; generic curvature $\rightarrow 4.44$), lifting v201/v284 to L^2 . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284_route_i_rp_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 $\text{RP} \rightarrow \text{DtN}$ (v194), L2 marks $\rightarrow$ pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov ($\chi_{\text{orb}}=0 \Rightarrow$ unique flat $\tau=i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.
- **v285_route_ii_seam_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector $\Leftrightarrow \det K=1 \Leftrightarrow$ holomorphic $\Leftrightarrow E_8$; the tower $16 \rightarrow 4 \rightarrow 1$), and **its open lemma coincides with Route (i)'s** — both are “the raw seam is the $(E_8)_1$ -at- $\tau=i$ object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. Build green, AUDIT OK.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E_8 -at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280_pillowcase_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat $\tau=i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281_holomorphy_modular_data.py** (QGAMB.MODULAR.01): #anyons = $|\det \text{Gram}|$ ($E_8 \rightarrow 1$, $\text{SO}(16) \rightarrow 4$), the anyon-condensation tower $16 \rightarrow 4 \rightarrow 1$, Gauss–Milgram $\rightarrow c=8$; holomorphy ($\det K=1$) is the single Tier-B discriminator.
- **v282_e8_tau_i_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \rightarrow 1$.
- **v283_live_killtest_scorecard.py** (PRED.LIVE.01): the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279_qgeo_obligation_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$ mark-locality $\rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so

the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Troyanov + OS).

- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278_lsz_bridge_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG T -products of S_{pert} (v269) are the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta = 6 \log(3/2) > 0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$ exactly = the two-body phase space, so $2 \text{Im } M = \sum \int d\Pi |M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector's Stelle ghost (v269/rt_F) is non-unitary $\rightarrow$ QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277_seam_calderon_e8_match.py** (QGAMB.TIERB.01) reduces Tier-B to ONE bit. [E] the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248 = \dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det \text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. [C] the seam side matches via v276 (flat $\tau=i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276_qgeo_flat_closes_commutator.py** (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate. The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]=0$. [E] if $H = f(\Delta_{\text{flat}})$ and the order-4 deck $\rho (z \mapsto iz)$ is a flat- $\tau=i$ isometry, then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ exactly, to all orders (verified on the discretised square torus: $\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H . [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat $\tau=i$ pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **redteam/rt_F_qft4d.py** (REDTEAM.F.01) stress-tests the just-published round (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity

R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the nonperturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the Λ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in tfpt_5_redteam + the red-team table; run_redteam.py green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the concrete one-loop quartic v271 and gauge β -coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν -mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, run_all.py green, AUDIT OK, npm build clean. The Zenodo description (zenodo_description.html) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272_nu_mass_scale.py (FLAV.NUSCALE.01) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo} , v153); [C] the observed $m_3=0.050$ eV with the TFPT PS→GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; [X] the NO floor $\Sigma m_\nu=0.0586$ eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **v273_eg_gauge_running.py (QFT4D.SPERT.03) derives the SM one-loop b_i via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.
- **v274_scale_overdetermination.py (SCALE.OVERDET.01) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to $\bar{M}_{\text{Pl}}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438 \times 10^{18}$ GeV, to 0.11%; the seam being a conformal $c=8$ CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- **v275_qgamb_roadmap.py (QGAMB.ROADMAP.01) gives the open gate QG.AMB.01 a module.** [E] Tier A gap decoupling ($\Delta_{\text{eff}} = 1.648 > 0$, v36/v76); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, run_all green, AUDIT OK, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **v271_eg_oneloop_quartic.py (QFT4D.SPERT.02) takes the v269 S_{pert} skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization

is the order-2 product T_2 ; the massless propagator has scaling degree $\text{sd}=2$ in $d=4$, the one-loop bubble has $\text{sd}=4=d$, so $\omega=\text{sd}-d=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d, d)=1$, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270_pmns_jarlskog_assembly.py** (FLAV.PMNS.03) **assembles the four TFPT channels** ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, $\sin^2 \theta_{23} = \frac{1}{2}$, $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$, $\delta = 4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength**. [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ ($J_{\text{max}} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative CP-violating sign reproduced. [O] δ is the μ_6 /triality phase (assembled, not from M_ν/D_F) and the absolute ν -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269_spert_paqft_skeleton.py** (QFT4D.SPERT.01) **builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: S_{top} (2d DHR braiding $|M|=1$, statistics, v243), S_{pert} (4d Epstein–Glaser/BV S-matrix of the spectral action), S_{phys} (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators $\text{dim} \leq 4$; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6 \log(3/2)$ gives an IR-safe adiabatic limit ($S_{\text{pert}} \rightarrow S_{\text{phys}}$ via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268_theta13_carrier_trace.py** (FLAV.TH13.01) **closes the θ_{13} exponent origin**. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \text{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$ -breaking $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) “ θ_{13} from M_ν ” route flagged by the new script atlas. [O] the CP magnitude and the absolute ν -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make_script_atlas.py** generates **verification/script_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (**status_ledger.csv** for $\text{claim} \rightarrow \text{status} \rightarrow \text{dependencies} \rightarrow \text{supersedes}$, **script_registry.csv/script_clusters.csv**

for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. [E] build green, AUDIT OK.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- `v267_qgeo_rigidity_minimal_axiom.py` (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks” ($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a ; cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on $y^2=x^3-x$; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]=0$ and $\omega \circ \rho = \omega$ all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal $j=0$ ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with $j=1728$ (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$) — the *reduction* is closed (axiom $\Rightarrow$ everything), the axiom is not. [E] build green, AUDIT OK.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- `v266_ps_threshold_proton.py` (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the **v265 fork**. The explicit two-step Pati–Salam thresholds give $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ($M_{GUT} \sim 2.4 \times 10^{15}$) gives $\tau_p \sim 4.4 \times 10^{33}$ yr $<$ Super-K and is *killed* even at the high hadronic band; [C] the E_8 -allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$ yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ($\sim 1.4 \times 10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be* $+(15, 1, 1)$), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- `v265_qft4d_fork_freeze.py` (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1=\alpha_2$ scale α_3^{-1} off by ~ 5.6 , spread ~ 8.8 at 2×10^{16} GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the E_8 -allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2 \theta_W = 3/8$ (v248), the $O(1)$ $\kappa = M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262_fqcd_mp_me.py (FR.MPME.02) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z)=0.1179$ with n_f thresholds (carrier $b_3=-7$, [E]) to $\alpha_s(2\text{ GeV})\sim 0.30$ and $\Lambda_{\overline{\text{MS}}}^{(3)}\sim 0.33\text{ GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e\sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263_mnu_from_df.py (FLAV.PMNS.02) makes M_ν a seesaw operator of D_F .** [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$ -symmetric inputs) $\mu\tau$ -symmetric; [C] it reproduces $\theta_{23}=45^\circ$ and $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; [O] $\theta_{13}=0$ in the leading texture, the CP *magnitude* (240°) is the μ_6 /triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **v264_raw_seam_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$ -invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in tfpt_4_frontier, tfpt_2_standard_model and tfpt_research_contracts. [E] AUDIT OK, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt_2 “ambient closure” section now states that the cutoff of its S_{rel} is the seam KMS weight ($f_2/f_0=1$, v259) and D_{rel} the covariance induction (v258); the tfpt_2 “(4) Constructive QFT closure” section adds the relative-object pointer; tfpt_4_frontier’s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and origin_theory’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG

separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)

- **v258_dirac_covariance_induction.py** (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of **v250/v252**. The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, $\beta=1$) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ .
- **v259_modular_cutoff_kappa.py** (PS.SPECACT.02) fixes the last open spectral-action input. The cutoff f is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa \approx 1.3$ pins $c_{PS}/c_{\text{grav}} \approx 1.7$).
- **v260_k3_kummer_unification.py** (ARCH.K3.01) unifies seam, carrier and lattice on one surface. The pillowcase is the elliptic-involution quotient $T_{\tau=i}^2/(z \mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1 , even, signature $(3, 19)$); [E] the seam is $T^2/(z \mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes = $\dim S^+$ (v197), $16=4 \times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 “glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.
- **v261_modular_spectral_closure.py** (QFT.MSC.01) is the assembly capstone. The whole QFT round (v238–v260) is certified as *one* relative object $\text{TFPT}_{\text{QFT}} = (A_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6 \log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)

- The canonical changelog now also drives a public /changelog website page. A new generator `verification/make_changelog_web.py` parses this file (dated \subsection*

entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers `[E]/[C]/[O]/[X]`, inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. `[E]` build (`npm run build green`, `/changelog` prerendered static) and `bash build.sh audit` (AUDIT OK).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- `v257_quasiorient_modpd.py` (PS.NCG.ORIENT.02) corrects the slightly too-generous **v256** typing. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit **v252** triple: `[E]` strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); `[E]` *quasi-orientability* holds (the left/right A -irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$); `[E]` strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- L/R -neutrino case); `[C]` modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- `v254_inner_fluctuations.py` (PS.NCG.FLUCT.01) closes **v250 #5** + the no-junk obligation. The inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator are computed explicitly: `[E]` they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1, 2)_{1/2}$ — no junk, no triplet, no $(10, 1, 3)/126$. `[E]` the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and `[E]` σ is an SM singlet $(1, 1, 0)$ with $B-L = 2$. `[C]` σ = scalaron.
- `v255_spectral_action_expansion.py` (PS.SPECACT.01) closes the spectral-action expansion (**v252 #11**). The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: `[E]` structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl² + R^2 +Higgs-quartic, $N = 96$); `[E]` $b/a^2 = 1/3$ (top-dominated); `[E]` $\sin^2 \theta_W = 3/8$; `[E]` the R^2 spin-0 mode = the Starobinsky scalaron; `[C]` Higgs quartic $\rightarrow \sim 125$ GeV with σ ; `[C]` $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ explicit; `[O]` exact decimal needs the cutoff moments.
- `v256_orientability_poincare.py` (PS.NCG.ORIENT.01) — honest status, no faked proof. KO-dimension 6 forces the intersection form $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be *antisymmetric*: `[E]` skew form (verified), `[E]` rank 2, corank 1, null $\sim (1, 1, -1)$ on the SM K-theory $\mathbb{Z}^3$; `[E]` the orientability local condition $[\gamma_F, \pi(a)] = 0$. `[C]` the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); `[O]` a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, v252–v253)

- **v252_full_finite_triple.py** (PS.DIRAC.02) advances the Dirac contract **v250** from a 7-obligation contract to a concrete object discharging 5 to [E]. The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma) — H_F$ the particle $\oplus$ antiparticle doubling of three $SO(10)$ 16-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2=+1$, $JD=DJ$), grading ($\gamma^2=1$, $[\gamma, a]=0$, $J\gamma=-\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}}=m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **v253_kappa_scalaron_ps.py** (PS.KAPPA.01) discharges residual 6(b) of **v249**. κ in $M_{PS} = \kappa M_s$ ($M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the 126-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ σ — the CCvS Pati–Salam mechanism, v251)

- **v251_first_order_sigma.py** (PS.DIRAC.01) advances the Dirac contract **v250** from a contract to an exhibited mechanism. On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2=+1$, $\gamma^2=1$, $D\gamma=-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R \nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ . Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] $\sigma = \text{scalaron}$ (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247_e8_branching_no126.py** (PS.E8BRANCH.01): the E_8 hull branches under $SO(10) \times SU(4)$ as $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\bar{16}, \bar{4})$, so the $SO(10)$ content is exactly $\{1, 10, 16, 45\}$ and the 126 is *absent* — the renormalisable 126_H is algebraically forbidden and $10+16+45$ is E_8 -supplied (only one $45 \Rightarrow$ proton-marginal). **v248_ps_algebra.py** (PS.ALGEBRA.01): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). **v249_ps_unification.py** (PS.RGTEST.01): the two-step $PS \rightarrow SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0\text{--}1.7$. **v250_dirac_triple_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ($H_F = 3 \times 16$, the seven NCG axioms, $\sigma = \text{scalaron}$) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **v246_unification_data_crosscheck.py** (QFT4D.RGTEST.01) — **the first hard data confrontation of the spectral-action prediction.** TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}–10^{17}$ GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed **[X]/[O]**: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft_skeleton.pdf**, **figures/qft_unification.pdf**) added via **make_figures.py** (PDF + website PNG) and embedded in **tfpt_research_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **v245_spectral_matter_content.py** (QFT4D.MATTER.01) **discharges the computable core of the 4d contract v244.** The carrier half-spinor (the $SO(10)$ 16) is verified, in exact rational arithmetic, to be one Standard-Model generation: **16** = $SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions + ν_R), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2 \theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3} g_1$), and the Higgs is the $(1, 2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of **CONTRACT.QFT4D.01** is now **[E]**; only the seam $\rightarrow$ Dirac realisation and the run-down couplings stay **[O]**. Wired into **run_all.py**, the ledger, the registry, cited in **tfpt_research_contracts.tex**, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** **v239_kms_thermal_time.py** (DYN.KMS.01): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. **v240_gns_os_reconstruction.py** (DYN.GNS.01): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{OS} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). **v241_dhr_sectors.py** (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \rightarrow 4 \rightarrow 1$ (v235/v237). **v242_spin_statistics_modular.py** (DHR.MODULAR.01): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. **v243_haag_ruelle_braiding.py** (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. **v244_spectral_action_contract.py** (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into **run_all.py**, the ledger and registry, cited in **tfpt_research_contracts.tex**, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238_modular_lindblad_dynamics.py** (DYN.SEMIGROUP.01, cluster registry). A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel’s generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time). [E] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real ≤ 0 spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments** (`experiments/`, all `data_limited`, not load-bearing). From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior ($w_{\text{in}}=-1$) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai $Q_{\text{geom}}=3/8$ equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}=3/8$ (shared de Sitter endpoint $1/2$); since $1/3 < 3/8 < 4/9 < 1/2$ this is a light-trapping ECO, yielding an echo template (round-trip delay ~ 0.7 ms at $62 M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\text{geom}}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES $\sim 3.3\sigma$ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- η_B — **full BDP Boltzmann ODE solve confirms the route at the frozen scale** (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10} , with *no* free M_R dial. This moves the leptogenesis route from “solve pending” to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically** (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221). The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected $\lambda=1$ (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain** (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts` search-targets). The

Stage-1 machinery (Kerr-ringdown subtraction $\rightarrow$ matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.

- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`).** The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with $c = 8$ ” ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$ glue realised at Lie level (`v143`) and its odd sectors twisted/Ramond ($E_8 = 120_{\text{NS}} + 128_{\text{R}}$, `v148`), while the $q(A_3)$ normalisation collapses into the one declared anchor (`v152`). Two exact adversarial checks added ($120+128 = 248$; glue index $= |\mu_4| = N_{\text{fam}} + 1 = 4$); status stays `[C]/[O]` (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/ (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, `[E]/[C]`).** Sharper than `v220/v225` (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j=0$ Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$. The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the `v88` misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays `[C]` (`CP.MU6.01`).
- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, `[E]/[C]`).** The du Val side of the McKay correspondence (`v219`) gives the open seam-realisation premise `QGeo.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}} + N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan ($\det 1$, even, (-2) -curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1=0$, $\pi_1=2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val

exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8 ; TOP0.E8.01, bedrock stays [O]).

- **CP is the universal family/triality phase, only sheet-split (v233_cp_triality_phase, tfpt_5_redteam, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1 \rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (v234_seam_holomorphy_selection, tfpt_research_contracts, [E]/[O]).** The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing E_8 : (i) holomorphy (μ -index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8 ; and holomorphic $c=8=g_{\text{car}}+N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8 . So P2, G_{net} , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk (v160)} \Rightarrow \text{holomorphic boundary} \Rightarrow E_8$; the single residual analytic step is “free bulk $\Rightarrow$ holomorphic boundary” (GATE.HOLO.01).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ (v235_seam_chern_simons, tfpt_research_contracts, [E]/[O]).** A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K -matrix with $\# \text{anyons} = |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|=1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ($\det 16$, 16 anyons) $\rightarrow SO(16)_1 = D_8$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$, holomorphic) — all rank 8, $c=8=g_{\text{car}}+N_{\text{fam}}$ — so holomorphic = E_8 = the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order- $|\mu_4|$ Lagrangian glue ($\det 16 \rightarrow 1$, vs. a partial order-2 isotropic $\rightarrow 4=D_8$). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue ($\det=1$)” = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton (v236_brieskorn_capstone, origin_theory, [E]).** Capstone of the icosahedral round: the singularity $x^2+y^3+z^5$, whose exponents are the atoms ($|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$, has Milnor number $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ (a *fifth* origin of the “8”), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8 , link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h \rangle = \mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (TOP0.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam is SRE (v237_seam_sre_closure, tfpt_research_contracts, [E]/[O]).** Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the K -matrix ground-state degeneracy on a genus- g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so “no topological degeneracy on any closed surface” $\Leftrightarrow \det K=1$; among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2, 3, 5\}$ (v219_icosahedral_mckay, origin_theory, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$, sum $30=h(E_8)$, sum of squares $120=|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex) (v222_cm_norm_duality, origin_theory, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j=1728$) gives $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3+2i)=13=\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j=0$) gives $N(3+2\omega)=7=\text{scalaron}$. The $(3, 2)$ split generates $(5, 6, 7, 13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (v223_coxeter_totative_clock, origin_theory, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7 \rangle = \{1, 7, 13, 19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h)=\text{rank}$ and $h=2 \cdot 3 \cdot 5$ (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227_degree_exponent_channel_split, tfpt_5_redteam, [E]).** Exponents sum $120=|R^+(E_8)|$ (magnitude channel), degrees sum $128=\text{rank} \cdot \dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228_rr_index_gate, origin_theory, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D=4=|\mu_4|$ on $\mathbb{P}^1$ forces $h^0=5=g_{\text{car}}$, $\text{rank } H_1=3=N_{\text{fam}}$, $h^0+H_1=8=\text{rank } E_8$, $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGEO.RR.01).
- **The seam as a finite recoverability code (v221_seam_qecc, origin_theory, [E]/[C]).** Code dimension $\dim S^+=16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1, (2/3)^6, (1/3)^6\}$, recovery bound $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220_cp_hexagonal_modulus, tfpt_5_redteam, [E]/[C]).** The seam deck is the square modulus $j=1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j=0$ ($\mathbb{Z}/6$, $\arg \rho = \pi/3$). The frozen $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$ has leading term $\pi/3 = \arg \rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (CP.MOD.01).
- **The dual normal frame (d, n) with CP as orientation (v225_dual_normal_frame, tfpt_5_redteam, [E]/[C]).** $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$ Nariai; $n = (5, -9, 6)$ reads $\det$ on the first-generation column; $\det(\mathbf{1}, d, n) = 21 = 3 \cdot 7$; $\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$ — CP as orientation (DUAL.FRAME.01).
- **F_{transfer} as one path on the Sheet Diamond (v224_diamond_ftransfer_path, tfpt_2_standard_model, [E]/[C]).** $M(s, t) = R + Q \text{diag}(s, t, t)$: K, C, F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1, 8, 10), (1, 8, 16)$; the Koide source→pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229_lepton_frobenius_algebra, tfpt_2_standard_model, [E]/[C]).** $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$, prod-

uct $\frac{32}{9}$; $A=\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (LEP.FROB.01).

- **The center budget (7, 11, 13) as three local norms (v230_center_budget_norms, tfpt_2_standard_model, [E]).** $C=R+Q \operatorname{diag}(1, 0, 0)$ row sums (7, 11, 13): $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (hex), $13=N_{\mathbb{Z}[i]}(3+2i)$ (square), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218_diamond_axis_geometry, tfpt_2_standard_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C+xU) = 14+6x$ (slope $6=|R^+(A_3)|$) and quadratic on the sheet axis $\det(C+yV) = 14+14y+4y^2$ (2nd difference $8=\operatorname{rank} E_8$); the anchor block has $\det B(C+xU) = 2\det(C+xU)$ and sheet curvature $6=|R^+(A_3)|$ — two curvatures $(8, 6) = (\operatorname{rank} E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\operatorname{Pl}(K) \rightarrow \operatorname{Pl}(C) \rightarrow \operatorname{Pl}(F)$ lifts in steps $(1, 8, 10) = (N_\Phi, \operatorname{rank} E_8, A_\Lambda)$ then $(1, 8, 16) = (N_\Phi, \operatorname{rank} E_8, \dim S^+)$ — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2 \operatorname{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3 \cdot 5 \cdot 7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28=2 \dim G_2$, $52=\dim F_4$ stay audit-only, not spine, as in v94/v95); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (v216_marks_gauss_bonnet, tfpt_research_contracts, [E]; ledger QGEO.MARKS.03).** The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), $\operatorname{rank} H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling v217_marks_emergence_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension $\rightarrow 248/248$); v217 numerical (Python-only). Suite $\rightarrow 217$ modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise (v215_seam_deck_killtest, tfpt_research_contracts, [E]/[O]; ledger QGEO.KILL.01).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4 -deck-invariant). Levers: **K1** exactly four marks ($b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$); **K2** the square

config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0 \mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f \mathbb{Z}_4$ -invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side **[E]** (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 **[O]** (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite $\rightarrow$ 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01 (v214_seam_pillowcase, tfpt_research_contracts, [E]/[O]; ledger QGEO.PILLOW.01).** The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2, 2, 2, 2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio(μ_4) = 2 (v168) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ j -invariant/CM arithmetic); Klein- J values mpmath-numerical. Suite $\rightarrow$ 214 modules.

2026-06-15 (the F_transfer transfer functor contract CONTRACT.F.01)

- **F_{transfer} formalised as a typed functor with four axioms (v213_ftransfer_functor, tfpt_research_contracts, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4 -deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\|\text{Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity ($\text{spec } T$ positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are F_{pole} (v183), $F_{\text{Boltzmann}}$ (v212), F_{relic} (v211), F_{QCD} (v164). **CONTRACT.F.01** is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger **CONTRACT.F.01**.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211_axion_spine_angle, tfpt_4_frontier, [C]).** A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite- T solver reaches Ω_{DM} at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, **DM.AXION.SPINE.01**); **[X]** $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.

- **Leptogenesis scalaron-decouple branch** (`v212_leptogenesis_decouple`, `tfpt_4_frontier`, `[C]`). Both Boltzmann inputs share the decouple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R \varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10} . The coupling mechanism is posited not derived, so η_B stays `[C]`; `[X]` a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, `[O]`). A new module machine-proves the COHO node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ ($k=1,2,3$) have pullback character $\rho^* \omega_k = i^k \omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since $i^4=1$), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3) =$ the A_3 exponents $= \text{Spec}(Q_+)$, rank $3 = N_{\text{fam}}$. The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^* \omega_1 = \omega_3$, $\sigma^* \omega_2 = \omega_2$, $\sigma^* \omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHO/MODULE-parity nodes are now `[E]`; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay `[O]`.

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`MobiusUniformisation.lean`, ledger `FORM.QGEO.02`, `[O]`). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the clock $\rho : z \mapsto iz$ has $\rho^4 = \text{id}$ and $\rho^2 \neq \text{id}$ (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma \rho \sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification “ $z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays `[O]`, *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, `[O]`). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201/v210` (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext`, `Classical.choice`, `Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j < 4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$ for $\zeta^4=1$ (so a μ_4 -mark sum is supported only on modes $\equiv 0 \pmod{4}$); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now `[E]` (formally proved); the *premise* (the physical seam being mark-local) stays `[O]` — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (v210_mark_local_dtn, tfpt_research_contracts, [O]).** A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (v204_muon_seam_g2, tfpt_4_frontier, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.
- **An independent gravitational $3/4$ (v205_xi_threequarter, tfpt_4_frontier, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative H_0 from the Λ branch (v206_h0_lambda_branch, tfpt_4_frontier, [C]).** $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1/3}} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5\text{--}67.1$ km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207_asymptotic_safety, tfpt_4_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric} . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208_bh_thermodynamics, tfpt_horizon_readouts, [E]/[C]).** Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209_bh_regular_core, tfpt_horizon_readouts, [C]).** No singularity ($\varphi \rightarrow \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt_3, [O]).** Added a paragraph to the “Möbius origin of π ” section: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2, \mathbb{C})$ rotation (v180) and the $\text{RP} \rightarrow \text{OS}$ causal cone is

the same Lorentz structure — so “Möbius origin of π ” and “one Lorentzian cone” are two faces of one boundary $\mathrm{PSL}(2, \mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to c_3 (tfpt_4_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\mathrm{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\mathrm{seam}}) = 170.4^\circ$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\mathrm{BH}}(r) = 16c_3^4 Q_e^{\mathrm{eff}} Q_m^{\mathrm{eff}} / r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\mathrm{top}}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\mathrm{top}} = 48c_3^4$ that fixes the α -kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\mathrm{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude $Q_e Q_m$ is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]:** the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into tfpt_horizon_readouts (new EHT section), the ledger, registry, Wolfram (243/243), and the website (predictions.ts). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C]).** The closed TFPT CKM point ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \rightarrow \pi^+ \nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \rightarrow \pi^0 \nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test [X]:** a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into tfpt_2_standard_model (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (predictions.ts). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGEO.SUBPRIN.01 (v201, [E]/[O]).** The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4 -character-block-diagonal iff f is $\mathbb{Z}_4$ -invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4 -mark sum $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g , hence is automatically $\mathbb{Z}_4$ -invariant ([E]). So the μ_4 -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks

($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGEO.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver** (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]). The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170^\circ$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6\text{--}2.3$ (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120\text{--}130^\circ$ or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is $O(1)$ -uncertain, so a full finite- T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite- T solve confirms it** (`axion_relic/full_finiteT_solve.py`). A converged nonlinear misalignment solve ($\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170^\circ$ — $\sim 5.5\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106^\circ$. The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- `v199_seam_state_invariance.py` [E]/[O] (claim QGEO.STATE.01, work package A). Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4 -character classes $\{n=r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- `v200_seam_variational_scan.py` [C]/[O] (claim QGEO.VARI.02, work package B). The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4 ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the F_{transfer} four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb_tfpt_v1.yaml) and its status semantics already exist.* Suite now 199 modules, highest ID v200; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198_modular_commutator_reduction.py** [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$ and the clock $\rho: z \mapsto iz = \text{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω) , so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4 -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is μ_4 -invariant” ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **v195_marks_lefschetz_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited): $\rho: z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ D_4 marks” to the milder premise “ H^1 carries the A_3 -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **v196_seam_energy_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the QGEO.SYM.01 conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4 -breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197_rr_carrier_clifford_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194_raw_seam_dtn_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] the residual relocates: the full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01

reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- `v193_qgeo_energy_commutator.py` [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, `v117`), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (QGEO.COHOH.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (`v150`) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7)**. `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- `v189_riemann_roch_carrier.py` [E]/[C] (claim ARCH.RRCAR.01). The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already QGEO.COHOH.01) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\text{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- `v190_nariai_entropy_bound.py` [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- `v191_universal_branch_line.py` [C] (claim HOR.BRANCHLINE.01). The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known $2/3$ ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- `v192_qgeo_energy_clock.py` [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (`v186` skipped).
- *All four are validated external-review proposals (problem_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). run_all.py green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **v184_etaB_anchored_boltzmann.py [C] (claim FR.ETAB.03).** An honest test of the external-review η_B proposal ($M_1 = M_R \varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{P1}$ both miss it (c_3/φ_0 -powers off $\sim 75\%/ \sim 40\%$), so $M_1 = M_R \varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in tfpt_4_frontier.
- **v185_axion_relic_solver.py [C] (claim FR.DM.03).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ($m_a \approx 23.8 \mu\text{eV}$) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim 21\%$ at $\theta \sim O(1)$), so the $\theta_i \approx 170^\circ$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in tfpt_4_frontier.
- **v187_ftransfer_laws.py [E] (claim FR.TRANSFER.01).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B , axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in tfpt_4_frontier.
- **v188_frontier_wording_guard.py [E] (claim AUDIT.WORDING.01).** A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording $\rightarrow$ v184 [C]; the old “open relic scale” dark-matter line $\rightarrow$ v185; the old “near, not exact” Koide line $\rightarrow$ v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the tfpt_4_frontier Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, v_{geo} as unit gauge, G_{net} a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183_koide_f_corner_transfer.py** [E]/[C] (claim FR.KOIDE.07). An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an F_{transfer} special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182_reviewer_residual_map.py** [E]/[C] (claim REVIEW.MAP.01). A careful external review raised four concerns (A mapping $2D \rightarrow 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv \text{QGEO.SYM.01}$), D (Möbius forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer} , v_{geo} , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ($\sin^2 \theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (AX.P1.01, AX.P2.01, QGEO.IS0.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”)), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single

source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.

- **G_{net} typed in three explicit levels** (website “What is open” card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~ 181 scripts (foundations $\rightarrow$ SM readouts $\rightarrow$ seam = horizon $\rightarrow$ horizon/flavor geometry $\rightarrow$ R1–R5/premise-(A) $\rightarrow$ PyR@TE cross-checks $\rightarrow$ AQFT bridges $\rightarrow$ AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181_clock_is_conformal_symmetry.py** [E]/[O] (claim QGEO.SYM.01). The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the μ_4 -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain **REALIZE(v175)** → ... → **SYM.01(v181)** ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, **tfpt_5_redteam** Target A, the **tfpt_research_contracts** central-theorem keybox, **origin_theory**; and on the website **OpenGates**, **VerificationDag**, **papers.ts** (Target A $\times 2$, G_{net}), the **glossary** G_{net} entry and the **faq** live-residual answer — all now end at QGEO.SYM.01.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180_clock_is_möbius.py** [E]/[O] (claim QGEO.ISO.01). Cracks QGEO.CONF.01 one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, *non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) *isometry* $\Rightarrow$ *conformal* $\Rightarrow$ *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). Hence QGEO.CONF.01 is *implied* by the milder premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in **tfpt_research_contracts**; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **v179_conformal_realisation.py** [E]/[O] (claim QGEO.CONF.01). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so “genus-0” is the same $\chi = 2$ that gives the “8” in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). KERNEL then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual = $H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So

the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism”. The two doors of `v177/v178` are one geometric realisation, left honestly `[O]`. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations `v178` — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` `[E]/[O]` (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite `[E]` core of each and reduces each to *one sharper* premise. *Marks core* `[E]`: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ , the marks are forced to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* `[E]`: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (`v162`) and the unique residue-normalised basis (`v177`) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/`v156`; no drift `v158`). Net: neither obligation closed; both sharpened to milder premises with their finite cores `[E]`. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree `v177` — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` `[E]/[O]` (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`). A second external residual-class audit pointed out that the `v176` statement takes the marks/ D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` `[E]` — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful- D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening `v168`’s cross-ratio to the full uniformisation); `COHOM` `[E]` — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; `MODULE` `[E]` — a *unique* μ_4 -equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); `NET` `[E]` (`v154/v175`). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4 -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via `v168/v137/v162/v154/v175`). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem `v176` — the one central proof obligation)

- `v176_seam_collar_realisation.py` `[E]/[O]` (claim `QGEO.REALIZE.02`). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented

one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c=8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4 -equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$ (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \rightarrow \frac{53}{54}u$ source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT_RECORD_ID was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175_net_existence_full_cone.py** [E]/[O] (claim QFT.NETRP.01). The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m :* the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\text{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) *A_2 assembled + verified:* the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174_seam_fock_readings.py** [E] (claim QFT.FOCK.01). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four μ_4 -corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL ($\deg \leq 2$) keeps $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in tfpt_2 (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170_diamond_slice_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3, 4, 6, 10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3, 4, 8, 14, 20, 32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in tfpt_3 (slice atlas). Exact; reading is audit (anti-numerology).
- **v171_os_moment_cluster.py** [E]/[C] (claim QFT.OSMOMENT.01). Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster $729/665$, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$. Cited in tfpt_research_contracts (G5).
- **v172_trace_anomaly_seed.py** [E] (claim SEED.ANOMALY.01). The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 = \text{scalaron exponent } 48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ factorisation. Cited in tfpt_2 (seed).
- **v173_pfaffian_cp_car.py** [E] (claim FLAV.STRONGCP.02). Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5 -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in tfpt_2 (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo**. Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **v168_qgeo_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz / (z^4 - 1)$ ($k = 1, 2, 3$) are μ_4 -eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4 , four parabolic marks and this μ_4 -decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **v169_etaB_boltzmann_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240^\circ$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10} , untuned) — the route is *viable*. But M_1 /washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166_higgs_free_seam.py** [E]/[C] (claim HIGGS.FREESEAM.01). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral $c=8$ fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 at one loop to ~ 0.002). The double condition predicts $m_H \sim 129\text{--}134$ GeV; the same condition at the scalaron scale gives ~ 107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ , reduced to top-only (verification/pyrate/sm_higgs_betas.txt). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163_rg_stability_flavor.py** [E]/[C] (claim FLAV.RGSTAB.01). Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2 -suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2} Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2 -driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **v164_qcd_scale.py** [E]/[O] (claim QCD.LAMBDA.01). The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-

loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/ Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.

- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β ’s, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py`** [E]/[C] (claim `EM.BUDGET.02`). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}^{2g_{\text{car}}-2} + 1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored)**. `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_{transfer} + irreducible core, v167)

- **New module `v167_inventory_post_closure.py`** [E]/[C]. The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16–19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1 = N_{\text{fam}} = 3$); the EH step (R3) folds into v_{geo} (`v152`) and the ambient measure into G_{net} (`v76`). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit `v153`); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer} , one functor with four typed conditional interfaces (Koide, η_B , axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping**. Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A_2 by assembly, R1 = `GATE.QGEO`, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py`** [C]/[O]. Pins the two residual gates of the `v160–v162` (A)-chain to their floor. A_2 (**net existence**) closes by assembly of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–

Mack–Todorov), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$, $248=120+128$, character $E_4/\eta^8=j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via $DtN = |k|$ (v156/v157). R_1 (**seam clock**) **reduces to GATE.QGEO [C]**: a μ_4 -equivariant flat gapped generator is *unique* ($= A_0^*$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification “seam boundary $= \mathbb{P} \setminus \mu_4$ ” = **GATE.QGEO** (cohomology established, v137, $b_1=3=N_{\text{fam}}$). **Final accounting**: by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) $= (A_2 \text{ assembly}) + (R_1=\text{GATE.QGEO}) + \text{the } \{\pi, v_{\text{geo}}\} \text{ core}$, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger **GATE.METRIC.19**). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module v162_seam_transport_identification.py [E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced**: a μ_4 -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$ for $U = \text{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2=6=|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of $\omega_k=|k|$ (v113/v156). **Closure**: with v160+v161, (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)

- **New module v161_one_particle_reduction.py [E]/[O].** *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\text{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m=0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence**: v160’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ — verified by the *signed* set-partition/Pfaffian moment $\leftrightarrow$ cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap}>0$, the value $(2/3)^6$ only sets the rate. **The honest residual:** **v56**’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}=1820$ and $\binom{16}{6}=8008$ higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of **v56**”, which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (**v160** is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection \{YYYY-MM-DD...\}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, **v158**)

- **The free chiral $c=8$ fixed point is provably stable (**v158**, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current $h=1$, the quartic $h=2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the $h=1$ currents are chiral (**v157**, not $(1,1)$ -marginal), and the lowest interaction (quartic, $h=2$) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the $b_1=41/10$ cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; `contracts / origin_theory / next.txt / website` moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, **v157**)

- **The free-bulk premise becomes a fixed-point property (**v157**, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee-Uhlmann — order-1 ΨDO , principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic $c=8$

theory is *rigid* — every field has $\bar{h}=0$, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving $\mu=1$ ($248=120_{\text{NS}}+128_{\text{R}}$). With $(E_8)_1$ unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ($\Delta=6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5+3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q -series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN = $|k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the $248=120+128$ split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- G_{net} ’s last premise reduces to a single shared physical premise (v155, Python-only). The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression PTP of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net} , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- v_{geo} re-typed: the No-Unit Theorem (v153). Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π } (ledger ANCHOR.VGEO.02).

- **G_{net} stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has $[B:A]=|L|=4$, $c=5+3=8$, $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ (`[E]/[C]`), and F_{transfer} as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/origin_theory/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d=2$ that ratio *is* the induced Newton constant $1/G$ (v68). So v_{geo} (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay $\{\text{one scale}, \pi\}$, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin_theory/next.txt/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (rev {git count}) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.

20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO_TOKEN secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ~270 files / ~3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (`v53–v56`, `v101`, `v105`, `v113`, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]/[I]/[L]` markers in running text replaced by the 4-class `[O]/[E]`.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` on failure, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the `v150` EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by `v150` needs no separate derivation.

SEAM.THEOREM.01 narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger SEAM.EHMODEL.02).

- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin_theory/next.txt moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot 1$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0 -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and t -independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ -regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi) \int \sqrt{g}R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, origin_theory, contracts, next.txt, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 (4+5=9, \sigma \perp w_0(a))$ and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$. R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z \mapsto iz, z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is $\det'$ -clean (v147).** Variance ratio $(1 - \alpha)$ (forced by norm = area) gives the Born-squared ratio $(1 - \alpha)^{p^2}$, the $\ln$ series is the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue is choosing the R-projection ($248 = 120_{\text{NS}} + 128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, tfpt_1, origin_theory third update, next.txt) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{P1}$ vs M_{P1} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies” $\rightarrow$ “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, $2/3$ motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + $c=8 \Leftrightarrow$ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net} : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ error was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A \Sigma$ (v97/v98) pairs the $Q_+=1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \rightarrow -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading = $Q_+ \circ \rho$. GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings $\rightarrow$ two (v142, frame integrality).** Integer covectors in the $(1, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \pmod{11}$; the index is $\|\text{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$ the σ -pairing is forced $\equiv 0 \pmod{11}$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).

- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$ -graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion = $\mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1's det-ratio step derived in-family (v144).** e_2 -rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays [C] with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (SEAM.THEOREM.01); the gate keeps its [O] typing untouched.
- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1`; `tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers

the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-`[E]` artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline vN references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline vN references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$, A_s -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single

source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.

- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$ rows are made consistent with v86/COSMO.NSTAR.01: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star=51.4$ ($n_s=0.9611$, $r=0.0045$), replacing the stale point values $N_\star=55$ (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [E] $\rightarrow$ [C] (conditional on $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$), matching the other two tables and the ledger [P] typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry author, date and v5.2 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes

deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.

- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven **keyboxes** are reduced to the single headline “In one sentence” card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a **winbox** (a result, not a claim), and the explanatory *plausible/follows next* **winboxes** become prose. The loud boxes now carry only genuinely load-bearing callouts (one **keybox**, five **winboxes**, one **gapbox**) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only **keybox** (claim), **winbox** (result) and **gapbox** (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the $2/3$ *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (leading c_3 , fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result (0.3092 ± 0.0087 , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence

hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.

- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$); F_{transfer} is named as one missing functor with $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$ as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv \text{tfpt_N}$). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\text{spec } A_0^*$); $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law $\rightarrow$ zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).

- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\text{rowsum}(V) = \text{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the A_0^* zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents and cusp weights $= \text{spec } A_0^*$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the $R4'$ residue is three pairings; the $H^1 \rightarrow$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the **GATE.QGEO** residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking **xltabular** (it had silently truncated ~ 110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared **tfpt_docset.tex**; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this **changelog.tex** introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest **-check** mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s, t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank = c) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue A_0^* (diag = $a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff$

$a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.

- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.